



RECURSIVE COMPUTATION OF INVARIANT DISTRIBUTION OF FELLER PROCESS: APPLICATIONS AND NUMERICAL EXPERIMENTS

EA MAP

March 14, 2025

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1

INTRODUCTION

This EA report focuses on studying a recursive algorithm which aims to approximate the invariant distribution (denoted ν) of a Feller process $(X_t)_{t \geq 0}$ and studying the convergence rate of the empirical measures. The study is primarily based on the work of Gilles Pagès and Clément Rey in [1] and [2], supplemented by additional theoretical studies and numerical extensions.

Invariant measures are fundamental in characterizing the long-term behavior of Feller process; however, in many cases, there is no explicit formula for ν . To address this issue, a core concept involves approaching the Feller process (X_t) with a simulable process $(\bar{X}_{\Gamma_n})_{n \in \mathbb{N}^*}$, built with transition functions $(\mathcal{Q}_{\gamma_n})_{n \in \mathbb{N}^*}$. The step sequence $(\gamma_n)_{n \in \mathbb{N}^*}$ ($\Gamma_n = \sum_{k=1}^n \gamma_k$) can be carefully considered to ensure both the accuracy and efficiency. Under the proposed framework, the invariant measure of ν can be approximated through a weighted empirical measures of $(\bar{X}_{\Gamma_n})$, denote by ν_n^η , where $(\eta_n)_{n \in \mathbb{N}^*}$ is the weight sequence

$$\nu_n^\eta(dx) = \frac{1}{H_n} \sum_{k=1}^n \eta_k \delta_{\bar{X}_{\Gamma_{k-1}}}(dx), \quad H_n = \sum_{k=1}^n \eta_k. \quad (1)$$

By imposing specific assumptions, we obtain the almost sure weak convergence of these measures (see [1]). Moreover, when the invariant measure ν is unique, it can be proved that $\nu_n^\eta(f) \xrightarrow{n \rightarrow \infty} \nu(f)$ for a class of test functions f .

In the second part, a q -order Central Limit Theorem is established for the empirical measures (see [2]). Specifically, we deal with a q -weak order stochastic approximation $(\bar{X}_{\Gamma_n})_{n \in \mathbb{N}^*}$ for $(X_t)_{t \geq 0}$. By considering a carefully chosen weight sequence $(\eta_n)_{n \in \mathbb{N}}$ (see (1)) and under certain assumptions, we can obtain the rate of convergence in the CLT raised to $n^{q/(2q+1)}$. However, as q increases, the complexity of computing the weight sequence grows significantly. In this paper, we propose an alternative framework for computing the weight sequence that provides a more straightforward and computationally efficient implementation. However, this approach comes at the cost of losing one order in the convergence rate.

The report is organized in the following way. Section 2 reviews the results from [1] that establish almost sure weak convergence of ν_n^η within the framework of the recursive algorithm. Section 3 applies the recursive algorithm for a new discretization scheme of Ito diffusion, where we conduct deep analysis for the properties of solutions of a stochastic differential equation. Section 4 presents the result from [2], introducing the idea of extending this abstract framework to derive a q -order CLT for $q = 1, 2$. And Section 5 introduces our computational framework, detailing the necessary assumptions and the new convergence rate result.

Finally, the last section presents numerical simulations for a simplified one-dimensional case using the Ornstein–Uhlenbeck process as an example. This section is divided into four parts. First, we validate the basic convergence of the empirical measure and demonstrate related applications from [2] and [3]. Next, we briefly discuss parameter selection and adjustments. After establishing the framework for the central limit theorem from [2], we directly verify the convergence rate using histograms and the log-log method we proposed. At the end, we validate the new method presented in Section 5 and perform comparisons and analysis.

Section 4, Section 5, Section 6.3 - 6.4 correspond to our work in period 2.

2

GENERAL FRAMEWORK OF RECURSIVE COMPUTATION

2.1 CONSTRUCTION OF EMPIRICAL MEASURE

In this part we present the general framework of recursive computation detailed in [1]. Let $(\Omega, \mathcal{G}, \mathbb{P})$ be a probability space. We consider a Feller process $(X_t)_{t \geq 0}$ on $(\Omega, \mathcal{G}, \mathbb{P})$ taking values in a locally compact and separable metric space E . We denote by $(P_t)_{t \geq 0}$ the associated Feller semi-group on $\mathcal{C}_0(E)$, i.e. a collection of positive linear maps from $\mathcal{C}_0(E)$ to itself such that $P_0 f = f$, $P_{t+s} f = P_t P_s f$, $t, s \geq 0$, and $\lim_{t \rightarrow 0} \|P_t f - f\|_\infty = 0$. We can then introduce the infinitesimal generator of $(X_t)_{t \geq 0}$ as a linear operator A defined on a subspace $\mathcal{D}(A)$ of $\mathcal{C}_0(E)$, such that

$$\forall f \in \mathcal{D}(A), Af = \lim_{t \rightarrow 0} \frac{P_t f - f}{t}$$

exists for the $\|\cdot\|_\infty$ -norm, and $\mathcal{D}(A)$ is called the domain of A .

We are interested in the computation of the invariant distribution of $(X_t)_{t \geq 0}$, which is characterized by Echeverria–Weiss theorem:

$$\mathcal{V} = \{\nu \in \mathcal{P}(E) : \forall t \geq 0, P_t \nu = \nu\} = \{\nu \in \mathcal{P}(E) : \forall f \in \mathcal{D}(A), \nu(Af) = 0\}.$$

When explicit computation of ν is infeasible or the real process is difficult to simulate, the approximation methods become crucial. We begin by introducing an empirical scheme $(\bar{X}_{\Gamma_n})_{n \in \mathbb{N}^*}$ defined on a time grid $\Gamma_n = \sum_{k=1}^n \gamma_k$ with step sizes $(\gamma_n)_{n \in \mathbb{N}^*}$, satisfying:

$$0 < \gamma_n \leq \gamma, \quad \lim_{n \rightarrow \infty} \gamma_n = 0, \quad \lim_{n \rightarrow \infty} \Gamma_n = +\infty,$$

which is generally a simulatable approximation of $(X_t)_{t \geq 0}$ at a reasonable computational cost. The pseudo-generator $(\tilde{A}_{\gamma_n})_{n \in \mathbb{N}^*}$ of $(\bar{X}_{\Gamma_n})_{n \in \mathbb{N}^*}$ is a family of linear operators from $\mathcal{C}_0(E)$ to itself defined by

$$\forall f \in \mathcal{C}_0(E), \forall n \in \mathbb{N}^*, \tilde{A}_{\gamma_n} f = \frac{\mathcal{Q}_{\gamma_n} f - f}{\gamma_n},$$

where $\mathcal{Q}_{\gamma_n}$ is the transition probability distributions given by $\mathbb{P}(\bar{X}_{\Gamma_n} \in dy | \bar{X}_{\Gamma_{n-1}}) = \mathcal{Q}_{\gamma_n}(\bar{X}_{\Gamma_{n-1}}, dy)$. We can also write explicitly

$$\tilde{A}_{\gamma_n} f(x) = \frac{1}{\gamma_n} \mathbb{E}[f(\bar{X}_{\Gamma_n}) - f(\bar{X}_{\Gamma_{n-1}}) | \bar{X}_{\Gamma_{n-1}} = x].$$

Based on $(\bar{X}_{\Gamma_n})_{n \in \mathbb{N}^*}$, we can build the empirical measures with a weight sequence $\eta := (\eta_n)_{n \in \mathbb{N}^*}$ such that

$$\forall n \in \mathbb{N}^*, \eta_n > 0, \lim_n H_n = +\infty \text{ with } H_n = \sum_{k=1}^n \eta_k \quad (2)$$

by defining the random weighted empirical random measures as follows:

$$\nu_n^\eta(dx) = \frac{1}{H_n} \sum_{k=1}^n \eta_k \delta_{\bar{X}_{\Gamma_{k-1}}}(dx), \quad (3)$$

where δ_x is the Dirac delta measure centered at x .

The following part is dedicated to show that under certain conditions, a.s. every weak limiting distribution of $(\nu_n^\eta)_{n \in \mathbb{N}^*}$ belongs to $\mathcal{V}$. In particular, when the invariant measure of $(X_t)_{t \geq 0}$ is unique, i.e. $\mathcal{V} = \{\nu\}$, we show that $\lim_n \nu_n^\eta f = \nu f$ $\mathbb{P}$ -a.s. for a particular class of continuous test functions f .

2.2 ASSUMPTIONS

We now present the necessary assumptions in order to prove the convergence of the empirical measures $(\nu_n^\eta)_{n \in \mathbb{N}}$.

Mean-Reverting Recursive Control Let $\bar{\gamma} > 0$, let $s \geq 1$, let $\psi, \phi : [v_*, \infty) \rightarrow (0, +\infty)$ be Borel functions, and let $\alpha > 0$ and $\beta \in \mathbb{R}$. Suppose that there exists a Borel function V , which we call Lyapunov function, if it satisfies

$$\mathcal{L}_V \equiv (V : E \rightarrow [v_*, +\infty), \quad v_* > 0, \quad \lim_{x \rightarrow \infty} V(x) = +\infty) \quad (4)$$

and

$$\mathcal{RC}_{Q,V}(\psi, \phi, \alpha, \beta, s) \equiv \begin{cases} \text{(i)} & \tilde{A}_\gamma \psi \circ V \text{ exists for every } \gamma \in (0, \bar{\gamma}], \\ \text{(ii)} & \forall x \in E, \sup_{\gamma \in (0, \bar{\gamma}]} \tilde{A}_\gamma \psi \circ V(x) \leq \frac{\psi \circ V(x)}{V(x)} (\beta - \alpha \phi \circ V(x)), \\ \text{(iii)} & \liminf_{v \rightarrow \infty} \phi(v) > \frac{\beta}{\alpha}, \quad \text{and} \quad \lim_{v \rightarrow +\infty} \frac{\phi(v) \psi(v)^{1/s}}{v} = +\infty. \end{cases} \quad (5)$$

The function ϕ thus controls the mean-reverting property, we call it strongly mean-reverting property when $\phi = Id$, and weakly mean-reverting property when $\lim_{v \rightarrow \infty} \phi(v)/v = 0$. We introduce next the sets of functions for which the a.s. convergence holds:

$$C_{\tilde{V}_{\psi, \phi, s}}(E) := \{f \in \mathcal{C}(E) : |f(x)| = o(\tilde{V}_{\psi, \phi, s}(x))\}, \text{ where } \tilde{V}_{\psi, \phi, s}(x) := \frac{\phi \circ V(x) \psi \circ V(x)^{1/s}}{V(x)}. \quad (6)$$

Infinitesimal Generator Approximation This assumption control the distance between $(\tilde{A}_\gamma)_{\gamma > 0}$ and A . We assume that there exists $\mathcal{D}(A)_0 \subset \mathcal{D}(A)$ with $\mathcal{D}(A)_0$ dense in $\mathcal{C}_0(E)$ such that:

$$\mathcal{E}(\tilde{A}, A, \mathcal{D}(A)_0) \equiv \forall \gamma \in (0, \bar{\gamma}], \forall f \in \mathcal{D}(A)_0, \forall x \in E, |\tilde{A}_\gamma f(x) - A f(x)| \leq \Lambda_f(x, \gamma), \quad (7)$$

where $\Lambda_f : E \times \mathbb{R}^+ \rightarrow \mathbb{R}^+$ can be represented in the following way: Let $(\Omega, \tilde{\mathcal{G}}, \tilde{\mathbb{P}})$ be a probability space. Let $g : E \rightarrow \mathbb{R}_+^q$, $q \in \mathbb{N}$, be a locally bounded Borel measurable function, and let: $\tilde{\Lambda}_f : (E \times \mathbb{R}^+ \times \tilde{\Omega}, \mathcal{B}(E) \otimes \mathcal{B}(\mathbb{R}^+) \otimes \tilde{\mathcal{G}}) \rightarrow \mathbb{R}_+^q$ be a measurable function such that:

$$\sup_{i \in \{1, \dots, q\}} \tilde{\mathbb{E}} \left[\sup_{x \in E} \sup_{\gamma \in (0, \bar{\gamma}]} \tilde{\Lambda}_{f,i}(x, \gamma, \tilde{\omega}) \right] < +\infty, \text{ and } \forall x \in E, \forall \gamma \in (0, \bar{\gamma}], \Lambda_f(x, \gamma) = \langle g(x), \tilde{\mathbb{E}}[\tilde{\Lambda}_f(x, \gamma, \tilde{\omega})] \rangle_{\mathbb{R}^q}.$$

We assume that for every $i \in \{1, \dots, q\}$, $\sup_{n \in \mathbb{N}^*} \nu_n^\eta(g_i, \omega) < +\infty$ $\tilde{\mathbb{P}}(d\omega)$ -a.s. and that there exists a measurable function $\gamma : (\Omega, \tilde{\mathcal{G}}) \rightarrow ((0, \bar{\gamma}], \mathcal{B}((0, \bar{\gamma}]))$ such that one of the following two conditions holds:

(I) $\tilde{\mathbb{P}}(d\tilde{\omega})$ -a.s.

$$\begin{cases} \text{(i)} & \forall K \in \mathcal{K}_E, \lim_{\gamma \rightarrow 0} \sup_{x \in K} \tilde{\Lambda}_{f,i}(x, \gamma, \tilde{\omega}) = 0, \\ \text{(ii)} & \lim_{x \rightarrow \infty} \sup_{\gamma \in (0, \gamma(\tilde{\omega})]} \tilde{\Lambda}_{f,i}(x, \gamma, \tilde{\omega}) = 0. \end{cases} \quad (8)$$

(II) $\tilde{\mathbb{P}}(d\tilde{\omega})$ -a.s.

$$\lim_{\gamma \rightarrow 0} \sup_{x \in E} \tilde{\Lambda}_{f,i}(x, \gamma, \tilde{\omega}) g_i(x) = 0.$$

Growth Control and Step Weight Assumption We conclude with a hypothesis concerning the steps in the approximation. Let $\rho \in [1, 2]$ and $\epsilon_I : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ be an increasing function. For $F \subset \{f : (E, \mathcal{B}(E)) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))\}$ and a Borel function $g : E \rightarrow \mathbb{R}^+$, we assume that, for every $n \in \mathbb{N}$,

$$\mathcal{GC}_Q(F, g, \rho, \epsilon_I) \equiv \mathbb{P}\text{-a.s.} \quad \forall f \in F, \quad \mathbb{E} [|f(\bar{X}_{\Gamma_{n+1}}) - \mathcal{Q}_{\gamma_{n+1}} f(\bar{X}_{\Gamma_n})|^\rho | \bar{X}_{\Gamma_n}] \leq C_f \epsilon_I(\gamma_{n+1}) g(\bar{X}_{\Gamma_n}), \quad (9)$$

where $C_f > 0$ is a finite constant that may depend on f . We suppose that:

$$\mathcal{SW}_{\mathcal{J},\gamma,\eta}(g, \rho, \epsilon_I) \equiv \mathbb{P}\text{-a.s.} \quad \sum_{n=1}^{\infty} \left| \frac{\eta_n}{H_n \gamma_n} \right|^{\rho} \epsilon_I(\gamma_n) g(\bar{X}_{\Gamma_n}) < +\infty, \quad (10)$$

and

$$\mathcal{SW}_{\mathcal{J}\mathcal{J},\gamma,\eta}(F) \equiv \mathbb{P}\text{-a.s.} \quad \forall f \in F, \sum_{n=0}^{\infty} \frac{\left(\frac{\eta_{n+1}}{\gamma_{n+1}} - \frac{\eta_n}{\gamma_n} \right)^+}{H_{n+1}} |f(\bar{X}_{\Gamma_n})| < +\infty. \quad (11)$$

Notice that this last assumption holds as soon as the sequence $(\eta_n/\gamma_n)_{n \in \mathbb{N}^*}$ is non-increasing. Finally we state a lemma which is useful for verifying the last two assumptions (see Lemma 2.3 in [1] for a proof).

Lemma 1. *Let $v^* > 0$, $V : E \rightarrow [v^*, \infty)$, and $\psi, \phi : [v^*, \infty) \rightarrow \mathbb{R}^+$ be such that $\tilde{A}_{\gamma_n} \psi \circ V$ exists for every $n \in \mathbb{N}^*$. Let $\alpha > 0$, $\beta \in \mathbb{R}$, and $s \geq 1$. Let $(\theta_n)_{n \in \mathbb{N}^*}$ be a non-increasing sequence such that $\sum_{n \geq 1} \theta_n \gamma_n < +\infty$. We assume that $\mathcal{RC}_{Q,V}(\psi, \phi, \alpha, \beta, s)$ (see (5)) holds and that $\mathbb{E}[\psi \circ V(\tilde{X}_{\Gamma_{n_0}})] < +\infty$ for every $n_0 \in \mathbb{N}^*$. Then*

$$\sum_{n=1}^{\infty} \theta_n \gamma_n \mathbb{E}[\tilde{V}_{\psi,\phi,1}(X_{\Gamma_{n-1}})] < +\infty,$$

with $\tilde{V}_{\psi,\phi,1}$ defined in (6). In particular, let $\rho \in [1, 2]$ and let $\epsilon_I : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ be an increasing function. If we also assume that

$$\mathcal{SW}_{\mathcal{J},\gamma,\eta}(\rho, \epsilon_I) \equiv \left(\frac{1}{\gamma_n} \epsilon_I(\gamma_n) \left(\frac{\eta_n}{H_n \gamma_n} \right)^{\rho} \right)_{n \in \mathbb{N}^*} \text{ is non-increasing and } \sum_{n=1}^{\infty} \left(\frac{\eta_n}{H_n \gamma_n} \right)^{\rho} \epsilon_I(\gamma_n) < +\infty, \quad (12)$$

then we have $\mathcal{SW}_{\mathcal{J},\gamma,\eta}(\tilde{V}_{\psi,\phi,1}, \rho, \epsilon_I)$ (see (10)). Finally, if

$$\mathcal{SW}_{\mathcal{J}\mathcal{J},\gamma,\eta} \equiv \left(\left(\frac{\eta_{n+1}}{\gamma_{n+1}} - \frac{\eta_n}{\gamma_n} \right)^+ \frac{1}{\gamma_n \times H_n} \right)_{n \in \mathbb{N}^*} \text{ is non-increasing and } \sum_{n=1}^{\infty} \left(\frac{\eta_{n+1}}{\gamma_{n+1}} - \frac{\eta_n}{\gamma_n} \right)^+ \frac{1}{H_n} < +\infty, \quad (13)$$

then we have $\mathcal{SW}_{\mathcal{J}\mathcal{J},\gamma,\eta}(\tilde{V}_{\psi,\phi,1})$ (see (11)).

2.3 CONVERGENCE OF EMPIRICAL MEASURE

In this section, we state the main theorems of recursive computation (see Theorem 2.3 and Theorem 2.4 in [1] for a proof). First, we show a tightness property which ensures the existence of a weak limiting distribution for $(\nu_n^{\eta})_{n \in \mathbb{N}}$. Then, in a second step, we show that every limiting distributions identified with an invariant distribution of the Feller process $(X_t)_{t \geq 0}$.

Theorem 1 (Almost sure tightness). *Let $s \geq 1$, $\rho \in [1, 2]$, $v^* > 0$, and $V : E \rightarrow [v^*, \infty)$, $g : E \rightarrow \mathbb{R}^+$, $\psi : [v^*, \infty) \rightarrow \mathbb{R}^+$, $\epsilon_I : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ be an increasing function.*

(A) *Assume that $\tilde{A}_{\gamma_n}(\psi \circ V)^{1/s}$ exists for every $n \in \mathbb{N}^*$, and $\mathcal{GC}_Q((\psi \circ V)^{1/s}, g, \rho, \epsilon_I)$ (see (9)), $\mathcal{SW}_{\mathcal{J},\gamma,\eta}(g, \rho, \epsilon_I)$ (see (10)), and $\mathcal{SW}_{\mathcal{J}\mathcal{J},\gamma,\eta}((\psi \circ V)^{1/s})$ (see (11)) hold. Then*

$$\mathbb{P}\text{-a.s.} \sup_{n \in \mathbb{N}^*} -\frac{1}{H_n} \sum_{k=1}^n \eta_k \tilde{A}_{\gamma_k}(\psi \circ V)^{1/s}(\bar{X}_{\Gamma_{k-1}}) < +\infty. \quad (14)$$

(B) *Let $\alpha > 0$ and $\beta \in \mathbb{R}$. Let $\phi : [v^*, \infty) \rightarrow \mathbb{R}_+^*$ be a continuous function such that $C_{\phi} := \sup_{v \in [v^*, \infty)} \phi(v)/v < +\infty$. Assume that (14), $\mathcal{RC}_{Q,V}(\psi, \phi, \alpha, \beta, s)$ (see (5)), $\mathcal{L}_V$ (see (4)) hold. Then*

$$\mathbb{P}\text{-a.s.} \sup_{n \in \mathbb{N}^*} \nu_n^{\eta}(\tilde{V}_{\psi,\phi,s}) < +\infty,$$

with $\tilde{V}_{\psi,\phi,s}$ defined in (6). Therefore, $(\nu_n^{\eta})_{n \in \mathbb{N}^*}$ is $\mathbb{P}$ -a.s. tight.

Theorem 2 (Identification of the limit). *Let $\rho \in [1, 2]$.*

(A) *Let $\mathcal{D}(A)_0 \subset \mathcal{D}(A)$ with $\mathcal{D}(A)_0$ dense in $\mathcal{C}_0(E)$. Assume that $\tilde{A}_{\gamma_n} f$ exists for every $f \in \mathcal{D}(A)_0$ and every $n \in \mathbb{N}^*$. Also assume that there exists a Borel function $g : E \rightarrow \mathbb{R}^+$ and an increasing function $\epsilon_I : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ such that $\mathcal{GC}_Q(\mathcal{D}(A)_0, g, \rho, \epsilon_I)$ (see (9)), and $\mathcal{SW}_{\mathcal{J}, \gamma, \eta}(g, \rho, \epsilon_I)$ (see (10)), hold and that*

$$\lim_{n \rightarrow +\infty} \frac{1}{H_n} \sum_{k=1}^n \left| \frac{\eta_{k+1}}{\gamma_{k+1}} - \frac{\eta_k}{\gamma_k} \right| = 0. \quad (15)$$

Then

$$\mathbb{P}\text{-a.s. } \forall f \in \mathcal{D}(A)_0, \lim_{n \rightarrow +\infty} \frac{1}{H_n} \sum_{k=1}^n \eta_k \tilde{A}_{\gamma_k} f(X_{\Gamma_{k-1}}) = 0. \quad (16)$$

(B) *Assume that (16) and $\mathcal{E}(\tilde{A}, A, \mathcal{D}(A)_0)$ (see (7)) hold. Then*

$$\mathbb{P}\text{-a.s. } \forall f \in \mathcal{D}(A)_0, \lim_{n \rightarrow +\infty} \nu_n^\eta(Af) = 0.$$

It follows that, $\mathbb{P}$ -a.s., every weak limiting distribution ν_∞^η belongs to $\mathcal{V}$. Finally, if the hypotheses from Theorem 1 (B) hold and $\mathcal{V} = \{\nu\}$, then

$$\mathbb{P}\text{-a.s. } \forall f \in \tilde{C}_{\tilde{V}_\psi, \phi, s}(E), \lim_{n \rightarrow +\infty} \nu_n^\eta(f) = \nu(f).$$

with $\tilde{C}_{V_\psi, \phi, s}(E)$ defined in (6).

3

APPLICATION TO THE DISCRETIZATION SCHEME OF ITO DIFFUSION

This part is mainly inspired by Section 3.1 of Pagès Gilles, and Clément Rey's work [4], we will treat a new case of the discretization scheme of Ito diffusion with the framework proposed in the previous part. We propose an approach based on polynomial test functions under weakly mean-reverting assumptions. In the first section, we introduce the problem and state a main theorem. Then in this section, we verify all the necessary assumptions and finally give a proof.

3.1 MAIN RESULT

We consider a d -dimensional Brownian motion $(W_t)_{t \geq 0}$. We are interested in the solution of the d -dimensional stochastic equation

$$X_t = x + \int_0^t b(X_s) ds + \int_0^t \sigma(X_s) dW_s,$$

where $b : \mathbb{R}^d \rightarrow \mathbb{R}^d$ and $\sigma : \mathbb{R}^d \rightarrow \mathbb{R}^{d \times d}$ are locally bounded functions. The infinitesimal generator of this process is given by

$$Af(x) = \langle b(x), \nabla f(x) \rangle + \frac{1}{2} \sum_{i,j=1}^d (\sigma \sigma^*)_{i,j}(x) \frac{\partial^2 f}{\partial x_i \partial x_j}(x) = \langle b(x), \nabla f(x) \rangle + \frac{1}{2} \text{Tr}(\sigma \sigma^*(x) \nabla^2 f(x)) \quad (17)$$

and its domain $\mathcal{D}(A)$ contains $\mathcal{D}(A)_0 = \mathcal{C}_K^3(\mathbb{R}^d)$. Notice that $\mathcal{D}(A)_0$ is dense in $\mathcal{C}^0(\mathbb{R}^d)$. Assume a Lipschitz condition holds for b and σ :

$$\exists L, \forall x, y \in \mathbb{R}^d, \|b(x) - b(y)\| + \|\sigma(x) - \sigma(y)\|_F \leq L\|x - y\|, \quad (18)$$

where $\|\cdot\|$ is the Euclidean norm, $\|\cdot\|_F$ is the Frobenius norm defined by

$$\|\sigma\|_F := \sqrt{\sum_{i=1}^d \sum_{j=1}^d |\sigma_{ij}|^2} = \text{Tr}(\sigma^* \sigma)^{\frac{1}{2}}, \quad \forall \sigma \in \mathbb{R}^{d \times d}.$$

If we pick $y = 0$, the Lipschitz condition implies the following linear growth condition:

$$\exists L', \forall x \in \mathbb{R}^d, \|b(x)\| + \|\sigma(x)\|_F \leq L'(1 + \|x\|). \quad (19)$$

The Lipschitz condition (18) ensures in fact the existence and uniqueness of the solution $(X_t)_{t \geq 0}$, and it is a Feller process with Feller semi-group $P_t f : x \mapsto \mathbb{E}[f(X_t)]$ (see Corollary 19.27 in [5]). We introduce its discretization scheme defined by $(X_{\Gamma_n})_{n \in \mathbb{N}}$, where the times grid $\Gamma_n = \sum_{k=1}^n \gamma_k, n \in \mathbb{N}$ satisfies

$$\forall n \in \mathbb{N}^*, 0 < \gamma_n \leq \bar{\gamma} := \sup_{n \in \mathbb{N}} \gamma_n < \infty, \lim_n \gamma_n = 0, \lim_n \Gamma_n = \infty. \quad (20)$$

The pseudo-generator of the scheme $(\tilde{A}_{\gamma_n})_{n \in \mathbb{N}^*}$ is defined by

$$\forall f \in \mathcal{C}_0(\mathbb{R}^d), \tilde{A}_{\gamma_n} f(x) = \frac{1}{\gamma_n} \mathbb{E}[f(X_{\Gamma_n}) - f(X_{\Gamma_{n-1}}) \mid X_{\Gamma_{n-1}} = x].$$

Let $v_* > 0$ and let $\phi : [v_*, \infty) \rightarrow \mathbb{R}^+$ be a continuous function. Let $p > 0$ and define $\psi_p(y) = y^p$. Let $\alpha > 0$ and $\beta \in \mathbb{R}$, assume that

$$\liminf_{y \rightarrow \infty} \phi(y) \geq \frac{\beta}{\alpha}, \quad C_\phi := \sup_{y \geq v_*} \frac{\phi(y)}{y} < \infty. \quad (21)$$

We assume that the Lyapunov function $V : \mathbb{R}^d \rightarrow [v_*, \infty)$, satisfies $\mathcal{L}_V$:

$$\mathcal{L}_V \equiv (V : E \rightarrow [v_*, +\infty), \lim_{x \rightarrow \infty} V(x) = +\infty), \quad (22)$$

and is essentially quadratic in the sense

$$\|\nabla V\|^2 \leq C_V V, \quad \|D^2 V\|_\infty < +\infty. \quad (23)$$

We define

$$\forall x \in \mathbb{R}^d, \quad \lambda_\psi(x) := \lambda_{D^2 V(x) + \nabla V(x) \otimes^2 \psi'' \circ V(x) \psi' \circ V(x)^{-1}}. \quad (24)$$

When $\psi(y) = \psi_p(y) = y^p$, we will also use the notation λ_p instead of λ_ψ . Assume the mean-reverting property of V :

$$\mathcal{R}_p(\alpha, \beta, \phi, V) \equiv \forall x \in \mathbb{R}^d, \langle \nabla V(x), b(x) \rangle + \frac{1}{2} \chi_p(x) \leq \beta - \alpha \phi \circ V(x), \quad (25)$$

$$\text{with } \chi_p(x) = \begin{cases} \|\lambda_1\|_\infty \|\sigma(x)\|_F^2 & \text{if } p \leq 1, \\ \|\lambda_p\|_\infty 2^{(2p-3)+} \|\sigma(x)\|_F^2 & \text{if } p > 1. \end{cases} \quad (26)$$

And assume that

$$\mathcal{B}(\phi) \equiv \forall x \in \mathbb{R}^d, \quad \|\nabla V(x)\| (\|b(x)\|^2 + \|\sigma(x)\|_F^2)^{\frac{1}{2}} + \|b(x)\|^2 + \|\sigma(x)\|_F^2 \leq C \phi \circ V(x). \quad (27)$$

Remark 1. When $\phi(y) = y^a$ for $a \in [\frac{1}{2}, 1]$, assumption $\mathcal{B}(\phi)$ (see (27)) can be implied by

$$\forall x \in \mathbb{R}^d, \quad \|b(x)\|^2 + \|\sigma(x)\|_F^2 \leq C' V(x)^{2a-1}. \quad (28)$$

In fact, since $\|\nabla V(x)\| \leq C_V^{\frac{1}{2}} V(x)^{\frac{1}{2}}$, we have

$$\|\nabla V(x)\| (\|b(x)\|^2 + \|\sigma(x)\|_F^2)^{\frac{1}{2}} + \|b(x)\|^2 + \|\sigma(x)\|_F^2 \leq (C_V C')^{\frac{1}{2}} V(x)^a + C' V(x)^{2a-1} \leq C V(x)^a.$$

Theorem 3. Let $p \geq 1$, $a \in (0, 1]$, $s \geq 1$, $\rho \in [1, 2]$, $\psi_p(y) = y^p$, $\phi(y) = y^a$ and $\epsilon_I(\gamma) = \gamma^{\frac{\rho}{2}}$. Let $\alpha > 0$ and $\beta \in \mathbb{R}$. Assume that Lipschitz condition (see (18)), (20), (21), $\mathcal{L}_V$ (see (22)), (23), $\mathcal{R}_p(\alpha, \beta, \phi, V)$ (see (25)), $\mathcal{B}(\phi)$ (see (27)), $\mathcal{SW}_{\mathcal{J}, \gamma, \eta}(\rho, \epsilon_I)$ (see (12)), $\mathcal{SW}_{\mathcal{J}, \gamma, \eta}(V^{p/s})$ (see (11)) and (15) are satisfied, and that $\text{app} \leq p + a - 1$. If $\frac{p}{s} + a - 1 > 0$, then $(\nu_n^\eta)_{n \in \mathbb{N}^*}$ is $\mathbb{P}$ -a.s. tight and

$$\mathbb{P} - \text{a.s.} \quad \sup_{n \in \mathbb{N}^*} \nu_n^\eta(V^{p/s+a-1}) < +\infty. \quad (29)$$

Moreover, assume that there is some $\epsilon > 0$, such that $g_\sigma \leq C V^{p/s+a-1}$, with $g_\sigma = (\|b\|^2 + \|\sigma\|_F^2)^{1+\epsilon}$. Then every weak limiting distribution ν of $(\nu_n^\eta)_{n \in \mathbb{N}^*}$ is an invariant distribution of $(X_t)_{t \geq 0}$, and when ν is unique, we have

$$\mathbb{P} - \text{a.s.} \quad \forall f \in \mathcal{C}_{\tilde{V}_{\psi_p, \phi, s}}(\mathbb{R}^d), \quad \lim_{n \rightarrow \infty} \nu_n^\eta(f) = \nu(f),$$

with $\mathcal{C}_{\tilde{V}_{\psi_p, \phi, s}}(\mathbb{R}^d)$ defined in (6).

Remark 2. 1. When $\frac{p}{s} \leq p + a - 1$, assumption $\mathcal{SW}_{\mathcal{J}, \gamma, \eta}(V^{p/s})$ (see (11)) can be replaced by $\mathcal{SW}_{\mathcal{J}, \gamma, \eta}$ (see (13)) by Lemma 1.

2. When $\frac{p}{s} > 1$, there is some $\epsilon > 0$, such that $g_\sigma \leq C V^{p/s+a-1}$, with $g_\sigma = (\|b\|^2 + \|\sigma\|_F^2)^{1+\epsilon}$ by assumption $\mathcal{B}(\phi)$ (see (21)).

3.2 RECURSIVE CONTROL

Proposition 1. *Under the assumptions of (18), (20), (21), and assume that (22), (23), (25), (27) are satisfied. Then, for every $\tilde{\alpha} \in (0, \alpha)$, there exists $n_0 \in \mathbb{N}^*$, such that*

$$\forall n \geq n_0, \forall x \in \mathbb{R}^d, \tilde{A}_{\gamma_n} \psi_p \circ V(x) \leq \frac{\psi_p \circ V(x)}{V(x)} p(\beta - \tilde{\alpha} \phi \circ V(x)). \quad (30)$$

Before getting into the main proof, we first introduce some lemmas which are curial in the proof.

Lemma 2. *Let $l \in \mathbb{N}^*$, we have the following inequality:*

$$\forall \alpha > 0, \forall u_i \in \mathbb{R}^d, i = 1, \dots, l, \left| \sum_{i=1}^l u_i \right|^\alpha \leq l^{(\alpha-1)+} \sum_{i=1}^l |u_i|^\alpha. \quad (31)$$

Lemma 3 (Burkholder inequality). *For all $p \geq 2$, there exists $C_p > 0$ s.t.*

$$\mathbb{E} \left[\sup_{s \leq T} \left\| \int_0^s \sigma(X_u) dW_u \right\|^p \right] \leq C_p \mathbb{E} \left[\left(\int_0^T \|\sigma(X_u)\|_F^2 du \right)^{\frac{p}{2}} \right] \quad (32)$$

Proof. Since all the matrix norms are equivalent, we obtain this result by a standard Burkholder inequality (see Theorem 18.17 in [5]). $\square$

Lemma 4. *With the notions above and assume $\gamma_{n+1} \leq 1$, we have*

$$\begin{aligned} \mathbb{E}[\|X_{\Gamma_{n+1}} - X_{\Gamma_n}\|^2 \mid X_{\Gamma_n}] &= \mathbb{E} \left[\left\| \int_{\Gamma_n}^{\Gamma_{n+1}} b(X_s) ds \right\|^2 + \int_{\Gamma_n}^{\Gamma_{n+1}} \|\sigma(X_s)\|_F^2 ds \mid X_{\Gamma_n} \right] \\ &\leq \mathbb{E} \left[\int_{\Gamma_n}^{\Gamma_{n+1}} \gamma_{n+1} \|b(X_s)\|^2 + \|\sigma(X_s)\|_F^2 ds \mid X_{\Gamma_n} \right]. \end{aligned} \quad (33)$$

In particular,

$$\begin{aligned} \mathbb{E} \left[\int_{\Gamma_n}^{\Gamma_{n+1}} \|X_s - X_{\Gamma_n}\|^2 ds \mid X_{\Gamma_n} \right] &= \int_{\Gamma_n}^{\Gamma_{n+1}} \mathbb{E}[\|X_s - X_{\Gamma_n}\|^2 \mid X_{\Gamma_n}] ds \\ &\leq \int_{\Gamma_n}^{\Gamma_{n+1}} \mathbb{E} \left[\int_{\Gamma_n}^s \|b(X_u)\|^2 + \|\sigma(X_u)\|_F^2 du \mid X_{\Gamma_n} \right] ds \leq \gamma_{n+1} \mathbb{E} \left[\int_{\Gamma_n}^{\Gamma_{n+1}} \|b(X_s)\|^2 + \|\sigma(X_s)\|_F^2 ds \mid X_{\Gamma_n} \right]. \end{aligned} \quad (34)$$

Proof. Since the solution of a stochastic differential equation is a homogeneous Markov process (see Section 2 in [6]), we have

$$\mathbb{E}[\Psi(X_{\bullet+s}) \mid \mathcal{F}_s] = \mathbb{E}[\Psi(X_{\bullet+s}) \mid X_s] = \mathbb{E}^{X_s}[\Psi(X_{\bullet})], \quad (35)$$

for all bounded measurable functionals Ψ of the sample paths (see Remark 6.3 in [5]). Hence

$$\begin{aligned} \mathbb{E}[\|X_{\Gamma_{n+1}} - X_{\Gamma_n}\|^2 \mid X_{\Gamma_n}] &= \mathbb{E}^{X_{\Gamma_n}}[\|X_{\gamma_{n+1}} - X_0\|^2] \\ &= \mathbb{E}^{X_{\Gamma_n}} \left[\left\| \int_0^{\gamma_{n+1}} b(X_s) ds \right\|^2 \right] + \mathbb{E}^{X_{\Gamma_n}} \left[\left\| \int_0^{\gamma_{n+1}} \sigma(X_s) dW_s \right\|^2 \right]. \end{aligned} \quad (36)$$

The second term is treated by multivariate Ito isometry (see Appendix D, Lemma 18 in [7]):

$$\mathbb{E}^{X_{\Gamma_n}} \left[\left\| \int_0^{\gamma_{n+1}} \sigma(X_s) dW_s \right\|^2 \right] = \mathbb{E}^{X_{\Gamma_n}} \left[\int_0^{\gamma_{n+1}} \|\sigma(X_s)\|_F^2 ds \right] = \mathbb{E} \left[\int_{\Gamma_n}^{\Gamma_{n+1}} \|\sigma(X_s)\|_F^2 ds \mid X_{\Gamma_n} \right]. \quad (37)$$

And we conclude by using the Markov property to the first term in (36). $\square$

Lemma 5. Suppose that $p \geq 1$, C_{2p} defined as in Lemma 3, let $0 < \delta < 1$ s.t. $1 - 3^{p-1}2^{2p}L^2C_{2p}\delta^p > 0$, let $K > 0$ s.t.

$$(1 - 3^{p-1}2^{2p}L^2C_{2p}\delta^p)K \geq 3^{p-1} + 3^{p-1}L^p2^{2p+\frac{1}{2}}C_{2p}^{\frac{1}{2}}\delta^{\frac{p}{2}}K^{\frac{1}{2}}, \quad (38)$$

Suppose $n_0 \in \mathbb{N}$ s.t. $\lambda_{n+1} \leq \delta, \forall n \geq n_0$, then for every $n \geq n_0$, we have

$$\mathbb{E} \left[\left(\int_{\Gamma_n}^{\Gamma_{n+1}} \|b(X_s)\|^2 ds \right)^p + \left(\int_{\Gamma_n}^{\Gamma_{n+1}} \|\sigma(X_s)\|_F^2 ds \right)^p \middle| X_{\Gamma_n} \right] \leq K\gamma_{n+1}^p (\|b(X_{\Gamma_n})\|^{2p} + \|\sigma(X_{\Gamma_n})\|_F^{2p}). \quad (39)$$

Proof. Using the fact that (X_t) is a Markov process as mentioned in Lemma 4, (39) is equivalent to

$$\mathbb{E}^{X_{\Gamma_n}} \left[\left(\int_0^{\gamma_{n+1}} \|b(X_s)\|^2 ds \right)^p + \left(\int_0^{\gamma_{n+1}} \|\sigma(X_s)\|_F^2 ds \right)^p \right] \leq K\gamma_{n+1}^p (\|b(X_{\Gamma_n})\|^{2p} + \|\sigma(X_{\Gamma_n})\|_F^{2p}). \quad (40)$$

For simplicity we note

$$P := \mathbb{E}^{X_{\Gamma_n}} \left[\left(\int_0^{\gamma_{n+1}} \|b(X_s)\|^2 ds \right)^p + \left(\int_0^{\gamma_{n+1}} \|\sigma(X_s)\|_F^2 ds \right)^p \right]. \quad (41)$$

$$Q := \gamma_{n+1}^p (\|b(X_{\Gamma_n})\|^{2p} + \|\sigma(X_{\Gamma_n})\|_F^{2p}). \quad (42)$$

We have

$$\mathbb{E}^{X_{\Gamma_n}} \left[\left(\int_0^{\gamma_{n+1}} \|\sigma(X_s)\|_F^2 ds \right)^p \right] = \mathbb{E}^{X_{\Gamma_n}} \left[\left(\int_0^{\gamma_{n+1}} (\|\sigma(X_s)\|_F^2 - \|\sigma(X_{\Gamma_n})\|_F^2) ds + \gamma_{n+1} \|\sigma(X_{\Gamma_n})\|_F^2 \right)^p \right] \quad (43)$$

Since σ is Lipschitz,

$$\begin{aligned} \int_0^{\gamma_{n+1}} (\|\sigma(X_s)\|_F^2 - \|\sigma(X_{\Gamma_n})\|_F^2) ds &= \int_0^{\gamma_{n+1}} \langle 2\sigma(X_{\Gamma_n}), \sigma(X_s) - \sigma(X_{\Gamma_n}) \rangle_F + \|\sigma(X_s) - \sigma(X_{\Gamma_n})\|_F^2 ds \\ &\leq 2L\|\sigma(X_{\Gamma_n})\|_F \int_0^{\gamma_{n+1}} \|X_s - X_{\Gamma_n}\| ds + L^2 \int_0^{\gamma_{n+1}} \|X_s - X_{\Gamma_n}\|^2 ds \end{aligned} \quad (44)$$

By using Lemma 2 and (44), we obtain from (43) that

$$\begin{aligned} \mathbb{E}^{X_{\Gamma_n}} \left[\left(\int_0^{\gamma_{n+1}} \|\sigma(X_s)\|_F^2 ds \right)^p \right] &\leq 3^{p-1} \left((2L)^p \|\sigma(X_{\Gamma_n})\|_F^p \mathbb{E}^{X_{\Gamma_n}} \left[\left(\int_0^{\gamma_{n+1}} \|X_s - X_{\Gamma_n}\| ds \right)^p \right] \right. \\ &\quad \left. + L^2 \mathbb{E}^{X_{\Gamma_n}} \left[\left(\int_0^{\gamma_{n+1}} \|X_s - X_{\Gamma_n}\|^2 ds \right)^p \right] + \gamma_{n+1}^p \|\sigma(X_{\Gamma_n})\|_F^{2p} \right) \end{aligned} \quad (45)$$

By applying Cauchy's inequality and Jensen's inequality, we have

$$\begin{aligned} \mathbb{E}^{X_{\Gamma_n}} \left[\left(\int_0^{\gamma_{n+1}} \|X_s - X_{\Gamma_n}\| ds \right)^p \right] &\leq \mathbb{E}^{X_{\Gamma_n}} \left[\left(\gamma_{n+1}^{\frac{1}{2}} \left(\int_0^{\gamma_{n+1}} \|X_s - X_{\Gamma_n}\|^2 ds \right)^{\frac{1}{2}} \right)^p \right] \\ &\leq \gamma_{n+1}^{\frac{p}{2}} \mathbb{E}^{X_{\Gamma_n}} \left[\left(\int_0^{\gamma_{n+1}} \|X_s - X_{\Gamma_n}\|^2 ds \right)^p \right]^{\frac{1}{2}}. \end{aligned} \quad (46)$$

We compute the last term in (46):

$$\begin{aligned} \mathbb{E}^{X_{\Gamma_n}} \left[\left(\int_0^{\gamma_{n+1}} \|X_s - X_{\Gamma_n}\|^2 ds \right)^p \right] &= \mathbb{E}^{X_{\Gamma_n}} \left[\left(\int_0^{\gamma_{n+1}} \left\| \int_0^s b(X_u) du + \int_0^s \sigma(X_u) dW_u \right\|^2 ds \right)^p \right] \\ &\leq \mathbb{E}^{X_{\Gamma_n}} \left[\left(\int_0^{\gamma_{n+1}} \left(2s \int_0^s \|b(X_u)\|^2 du + 2 \left\| \int_0^s \sigma(X_u) dW_u \right\|^2 \right) ds \right)^p \right] \\ &\leq \mathbb{E}^{X_{\Gamma_n}} \left[\gamma_{n+1}^{p-1} \int_0^{\gamma_{n+1}} \left(2s \int_0^s \|b(X_u)\|^2 du + 2 \left\| \int_0^s \sigma(X_u) dW_u \right\|^2 \right)^p ds \right] \quad (\text{Jensen's inequality}) \\ &\leq \gamma_{n+1}^{2p} 2^{2p-1} \mathbb{E}^{X_{\Gamma_n}} \left[\left(\int_0^{\gamma_{n+1}} \|b(X_u)\|^2 du \right)^p \right] + \gamma_{n+1}^p 2^{2p-1} \mathbb{E}^{X_{\Gamma_n}} \left[\sup_{s \leq \gamma_{n+1}} \left\| \int_0^s \sigma(X_u) dW_u \right\|^{2p} \right] \\ &\leq \gamma_{n+1}^p 2^{2p-1} C_{2p} P \quad (\text{Burkholder inequality, see Lemma 3}) \end{aligned} \quad (47)$$

From (45), (46) and (47) we have

$$\begin{aligned} \mathbb{E}^{X_{\Gamma_n}} \left[\left(\int_0^{\gamma_{n+1}} \|\sigma(X_s)\|_F^2 ds \right)^p \right] &\leq 3^{p-1} (2L)^p 2^{p-\frac{1}{2}} C_{2p}^{\frac{1}{2}} \gamma_{n+1}^{\frac{p}{2}} Q^{\frac{1}{2}} P^{\frac{1}{2}} \\ &\quad + 3^{p-1} L^2 2^{2p-1} C_{2p} \gamma_{n+1}^p P + 3^{p-1} \gamma_{n+1}^p \|\sigma(X_{\Gamma_n})\|_F^2. \end{aligned} \quad (48)$$

Similarly, by replacing the Frobenius norm of $\sigma(X)$ in (48) by the Euclidean norm of $b(X)$ and adding two inequalities, we get

$$P \leq 3^{p-1} L^p 2^{2p+\frac{1}{2}} C_{2p}^{\frac{1}{2}} \gamma_{n+1}^{\frac{p}{2}} Q^{\frac{1}{2}} P^{\frac{1}{2}} + 3^{p-1} L^2 2^{2p} C_{2p} \gamma_{n+1}^p P + 3^{p-1} Q. \quad (49)$$

Since $\gamma_{n+1} \leq \delta$, from (38) we get

$$(1 - 3^{p-1} 2^{2p} L^2 C_{2p} \gamma_{n+1}^p) K \geq 3^{p-1} + 3^{p-1} L^p 2^{2p+\frac{1}{2}} C_{2p}^{\frac{1}{2}} \gamma_{n+1}^{\frac{p}{2}} K^{\frac{1}{2}},$$

and with (49), we easily see that $P \leq KQ$. $\square$

Corollary 1. Suppose that $p > 0$, let δ, K, n_0 be defined as in Lemma 5, then for all $n \geq n_0$, there exists $M_p > 0$ which depends on p s.t.

$$\mathbb{E}[\|X_{\Gamma_{n+1}} - X_{\Gamma_n}\|^{2p} \mid X_{\Gamma_n}] \leq M_p \gamma_{n+1}^p (\|b(X_{\Gamma_n})\|^2 + \|\sigma(X_{\Gamma_n})\|_F^2)^p. \quad (50)$$

Proof. Case $p \geq 1$. By applying Lemma 2, we have

$$\begin{aligned} \mathbb{E}[\|X_{\Gamma_{n+1}} - X_{\Gamma_n}\|^{2p} \mid X_{\Gamma_n}] &\leq 2^{2p-1} \mathbb{E} \left[\left\| \int_{\Gamma_n}^{\Gamma_{n+1}} b(X_s) ds \right\|^{2p} \mid X_{\Gamma_n} \right] \\ &\quad + 2^{2p-1} \mathbb{E} \left[\left\| \int_{\Gamma_n}^{\Gamma_{n+1}} \sigma(X_s) dW_s \right\|^{2p} \mid X_{\Gamma_n} \right]. \end{aligned}$$

The first term can be controlled using Cauchy's inequality:

$$\mathbb{E} \left[\left\| \int_{\Gamma_n}^{\Gamma_{n+1}} b(X_s) ds \right\|^{2p} \mid X_{\Gamma_n} \right] \leq \gamma_{n+1}^p \mathbb{E} \left[\left(\int_{\Gamma_n}^{\Gamma_{n+1}} \|b(X_s)\|^2 ds \right)^p \mid X_{\Gamma_n} \right].$$

And we can treat the second term with Burkholder inequality (see Lemma 3):

$$\begin{aligned} \mathbb{E} \left[\left\| \int_{\Gamma_n}^{\Gamma_{n+1}} \sigma(X_s) dW_s \right\|^{2p} \mid X_{\Gamma_n} \right] &= \mathbb{E}^{X_{\Gamma_n}} \left[\left\| \int_0^{\gamma_{n+1}} \sigma(X_s) dW_s \right\|^{2p} \right] \\ &\leq C_{2p} \mathbb{E}^{X_{\Gamma_n}} \left[\left(\int_{\Gamma_n}^{\Gamma_{n+1}} \|\sigma(X_s)\|_F^2 ds \right)^p \right] = C_{2p} \mathbb{E} \left[\left(\int_{\Gamma_n}^{\Gamma_{n+1}} \|\sigma(X_s)\|_F^2 ds \right)^p \mid X_{\Gamma_n} \right]. \end{aligned}$$

Then (50) is a direct consequence of Lemma 5.

Case $p < 1$. Since $x \mapsto x^p$ is concave, we can use Jensen's inequality to conclude that

$$E[(\|X_{\Gamma_{n+1}} - X_{\Gamma_n}\|^2)^p \mid X_{\Gamma_n}] \leq E[\|X_{\Gamma_{n+1}} - X_{\Gamma_n}\|^2 \mid X_{\Gamma_n}]^p \leq M_1^p \gamma_{n+1}^p (\|b(X_{\Gamma_n})\|^2 + \|\sigma(X_{\Gamma_n})\|_F^2)^p.$$

$\square$

Proof of Proposition 1. By definition of $\tilde{A}_{\gamma_n}$, (30) is equivalent to

$$\mathbb{E}[\psi \circ V(X_{\Gamma_{n+1}}) - \psi \circ V(X_{\Gamma_n}) \mid X_{\Gamma_n}] \leq \frac{\gamma_{n+1} \psi \circ V(X_{\Gamma_n})}{V(X_{\Gamma_n})} (\beta - \alpha \phi \circ V(X_{\Gamma_n})). \quad (51)$$

Case $p \geq 1$. From Taylor's formula and the definition of λ_p (see (24)) we have

$$\begin{aligned} \psi_p \circ V(X_{\Gamma_{n+1}}) &= \psi_p \circ V(X_{\Gamma_n}) + \langle X_{\Gamma_{n+1}} - X_{\Gamma_n}, \nabla V(X_{\Gamma_n}) \rangle \psi'_p \circ V(X_{\Gamma_n}) \\ &\quad + \frac{1}{2} (D^2 V(\Upsilon_{n+1}) \psi'_p \circ V(\Upsilon_{n+1}) + \nabla V(\Upsilon_{n+1})^{\otimes 2} \psi''_p \circ V(\Upsilon_{n+1})) (X_{\Gamma_{n+1}} - X_{\Gamma_n})^{\otimes 2} \\ &\leq \psi_p \circ V(X_{\Gamma_n}) + \langle X_{\Gamma_{n+1}} - X_{\Gamma_n}, \nabla V(X_{\Gamma_n}) \rangle \psi'_p \circ V(X_{\Gamma_n}) \\ &\quad + \frac{1}{2} \lambda_p(\Upsilon_{n+1}) \psi'_p \circ V(\Upsilon_{n+1}) |X_{\Gamma_{n+1}} - X_{\Gamma_n}|^2, \end{aligned} \quad (52)$$

with $\Upsilon_{n+1} \in (X_{\Gamma_n}, X_{\Gamma_{n+1}})$. First we deduce from (23) that $\sup_{x \in \mathbb{R}^d} \lambda_p(x) < \infty$. And we have

$$\begin{aligned} &\mathbb{E}[\langle X_{\Gamma_{n+1}} - X_{\Gamma_n}, \nabla V(X_{\Gamma_n}) \rangle \mid X_{\Gamma_n}] - \gamma_{n+1} \langle b(X_{\Gamma_n}), \nabla V(X_{\Gamma_n}) \rangle \\ &= \mathbb{E}[\int_{\Gamma_n}^{\Gamma_{n+1}} \langle b(X_s) - b(X_{\Gamma_n}), \nabla V(X_{\Gamma_n}) \rangle ds \mid X_{\Gamma_n}] \leq L \|\nabla V(X_{\Gamma_n})\| \mathbb{E}[\int_{\Gamma_n}^{\Gamma_{n+1}} \|X_s - X_{\Gamma_n}\| ds \mid X_{\Gamma_n}] \\ &\leq L \gamma_{n+1}^{\frac{1}{2}} \|\nabla V(X_{\Gamma_n})\| \mathbb{E}[\int_{\Gamma_n}^{\Gamma_{n+1}} \|X_s - X_{\Gamma_n}\|^2 ds \mid X_{\Gamma_n}]^{\frac{1}{2}}, \end{aligned} \quad (53)$$

where we used (46) with $p = 1$ for the last inequality. Using Lemma 4 (see (34)) and Lemma 5 (see (39) with $p = 1$), we deduce from (53) that

$$\begin{aligned} &\mathbb{E}[\langle X_{\Gamma_{n+1}} - X_{\Gamma_n}, \nabla V(X_{\Gamma_n}) \rangle \mid X_{\Gamma_n}] - \gamma_{n+1} \langle b(X_{\Gamma_n}), \nabla V(X_{\Gamma_n}) \rangle \\ &\leq L \gamma_{n+1}^{\frac{3}{2}} \|\nabla V(X_{\Gamma_n})\| \mathbb{E}[\frac{1}{\gamma_{n+1}} \int_{\Gamma_n}^{\Gamma_{n+1}} \|b(X_s)\|^2 + \|\sigma(X_s)\|_F^2 ds \mid X_{\Gamma_n}]^{\frac{1}{2}} \\ &\leq L K^{\frac{1}{2}} \gamma_{n+1}^{\frac{3}{2}} \|\nabla V(X_{\Gamma_n})\| (\|b(X_{\Gamma_n})\|^2 + \|\sigma(X_{\Gamma_n})\|_F^2)^{\frac{1}{2}}, \end{aligned} \quad (54)$$

for all $n \geq n_0$ with n_0, K defined as in Lemma 5. From Lemma 4 (see (33)) we have

$$\begin{aligned} \mathbb{E}[\|X_{\Gamma_{n+1}} - X_{\Gamma_n}\|^2 \mid X_{\Gamma_n}] &\leq \mathbb{E}[\int_{\Gamma_n}^{\Gamma_{n+1}} \gamma_{n+1} \|b(X_s)\|^2 + \|\sigma(X_s)\|_F^2 ds \mid X_{\Gamma_n}] \\ &\leq \gamma_{n+1}^2 \mathbb{E}[\frac{1}{\gamma_{n+1}} \int_{\Gamma_n}^{\Gamma_{n+1}} \|b(X_s)\|^2 + \|\sigma(X_s)\|_F^2 ds \mid X_{\Gamma_n}] + \mathbb{E}[\int_{\Gamma_n}^{\Gamma_{n+1}} \|\sigma(X_s)\|_F^2 ds \mid X_{\Gamma_n}] \end{aligned} \quad (55)$$

For the last term of (55), we use (48) and Lemma 5 (see (39)) with $p = 1$:

$$\begin{aligned} &\mathbb{E}[\int_{\Gamma_n}^{\Gamma_{n+1}} \|\sigma(X_s)\|_F^2 ds \mid X_{\Gamma_n}] - \gamma_{n+1} \|\sigma(X_{\Gamma_n})\|_F^2 \leq 2^{\frac{3}{2}} L C_2^{\frac{1}{2}} \gamma_{n+1}^{\frac{1}{2}} Q^{\frac{1}{2}} P^{\frac{1}{2}} + 2L^2 C_2 \gamma_{n+1} P \\ &\leq (2^{\frac{3}{2}} L C_2^{\frac{1}{2}} K^{\frac{1}{2}} \gamma_{n+1}^{\frac{1}{2}} + 2L^2 C_2 \gamma_{n+1} K) Q = \gamma_{n+1}^{\frac{3}{2}} C (\|b(X_{\Gamma_n})\|^2 + \|\sigma(X_{\Gamma_n})\|_F^2), \end{aligned} \quad (56)$$

with $C = 2^{\frac{3}{2}} L C_2^{\frac{1}{2}} K^{\frac{1}{2}} + (2L^2 C_2 + 1) \gamma_{n+1}^{\frac{1}{2}} K$, where we adapt the notations P and Q defined in (41) and (42) for a more concise writing. Together with (55) and (56), we obtain

$$\mathbb{E}[\|X_{\Gamma_{n+1}} - X_{\Gamma_n}\|^2 \mid X_{\Gamma_n}] \leq \gamma_{n+1} \|\sigma(X_{\Gamma_n})\|_F^2 + \gamma_{n+1}^{\frac{3}{2}} C (\|b(X_{\Gamma_n})\|^2 + \|\sigma(X_{\Gamma_n})\|_F^2), \quad (57)$$

Assume first that $p = 1$. From (52), (54) and (57) we have

$$\begin{aligned} \mathbb{E}[V(X_{\Gamma_{n+1}}) - V(X_{\Gamma_n}) \mid X_{\Gamma_n}] &\leq \gamma_{n+1} \left(\langle b(X_{\Gamma_n}), \nabla V(X_{\Gamma_n}) \rangle + \frac{1}{2} \|\lambda_1\|_{\infty} \|\sigma(X_{\Gamma_n})\|_F^2 \right) \\ &\quad + \gamma_{n+1}^{\frac{3}{2}} \left(L K^{\frac{1}{2}} \|\nabla V(X_{\Gamma_n})\| (\|b(X_{\Gamma_n})\|^2 + \|\sigma(X_{\Gamma_n})\|_F^2)^{\frac{1}{2}} + \frac{1}{2} \|\lambda_1\|_{\infty} C (\|b(X_{\Gamma_n})\|^2 + \|\sigma(X_{\Gamma_n})\|_F^2) \right). \end{aligned}$$

By using $\mathcal{B}(\phi)$ (see (27)), for every $\tilde{\alpha} \in (0, \alpha)$, there exists $n_0(\tilde{\alpha}) > n_0$ such that for every $n \geq n_0(\tilde{\alpha})$,

$$\begin{aligned} \gamma_{n+1}^{\frac{3}{2}} (LK^{\frac{1}{2}} \|\nabla V(X_{\Gamma_n})\| (\|b(X_{\Gamma_n})\|^2 + \|\sigma(X_{\Gamma_n})\|_F^2)^{\frac{1}{2}} + \frac{1}{2} \|\lambda_1\|_{\infty} C(\|b(X_{\Gamma_n})\|^2 + \|\sigma(X_{\Gamma_n})\|_F^2)) \\ \leq \gamma_{n+1}(\alpha - \tilde{\alpha}) \phi \circ V(X_{\Gamma_n}). \end{aligned}$$

From assumption $\mathcal{R}_p(\alpha, \beta, \phi, V)$ (see (25)), we conclude that

$$\tilde{A}_{\gamma_n} \psi \circ V(x) \leq \beta - \tilde{\alpha} \phi \circ V(x).$$

Assume now that $p > 1$. Since $\|\nabla V\|^2 \leq C_V V$, $\sqrt{V}$ is Lipschitz. We use Lemma 2 to obtain that

$$V^{p-1}(\Upsilon_{n+1}) \leq (\sqrt{V}(X_{\Gamma_n}) + [\sqrt{V}]_1 \|X_{\Gamma_{n+1}} - X_{\Gamma_n}\|)^{2p-2} \leq 2^{(2p-3)+} V^{p-1}(X_{\Gamma_n}) + [\sqrt{V}]_1^{2p-2} \|X_{\Gamma_{n+1}} - X_{\Gamma_n}\|^{2p-2}. \quad (58)$$

Then by using (54), (57), (58) and Corollary 1 (see (50)), we derive from (52) that

$$\begin{aligned} \mathbb{E}[V^p(X_{\Gamma_{n+1}}) - V^p(X_{\Gamma_n}) \mid X_{\Gamma_n}] &\leq \gamma_{n+1} \langle b(X_{\Gamma_n}), \nabla V(X_{\Gamma_n}) \rangle p V^{p-1}(X_{\Gamma_n}) \\ &\quad + \frac{1}{2} \gamma_{n+1} \|\lambda_p\|_{\infty} 2^{(2p-3)+} p V^{p-1}(X_{\Gamma_n}) \|\sigma(X_s)\|_F^2 + R, \end{aligned}$$

$$\begin{aligned} \text{with } R &= \gamma_{n+1}^{\frac{3}{2}} LK^{\frac{1}{2}} \|\nabla V(X_{\Gamma_n})\| (\|b(X_{\Gamma_n})\|^2 + \|\sigma(X_{\Gamma_n})\|_F^2)^{\frac{1}{2}} p V^{p-1}(X_{\Gamma_n}) \\ &\quad + \gamma_{n+1}^{\frac{3}{2}} \frac{1}{2} \|\lambda_p\|_{\infty} 2^{(2p-3)+} p V^{p-1}(X_{\Gamma_n}) C(\|b(X_{\Gamma_n})\|^2 + \|\sigma(X_{\Gamma_n})\|_F^2) \\ &\quad + \gamma_{n+1}^p \frac{p}{2} \|\lambda_p\|_{\infty} [\sqrt{V}]_1^{2p-2} M (\|b(X_{\Gamma_n})\|^2 + \|\sigma(X_{\Gamma_n})\|_F^2)^p. \end{aligned}$$

Notice that $\phi \circ V(x) < C_{\phi} V(x)$ (see (21)) and by using $\mathcal{B}(\phi)$ (see (27)), for every $\tilde{\alpha} \in (0, \alpha)$, there exists $n_0(\tilde{\alpha}) > n_0$ such that for every $n \geq n_0(\tilde{\alpha})$,

$$R \leq \gamma_{n+1}(\alpha - \tilde{\alpha}) p V^{p-1}(X_{\Gamma_n}) \phi \circ V(X_{\Gamma_n}),$$

which leads to the recursive control for $p > 1$ from assumption $\mathcal{R}_p(\alpha, \beta, \phi, V)$ (see (25)).

Case $p < 1$. Since $x \mapsto x^p$ is concave, we have

$$V^p(X_{\Gamma_{n+1}}) - V^p(X_{\Gamma_n}) \leq p V^{p-1}(X_{\Gamma_n}) (V(X_{\Gamma_{n+1}}) - V(X_{\Gamma_n})).$$

The recursive control for $p < 1$ follows immediately the case $p = 1$ which we have just proved. $\square$

3.3 PROOF OF THE INFINITESIMAL ESTIMATION

Proposition 2. Assume the Lipschitz condition (18) and that there exists $\epsilon > 0$ s.t.

$$\sup_{n \in \mathbb{N}^*} \nu_n^{\eta} ((1 + \|b\|^2 + \|\sigma\|_F^2)^{\epsilon} (\|b\|^2 + \|\sigma\|_F^2)) < +\infty \quad (59)$$

Then $\mathcal{E}(\tilde{A}, A, \mathcal{D}_0(A))$ (see (7)) holds, i.e. there is $\bar{\gamma}$ such that

$$\forall \gamma \in (0, \bar{\gamma}], \forall f \in \mathcal{C}_K^3(\mathbb{R}^d), \forall x \in \mathbb{R}^d, \|\tilde{A}_{\gamma} f(x) - A f(x)\| \leq \Lambda_f(x, \gamma),$$

where $\Lambda_f(x, \gamma) = g(x) \tilde{\Lambda}_f(x, \gamma)$ with $g : \mathbb{R}^d \rightarrow \mathbb{R}_+$ a locally bounded Borel measurable function. Moreover, g and $\tilde{\Lambda}_f$ satisfy $\sup_{n \in \mathbb{N}^*} \nu_n^{\eta}(g) < +\infty$ and

$$\begin{cases} (i) \forall K \in \mathcal{K}_{\mathbb{R}^d}, \lim_{\gamma \rightarrow 0} \sup_{x \in K} \tilde{\Lambda}_f(x, \gamma) = 0, \\ (ii) \lim_{x \rightarrow \infty} \sup_{\gamma \in (0, \bar{\gamma}]} \tilde{\Lambda}_f(x, \gamma) = 0. \end{cases} \quad (60)$$

We remark that (60) is in fact a special case of (8).

Proof. For $f \in \mathcal{C}_K^3(\mathbb{R}^d)$, we note d the diameter of $\text{supp}(f)$, i.e. $d = \max_{x \in \text{supp}(f)} \|x\|$. To treat the pseudo-generator $\tilde{A}_\gamma$, we first apply an Itô formula:

$$\mathbb{E}[f(X_{\Gamma_{n+1}}) - f(X_{\Gamma_n}) \mid X_{\Gamma_n}] = \mathbb{E}\left[\int_{\Gamma_n}^{\Gamma_{n+1}} \langle b(X_s), \nabla f(X_s) \rangle ds + \int_{\Gamma_n}^{\Gamma_{n+1}} \frac{1}{2} \text{Tr}(\sigma \sigma^*(X_s) \nabla^2 f(X_s)) ds \mid X_{\Gamma_n}\right]. \quad (61)$$

We begin by studying the first term of (61):

$$\begin{aligned} & \mathbb{E}\left[\int_{\Gamma_n}^{\Gamma_{n+1}} \langle b(X_s), \nabla f(X_s) \rangle ds \mid X_{\Gamma_n}\right] - \gamma_{n+1} \langle b(X_{\Gamma_n}), \nabla f(X_{\Gamma_n}) \rangle \\ &= \mathbb{E}\left[\int_{\Gamma_n}^{\Gamma_{n+1}} \langle \nabla f(X_s) - \nabla f(X_{\Gamma_n}), b(X_{\Gamma_n}) \rangle + \langle \nabla f(X_s), b(X_s) - b(X_{\Gamma_n}) \rangle ds \mid X_{\Gamma_n}\right]. \end{aligned} \quad (62)$$

We observe that, if $\|X_s - X_{\Gamma_n}\| < \|X_{\Gamma_n}\| - d$, which also implies $\|X_{\Gamma_n}\| > d$, we can deduce that X_{Γ_n}, X_s do not belong to $\text{supp}(f)$. Thus we can add the indicator function $\mathbb{1}_{\|X_s - X_{\Gamma_n}\| \geq \|X_{\Gamma_n}\| - d}$ in the RHS of (62). Let $\alpha > 0$ such that $\frac{\alpha}{2} < \epsilon$, since $\nabla f \in \mathcal{C}_K^2(\mathbb{R}^d)$, it is α -Hölder continuous, and we define $|\nabla f|_{C^{0,\alpha}}$ to be its Hölder coefficient. Then the first term in the RHS of (62) can be controlled as:

$$\begin{aligned} & \mathbb{E}\left[\int_{\Gamma_n}^{\Gamma_{n+1}} \langle \nabla f(X_s) - \nabla f(X_{\Gamma_n}), b(X_{\Gamma_n}) \rangle ds \mid X_{\Gamma_n}\right] \\ & \leq |\nabla f|_{C^{0,\alpha}} \|b(X_{\Gamma_n})\| \mathbb{E}\left[\int_{\Gamma_n}^{\Gamma_{n+1}} \|X_s - X_{\Gamma_n}\|^\alpha \mathbb{1}_{\|X_s - X_{\Gamma_n}\| \geq \|X_{\Gamma_n}\| - d} ds \mid X_{\Gamma_n}\right] \\ & \stackrel{\text{Hölder}}{\leq} |\nabla f|_{C^{0,\alpha}} \|b(X_{\Gamma_n})\| \mathbb{E}\left[\|X_s - X_{\Gamma_n}\|^2 \mid X_{\Gamma_n}\right]^{\frac{\alpha}{2}} \mathbb{E}\left[\mathbb{1}_{\|X_s - X_{\Gamma_n}\| \geq \|X_{\Gamma_n}\| - d} ds \mid X_{\Gamma_n}\right]^{\frac{2-\alpha}{2}} \\ & \leq \gamma_{n+1} |\nabla f|_{C^{0,\alpha}} \|b(X_{\Gamma_n})\| \sup_{\Gamma_n \leq s \leq \Gamma_{n+1}} \mathbb{E}\left[\|X_s - X_{\Gamma_n}\|^2 \mid X_{\Gamma_n}\right]^{\frac{\alpha}{2}} \sup_{\Gamma_n \leq s \leq \Gamma_{n+1}} \mathbb{E}\left[\mathbb{1}_{\|X_s - X_{\Gamma_n}\| \geq \|X_{\Gamma_n}\| - d} \mid X_{\Gamma_n}\right]^{\frac{2-\alpha}{2}}. \end{aligned}$$

For the last term, we can use Markov's inequality:

$$\mathbb{E}[\mathbb{1}_{\|X_s - X_{\Gamma_n}\| \geq \|X_{\Gamma_n}\| - d} \mid X_{\Gamma_n}] \leq \min\left(\frac{\mathbb{E}[\|X_s - X_{\Gamma_n}\|^2 \mid X_{\Gamma_n}]}{(\|X_{\Gamma_n}\| - d)^2}, 1\right) \mathbb{1}_{\|X_{\Gamma_n}\| > d} + \mathbb{1}_{\|X_{\Gamma_n}\| \leq d}$$

By applying Corollary 1 we conclude that

$$\frac{1}{\gamma_{n+1}} \mathbb{E}\left[\int_{\Gamma_n}^{\Gamma_{n+1}} \langle \nabla f(X_s) - \nabla f(X_{\Gamma_n}), b(X_{\Gamma_n}) \rangle ds \mid X_{\Gamma_n}\right] \leq g_1(X_{\Gamma_n}) \tilde{\Lambda}_{f,1}(X_{\Gamma_n}, \gamma_{n+1}),$$

with

$$\begin{aligned} g_1(x) &= |\nabla f|_{C^{0,\alpha}} \|b(x)\| M_1^{\frac{\alpha}{2}} (\|b(x)\|^2 + \|\sigma(x)\|_F^2)^{\frac{\alpha}{2}} (1 + \|b(x)\|^2 + \|\sigma(x)\|_F^2)^\delta, \\ \tilde{\Lambda}_{f,1}(x, \gamma) &= \frac{\gamma^{\frac{\alpha}{2}}}{(1 + \|b(x)\|^2 + \|\sigma(x)\|_F^2)^\delta} \left(\min\left(\frac{(M_1 \gamma)^{\frac{2-\alpha}{2}} (\|b(x)\|^2 + \|\sigma(x)\|_F^2)^{\frac{2-\alpha}{2}}}{(\|x\| - d)^{2-\alpha}}, 1\right) \mathbb{1}_{\|x\| > d} + \mathbb{1}_{\|x\| \leq d} \right), \end{aligned}$$

where M_1 is defined as in Corollary 1, $\delta > 0$ s.t. $\frac{\alpha}{2} + \delta < \epsilon$. Since b and σ have sub-linear growth (see (19)), it's easy to see that $\tilde{\Lambda}_{f,1}$ satisfies the condition (60). And from (59), we obtain that $\sup_{n \in \mathbb{N}_*} \nu_n^\eta(g_1) < +\infty$. Similarly, we use the fact that b is L-Lipschitz to the second term in the RHS of (62), and we obtain

$$\frac{1}{\gamma_{n+1}} \mathbb{E}\left[\int_{\Gamma_n}^{\Gamma_{n+1}} \langle \nabla f(X_s), b(X_s) - b(X_{\Gamma_n}) \rangle ds \mid X_{\Gamma_n}\right] \leq g_2(X_{\Gamma_n}) \tilde{\Lambda}_{f,2}(X_{\Gamma_n}, \gamma_{n+1}),$$

with

$$g_2(x) = L \|\nabla f\|_\infty M_1^{\frac{1}{2}} (\|b(x)\|^2 + \|\sigma(x)\|_F^2)^{\frac{1}{2}} (1 + \|b(x)\|^2 + \|\sigma(x)\|_F^2)^\delta,$$

$$\tilde{\Lambda}_{f,2}(x, \gamma) = \frac{\gamma^{\frac{1}{2}}}{(1 + \|b(x)\|^2 + \|\sigma(x)\|_F^2)^\delta} \left(\min \left(\frac{M_1^{\frac{1}{2}} \gamma^{\frac{1}{2}} (\|b(x)\|^2 + \|\sigma(x)\|_F^2)^{\frac{1}{2}}}{\|x\| - d}, 1 \right) \mathbb{1}_{\|x\| > d} + \mathbb{1}_{\|x\| \leq d} \right).$$

For the second term of (61), we write the matrix trace as a Frobenius inner product, and obtain a similar equation as (61):

$$\begin{aligned} \mathbb{E} \left[\int_{\Gamma_n}^{\Gamma_{n+1}} \frac{1}{2} \langle \sigma \sigma^*(X_s), \nabla^2 f(X_s) \rangle_F ds \mid X_{\Gamma_n} \right] - \gamma_{n+1} \frac{1}{2} \langle \sigma \sigma^*(X_{\Gamma_n}), \nabla^2 f(X_{\Gamma_n}) \rangle_F \\ = \frac{1}{2} \mathbb{E} \left[\int_{\Gamma_n}^{\Gamma_{n+1}} \langle \sigma \sigma^*(X_{\Gamma_n}), \nabla^2 f(X_s) - \nabla^2 f(X_{\Gamma_n}) \rangle_F + \langle \sigma \sigma^*(X_s) - \sigma \sigma^*(X_{\Gamma_n}), \nabla^2 f(X_s) \rangle_F ds \mid X_{\Gamma_n} \right]. \end{aligned}$$

Since σ is L -Lipschitz, and using the fact that $\|\sigma \sigma^*\|_F \leq \|\sigma\|_F^2$, we have

$$\|\sigma \sigma^*(X_s) - \sigma \sigma^*(X_{\Gamma_n})\|_F \leq L^2 \|X_s - X_{\Gamma_n}\|^2 + 2L \|\sigma(X_{\Gamma_n})\|_F \|X_s - X_{\Gamma_n}\|.$$

Therefore, from a similar reasoning, by using Cauchy's inequality to the term

$$\mathbb{E}[\|X_s - X_{\Gamma_n}\|^2 \mathbb{1}_{\|X_s - X_{\Gamma_n}\| > \|X_{\Gamma_n}\| - d} \mid X_{\Gamma_n}]$$

and Corollary 1 with $p = 2$, we obtain

$$\frac{1}{\gamma_{n+1}} \mathbb{E} \left[\int_{\Gamma_n}^{\Gamma_{n+1}} \langle \sigma \sigma^*(X_s) - \sigma \sigma^*(X_{\Gamma_n}), \nabla^2 f(X_s) \rangle_F ds \mid X_{\Gamma_n} \right] \leq g_3(X_{\Gamma_n}) \tilde{\Lambda}_{f,3}(X_{\Gamma_n}, \gamma_{n+1}),$$

with

$$\begin{aligned} g_3(x) &= \|\nabla^2 f\|_{F,\infty} (2LM_1^{\frac{1}{2}} + L^2 M_2^{\frac{1}{2}} \gamma^{\frac{1}{2}}) (\|b(x)\|^2 + \|\sigma(x)\|_F^2) (1 + \|b(x)\|^2 + \|\sigma(x)\|_F^2)^\delta, \\ \tilde{\Lambda}_{f,3}(x, \gamma) &= \frac{\gamma^{\frac{1}{2}}}{(1 + \|b(x)\|^2 + \|\sigma(x)\|_F^2)^\delta} \left(\min \left(\frac{M_1^{\frac{1}{2}} \gamma^{\frac{1}{2}} (\|b(x)\|^2 + \|\sigma(x)\|_F^2)^{\frac{1}{2}}}{\|x\| - d}, 1 \right) \mathbb{1}_{\|x\| > d} + \mathbb{1}_{\|x\| \leq d} \right), \end{aligned}$$

and

$$\frac{1}{\gamma_{n+1}} \mathbb{E} \left[\int_{\Gamma_n}^{\Gamma_{n+1}} \langle \sigma \sigma^*(X_{\Gamma_n}), \nabla^2 f(X_s) - \nabla^2 f(X_{\Gamma_n}) \rangle_F ds \mid X_{\Gamma_n} \right] \leq g_4(X_{\Gamma_n}) \tilde{\Lambda}_{f,4}(X_{\Gamma_n}, \gamma_{n+1}),$$

with

$$\begin{aligned} g_4(x) &= |\nabla^2 f|_{C^{0,\alpha}} \|\sigma(x)\|_F^2 M_1^{\frac{\alpha}{2}} (\|b(x)\|^2 + \|\sigma(x)\|_F^2)^{\frac{\alpha}{2}} (1 + \|b(x)\|^2 + \|\sigma(x)\|_F^2)^\delta, \\ \tilde{\Lambda}_{f,4}(x, \gamma) &= \frac{\gamma^{\frac{\alpha}{2}}}{(1 + \|b(x)\|^2 + \|\sigma(x)\|_F^2)^\delta} \left(\min \left(\frac{(M_1 \gamma)^{\frac{2-\alpha}{2}} (\|b(x)\|^2 + \|\sigma(x)\|_F^2)^{\frac{2-\alpha}{2}}}{(\|x\| - d)^{2-\alpha}}, 1 \right) \mathbb{1}_{\|x\| > d} + \mathbb{1}_{\|x\| \leq d} \right). \end{aligned}$$

To conclude, we have $\|\tilde{\Lambda}_\gamma f(x) - A f(x)\| \leq \sum_{i=1}^4 g_i(x) \tilde{\Lambda}_{f,i}(x, \gamma)$. Since there is some constant C s.t. $g_i \leq C(1 + \|b\|^2 + \|\sigma\|_F^2)^\epsilon (\|b\|^2 + \|\sigma\|_F^2)$ and $\tilde{\Lambda}_{f,i}(x, \gamma)$ satisfies (60) for all $i = 1, 2, 3, 4$, we have completed the proof. $\square$

3.4 PROOF OF GROWTH CONTROL AND STEP WEIGHT ASSUMPTIONS

Proposition 3. Let $p > 0$, $a \in (0, 1]$, $s \geq 1$, $\rho \in [1, 2]$, $\psi(y) = y^p$ and $\phi(y) = y^a$. We assume (18) and (20). Then for every $n \in \mathbb{N}$, for every $f \in \mathcal{D}(A)_0 = \mathcal{C}_K^3(\mathbb{R}^d)$,

$$\mathbb{E}[\|f(X_{\Gamma_{n+1}}) - f(X_{\Gamma_n})\|^\rho \mid X_{\Gamma_n}] \leq C_f \gamma_{n+1}^{\frac{\rho}{2}} (\|b(X_{\Gamma_n})\|^2 + \|\sigma(X_{\Gamma_n})\|_F^2)^{\frac{\rho}{2}}. \quad (63)$$

In other words, we have $\mathcal{GC}_Q(\mathcal{D}(A)_0, g_\sigma, \rho, \epsilon_I)$ (see (9)) with $g_\sigma = (\|b\|^2 + \|\sigma\|_F^2)^{\frac{\rho}{2}}$ and $\epsilon_I(\gamma) = \gamma^{\frac{\rho}{2}}$. Moreover, if (23) and $\mathcal{B}(\phi)$ (see (27)) hold and $\text{app}/s \leq p + a - 1$. Then for every $n \in \mathbb{N}$, we have

$$\mathbb{E}[\|V^{p/s}(X_{\Gamma_{n+1}}) - V^{p/s}(X_{\Gamma_n})\|^\rho \mid X_{\Gamma_n}] \leq C \gamma_{n+1}^{\frac{\rho}{2}} V^{p+a-1}(X_{\Gamma_{n+1}}). \quad (64)$$

In other words, we have $\mathcal{GC}_Q(V^{p/s}, V^{p+a-1}, \rho, \epsilon_I)$ (see (9)) with $\epsilon_I(\gamma) = \gamma^{\frac{\rho}{2}}$.

Proof. By using Corollary 1 with $p = \frac{1}{2}$, we know that there is $n_0 \in \mathbb{N}$ s.t.

$$\mathbb{E}[\|X_{\Gamma_{n+1}} - X_{\Gamma_n}\| \mid X_{\Gamma_n}] \leq \gamma_{n+1}^{\frac{1}{2}} M_{\frac{1}{2}}(\|b(X_{\Gamma_n})\|^2 + \|\sigma(X_{\Gamma_n})\|_F^2)^{\frac{1}{2}}.$$

The rest of proof is the same as in Lemma 3.8 in [4]. □

3.5 PROOF OF THEOREM 3

The proof consists of verifying all the assumption introduced in Theorem 1 and Theorem 2.

Step 1. Mean-reverting recursive control. Since (23), $\mathcal{B}(\phi)$ (see (27)) and $\mathcal{R}_p(\alpha, \beta, \phi, V)$ (see (25)) hold, it follows from Proposition 1 that $\mathcal{RC}_{Q,V}(\psi_p, \phi, p\tilde{\alpha}, p\beta, s)$ (see (5)) is satisfied for every $\tilde{\alpha} \in (0, \alpha)$ and every $s \geq 1$ such that $p/s + a - 1 > 0$.

Step 2. Step weight assumption. We have $\mathcal{SW}_{\mathcal{J},\gamma,\eta}(V^{p/s})$ (see (11)) from assumption, and we can show that $\mathcal{SW}_{\mathcal{J},\gamma,\eta}(V^{p+a-1}, \rho, \epsilon_I)$ (see (10)) by Lemma 1.

Step 3. Growth control assumption. Now, we prove $\mathcal{GC}_{\mathcal{Q}}(F, V^{p+a-1}, \rho, \epsilon_I)$ (see (9)) for $F = \mathcal{D}(A)_0$ and $F = \{V^{p/s}\}$. We notice that $\frac{p}{2} \leq 1$, thus $\frac{ap}{2} \leq p + a - 1$. From $\mathcal{B}(\phi)$ (see (21)), we obtain that

$$(\|b(X_{\Gamma_n})\|^2 + \|\sigma(X_{\Gamma_n})\|_F^2)^{\frac{p}{2}} \leq C^{\frac{p}{2}} V^{\frac{ap}{2}}(X_{\Gamma_n}) \leq C^{\frac{p}{2}} V^{a+p-1}(X_{\Gamma_n}).$$

Then Proposition 3 shows that $\mathcal{GC}_{\mathcal{Q}}(F, V^{p+a-1}, \rho, \epsilon_I)$ (see (9)) for $F = \mathcal{D}(A)_0$ and $F = \{V^{p/s}\}$.

Step 4. Conclusion.

1. The first part of Theorem 3 (see (29)) is a consequence of Theorem 1. We obtain from Steps 2 and 3 that assumptions $\mathcal{GC}_{\mathcal{Q}}(V^{p/s}, V^{p+a-1}, \rho, \epsilon_I)$ (see (9)), $\mathcal{SW}_{\mathcal{J},\gamma,\eta}(V^{p+a-1}, \rho, \epsilon_I)$ (see (10)) and $\mathcal{SW}_{\mathcal{J},\gamma,\eta}(V^{p/s})$ (see (11)) hold, which are the hypotheses from Theorem 1 (A) with $g = V^{p+a-1}$.

Due to Step 1, assumption $\mathcal{RC}_{Q,V}(\psi_p, \phi, p\tilde{\alpha}, p\beta, s)$ (see (5)) is satisfied for every $\tilde{\alpha} \in (0, \alpha)$ and every $s \geq 1$ such that $p/s + a - 1 > 0$. Moreover, since $\mathcal{L}_V$ (see (22)) holds, the hypotheses from Theorem 1 (B) are satisfied. We thus conclude from Theorem 1 that $(\nu_n^\eta)_{n \in \mathbb{N}^*}$ is $\mathbb{P}$ - a.s.tight and (29) holds.

2. The second part of Theorem 3 comes from Theorem 2. We obtain from Steps 2 and 3 that assumptions $\mathcal{GC}_{\mathcal{Q}}(\mathcal{D}(A)_0, V^{p+a-1}, \rho, \epsilon_I)$ (see (9)) and $\mathcal{SW}_{\mathcal{J},\gamma,\eta}(V^{p+a-1}, \rho, \epsilon_I)$ (see (10)) hold, which are the hypotheses from Theorem 2 (A) with $g = V^{p+a-1}$.

Since there is some $\epsilon > 0$, such that $g_\sigma \leq CV^{p/s+a-1}$, with $g_\sigma = (\|b\|^2 + \|\sigma\|_F^2)^{1+\epsilon}$, we have $\sup_{n \in \mathbb{N}^*} \nu_n^\eta(g_\sigma) < +\infty$. It follows from Proposition 2 that $\mathcal{E}(\tilde{A}, A, \mathcal{D}_0(A))$ (see (7)) holds. Then the hypotheses from Theorem 2 (B) hold, which leads to the conclusion.

4

RATE OF CONVERGENCE: AN INITIAL APPROACH

In this section, we introduce the idea of extending the framework from Section 2 to establish a central limit theorem of order $q = 1, 2$ for the empirical measures (see details in [2]). The approach involves decomposing the terms in the construction of the empirical measure (3) into a martingale component and a residual term, followed by the application of a martingale central limit theorem. To achieve this, we need to modify the original assumptions to give stronger restrictions.

4.1 ASSUMPTIONS ON THE RANDOM MEASURE

Weak Approximation Assumption Let $q = 1, 2$ be the weak order of the approximation. The purpose is to propose a sharper version of the infinitesimal generator approximation $\mathcal{E}(\tilde{A}, A, \mathcal{D}(A)_0)$ (see (7)), which concerns the dominating terms of $\tilde{A}_\gamma - \sum_{i=1}^q \frac{\gamma^i}{i!} A^i$. Precisely, we consider a more restricted set of test functions $F \subset \{f, f : (E, \mathcal{B}(E)) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))\}$ and the linear operator $\mathfrak{M}_q$ defined on F . Let $\tilde{\eta}_q : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ be an increasing function such that the weight sequence $(\tilde{\eta}_{q,n})_{n \in \mathbb{N}^*} = (\tilde{\eta}_q(\gamma_n))_{n \in \mathbb{N}^*}$ satisfies (2) and such that $\mathbb{P}$ -a.s., $\lim_{n \rightarrow \infty} \nu_n^{\tilde{\eta}_q}(\mathfrak{M}_q f) = \nu(\mathfrak{M}_q f)$ for every $f \in F$. We suppose that

$$\mathcal{E}_q(F, \tilde{A}, A, \mathfrak{M}_q, \tilde{\eta}_q) \equiv \forall f \in F, \forall x \in E, \forall \gamma \in (0, \bar{\gamma}], |\mathcal{R}_q f(x, \gamma) - \tilde{\eta}_q(\gamma) \mathfrak{M}_q f(x)| \leq \tilde{\eta}_q(\gamma) \Lambda_{f,q}(x, \gamma), \quad (65)$$

with

$$\begin{aligned} \mathcal{R}_1 f(x, \gamma) &= \gamma A f(x) - \gamma \tilde{A}_\gamma f(x), \\ \mathcal{R}_2 f(x, \gamma) &= \gamma A f(x) - \gamma \tilde{A}_\gamma f(x) + \frac{\gamma^2}{2} \tilde{A}_\gamma A f(x). \end{aligned}$$

In addition, we also assume that $\Lambda_{f,q} : E \times \mathbb{R}_+ \rightarrow \mathbb{R}_+$ can be represented in the following way: Let $(\tilde{\Omega}, \tilde{\mathcal{G}}, \tilde{\mathbb{P}})$ be a probability space. Let $g : E \rightarrow \mathbb{R}_+^l$, $l \in \mathbb{N}$, be a locally bounded Borel measurable function, and let: $\tilde{\Lambda}_f : (E \times \mathbb{R}_+ \times \tilde{\Omega}, \mathcal{B}(E) \otimes \mathcal{B}(\mathbb{R}_+) \otimes \tilde{\mathcal{G}}) \rightarrow \mathbb{R}_+^l$ be a measurable function such that:

$$\sup_{i \in \{1, \dots, l\}} \tilde{\mathbb{E}} \left[\sup_{x \in E} \sup_{\gamma \in (0, \bar{\gamma}]} \tilde{\Lambda}_{f,i}(x, \gamma, \tilde{\omega}) \right] < +\infty, \text{ and } \forall x \in E, \forall \gamma \in (0, \bar{\gamma}], \Lambda_f(x, \gamma) = \langle g(x), \tilde{\mathbb{E}}[\tilde{\Lambda}_f(x, \gamma, \tilde{\omega})] \rangle_{\mathbb{R}^l}.$$

We assume that for every $i \in \{1, \dots, l\}$, $\sup_{n \in \mathbb{N}^*} \nu_n^\eta(g_i, \omega) < +\infty$ $\tilde{\mathbb{P}}(d\omega)$ -a.s. and that there exists a measurable function $\gamma : (\Omega, \tilde{\mathcal{G}}) \rightarrow ((0, \bar{\gamma}], \mathcal{B}((0, \bar{\gamma}]))$ such that one of the following two conditions holds:

(I) $\tilde{\mathbb{P}}(d\tilde{\omega})$ -a.s.

$$\begin{cases} \text{(i)} & \forall K \in \mathcal{K}_E, \lim_{\gamma \rightarrow 0} \sup_{x \in K} \tilde{\Lambda}_{f,i}(x, \gamma, \tilde{\omega}) = 0, \\ \text{(ii)} & \lim_{x \rightarrow \infty} \sup_{\gamma \in (0, \gamma(\tilde{\omega})]} \tilde{\Lambda}_{f,i}(x, \gamma, \tilde{\omega}) = 0. \end{cases}$$

(II) $\tilde{\mathbb{P}}(d\tilde{\omega})$ -a.s.

$$\lim_{\gamma \rightarrow 0} \sup_{x \in E} \tilde{\Lambda}_{f,i}(x, \gamma, \tilde{\omega}) g_i(x) = 0.$$

Remark 3. For $q = 2$, if $\mathcal{E}_1(F_1, \tilde{A}, A, \mathfrak{M}_1, \tilde{\eta}_1)$ (see (65)) holds for some F_1 , $\mathfrak{M}_1$, $\tilde{\eta}_1$, if $\tilde{\eta}_2(\gamma) = \gamma \tilde{\eta}_1(\gamma)$ and $\forall f \in F_2$, $Af \in F_1$. Notice that

$$\mathcal{R}_2 f(x, \gamma) = \gamma A f(x) + \frac{\gamma^2}{2} A^2 f(x) - \gamma \tilde{A}_\gamma f(x) - \frac{\gamma}{2} \mathcal{R}_1 A f(x, \gamma),$$

we have $\mathcal{E}_2(F_2, \tilde{A}, A, \mathfrak{M}_2, \tilde{\eta}_2)$ (see (65)) holds as soon as there is a linear operator $\tilde{\mathfrak{M}}_2$ on F such that

$$\forall f \in F_2, \forall x \in E, \forall \gamma \in (0, \bar{\gamma}], \left| \gamma A f(x) + \frac{\gamma^2}{2} A^2 f(x) - \gamma \tilde{A}_\gamma f(x) - \tilde{\eta}_2(\gamma) \tilde{\mathfrak{M}}_2 f(x) \right| \leq \tilde{\eta}_2(\gamma) \Lambda_{f,2}(x, \gamma),$$

with $\Lambda_{f,2}(x, \gamma)$ satisfies the conditions in (65). This corresponds to a 2-order approximation of A by $\tilde{A}_\gamma$.

Growth Assumption We introduce a new version of $\mathcal{GC}_Q(F, g, \rho, \epsilon_I)$ (see (9)), which is crucial to identify the asymptotic variance in the CLT. We denote by $\mathcal{P}_{\tilde{X},2}$ the set of $\mathcal{F}_n^{\tilde{X}} := \sigma(\tilde{X}_{\Gamma_k}, k \leq n)$ - progressively measurable processes $(\mathcal{X}_n)_{n \in \mathbb{N}^*}$ with $\mathcal{X}_{n+1} \in L^2(\mathcal{F}_n^{\tilde{X}})$ and $\mathbb{E}[\mathcal{X}_{n+1} | \tilde{X}_{\Gamma_n}] = 0$ for every $n \in \mathbb{N}$. Let $\rho \in [1, 2]$ and let $\epsilon_{\mathcal{X}}, \epsilon_{\mathcal{GC}} : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ be two increasing functions such that the weight sequence $(\epsilon_{\mathcal{X},n})_{n \in \mathbb{N}^*} = (\epsilon_{\mathcal{X}}(\gamma_n))_{n \in \mathbb{N}^*}$ satisfies (2). Let $F \subset \{f, f : (E, \mathcal{B}(E)) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))\}$ and $g : E \rightarrow \mathbb{R}_+$ be a Borel measurable function. Finally, we introduce a linear operator $\mathfrak{D}$ defined on F . We assume that

$$\begin{aligned} \mathcal{GC}_{Q,q}(F, g, \rho, \epsilon_{\mathcal{X}}, \epsilon_{\mathcal{GC}}, \mathfrak{D}) &\equiv \mathbb{P} - a.s. \forall f \in F, \exists \mathcal{X}_f \in \mathcal{P}_{\tilde{X},2}, \\ \mathbb{E}[|B_q f(\tilde{X}_{\Gamma_n}, \tilde{X}_{\Gamma_{n+1}}, \gamma_{n+1}) - \mathcal{X}_{f,n+1}|^\rho | \tilde{X}_{\Gamma_n}] &\leq C_f \epsilon_{\mathcal{GC}}(\gamma_{n+1}) g(\tilde{X}_{\Gamma_n}), \end{aligned} \quad (66)$$

with $C_f > 0$ a finite constant that may depend on f and for every $x, y \in E, \gamma \geq 0$,

$$\begin{aligned} B_1 f(x, y, \gamma) &= \mathcal{Q}_\gamma f(x) - f(y), \\ B_2 f(x, y, \gamma) &= \mathcal{Q}_\gamma f(x) - f(y) - \frac{\gamma}{2} (\mathcal{Q}_\gamma A f(x) - A f(y)), \end{aligned}$$

and also $\mathbb{E}[|\mathcal{X}_{f,n+1}|^2 | \tilde{X}_{\Gamma_n}] = \epsilon_{\mathcal{X}}(\gamma_{n+1}) \mathfrak{D} f(\tilde{X}_{\Gamma_n})$ with for every $f \in F, \lim_{n \in \mathbb{N}^*} \nu_n^{\epsilon_{\mathcal{X}}}(\mathfrak{D} f, w) = \nu(\mathfrak{D} f)$, $\mathbb{P}$ -a.s., and

$$\forall \mathfrak{C} > 0, \lim_{n \rightarrow \infty} \frac{1}{H_{\epsilon_{\mathcal{X}},n}} \sum_{k=1}^{n-1} \mathbb{E}[|\mathcal{X}_{f,k+1}|^2 \mathbb{1}_{|\mathcal{X}_{f,k+1}| > \sqrt{H_{\epsilon_{\mathcal{X}},n}} \mathfrak{C}} | \tilde{X}_{\Gamma_k}] \stackrel{\mathbb{P}}{=} 0. \quad (67)$$

Finally, we combine this assumption with

$$\mathcal{SW}_{\mathcal{GC},\gamma}(g, \rho, \epsilon_{\mathcal{X}}, \epsilon_{\mathcal{GC}}) \equiv \mathbb{P} - a.s. \sum_{n=1}^{\infty} \frac{\epsilon_{\mathcal{GC}}(\gamma_n)}{H_{\epsilon_{\mathcal{X}},n}^{\frac{\rho}{2}}} g(\tilde{X}_{\Gamma_n}) < \infty. \quad (68)$$

Proposition 4. Under the above assumptions, $\mathbb{P} - a.s.$, we have

$$\lim_{n \rightarrow \infty} \frac{1}{\sqrt{H_{\epsilon_{\mathcal{X}},n}}} \sum_{k=1}^n \mathcal{X}_{f,k} \stackrel{law}{=} \mathcal{N}(0, \nu(\mathfrak{D} f)).$$

Proof. We will use a general CLT result from [9] (Corollary 3.1): Let $(\tilde{M}_{k,n})_{k \in \{1, \dots, n\}, n \in \mathbb{N}}$ be a $\mathbb{R}$ -valued martingale array and define $\mathcal{F}_{k,n}^{\tilde{M}} = \sigma(\tilde{M}_{i,n}, i \in \{0, \dots, k\})$. We assume that $(\tilde{M}_n)_{n \in \mathbb{N}}$ satisfies the Lindeberg condition:

$$\forall \mathfrak{C} > 0, \lim_{n \rightarrow \infty} \sum_{k=0}^{n-1} \mathbb{E}[|\tilde{M}_{k+1,n} - \tilde{M}_{k,n}|^2 \mathbb{1}_{|\tilde{M}_{k+1,n} - \tilde{M}_{k,n}| > \mathfrak{C}} | \mathcal{F}_{k,n}^{\tilde{M}}] \stackrel{\mathbb{P}}{=} 0$$

and that

$$\lim_{n \rightarrow \infty} \sum_{k=0}^{n-1} \mathbb{E}[|\tilde{M}_{k+1,n} - \tilde{M}_{k,n}|^2 | \mathcal{F}_{k,n}^{\tilde{M}}] \stackrel{\mathbb{P}}{=} \zeta_M^2$$

with ζ_M^2 an almost sure finite random variable. Then

$$\lim_{n \rightarrow \infty} \tilde{M}_{n,n} \stackrel{law}{=} \tilde{\mathcal{N}}(\zeta_M^2),$$

where $\tilde{\mathcal{N}}(\zeta_M^2)$ is a random variable with Laplace transform $\mathbb{E}[\exp(v \tilde{\mathcal{N}}(\zeta_M^2))] = \mathbb{E}[\exp(v^2 \zeta_M^2 / 2)]$ for every $v \in \mathbb{R}$.

Here, since (67) holds and also $\mathbb{E}[|\mathcal{X}_{f,n+1}|^2 | \tilde{X}_{\Gamma_n}] = \epsilon_{\mathcal{X}}(\gamma_{n+1}) \mathfrak{D} f(\tilde{X}_{\Gamma_n})$, $\lim_{n \in \mathbb{N}^*} \nu_n^{\epsilon_{\mathcal{X}}}(\mathfrak{D} f, w) = \nu(\mathfrak{D} f)$, we conclude using the above result. $\square$

4.2 CONVERGENCE RATE RESULTS

When we consider the q -order weak approximation $(\bar{X}_{\Gamma_n})_{n \in \mathbb{N}^*}$ of a Feller process $(X_t)_{t \geq 0}$, we can obtain convergence of some weighted empirical measures. A crucial point here is to consider a specific weight sequence and a decomposition of type martingale - residual.

Proposition 5. *For all $f \in \mathcal{D}(A)$, the direct calculations lead to*

$$\begin{aligned} (q = 1) \quad & \gamma_{n+1} Af(\bar{X}_{\Gamma_n}) = f(\bar{X}_{\Gamma_{n+1}}) - f(\bar{X}_{\Gamma_n}) + B_1 f(\bar{X}_{\Gamma_n}, \bar{X}_{\Gamma_{n+1}}, \gamma_{n+1}) + \mathcal{R}_1(\bar{X}_{\Gamma_n}, \gamma_{n+1}), \\ (q = 2) \quad & \frac{\gamma_{n+1}}{2} (Af(\bar{X}_{\Gamma_n}) + Af(\bar{X}_{\Gamma_{n+1}})) = f(\bar{X}_{\Gamma_{n+1}}) - f(\bar{X}_{\Gamma_n}) + B_2 f(\bar{X}_{\Gamma_n}, \bar{X}_{\Gamma_{n+1}}, \gamma_{n+1}) + \mathcal{R}_2(\bar{X}_{\Gamma_n}, \gamma_{n+1}). \end{aligned}$$

We can now state the q -order CLT. Before that, we state a theorem that we need to use.

Theorem 4. (*Chow* (see [9], Theorem 2.17)). *Let $(M_n)_{n \in \mathbb{N}^*}$ be a real valued martingale with respect to some filtration $\mathcal{F} = (\mathcal{F}_n)_{n \in \mathbb{N}}$. Then*

$$\lim_{n \rightarrow +\infty} M_n = M_\infty \in \mathbb{R} \quad \text{a.s. on the event } \bigcup_{r \in [0,1]} \left\{ \sum_{n=1}^{\infty} \mathbb{E}[|M_n - M_{n-1}|^{1+r} | \mathcal{F}_{n-1}] < +\infty \right\}.$$

Finally, we introduce the weight sequences

$$\forall n \in \mathbb{N}, \quad \eta_n = \begin{cases} C_{\gamma, \eta} \gamma_n & \text{if } q = 1, \\ \frac{C_{\gamma, \eta}}{2} (\gamma_n + \gamma_{n+1}) & \text{if } q = 2, \end{cases} \quad (69)$$

with $C_{\gamma, \eta} \in \mathbb{R}$.

Theorem 5. *Let $q \in \{1, 2\}$, $F \subset \{f : (E, \mathcal{B}(E)) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))\}$, $Af \in \mathcal{C}_b(E)\}$, $g : E \rightarrow \mathbb{R}_+$ a Borel function, $\epsilon_{\mathcal{X}}, \epsilon_{GC} : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ be two increasing functions, let $\tilde{\eta}_q : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ and let $\mathfrak{M}_q$ and $\mathfrak{D}$ be two linear operators defined on F . Finally let γ_n and η_n , $n \in \mathbb{N}$, be the time step and the weight sequences satisfying (69). Assume that $\mathcal{E}_q(F, \tilde{A}, A, \mathfrak{M}_q, \tilde{\eta}_q)$ (see (65)), $\mathcal{GC}_{Q,q}(F, g, \rho, \epsilon_{\mathcal{X}}, \epsilon_{GC}, \mathfrak{D})$ (see (66)) and $\mathcal{SW}_{GC, \gamma}(g, \rho, \epsilon_{\mathcal{X}}, \epsilon_{GC})$ (see (68)) hold.*

Then, for every $f \in F$, we have the following properties:

A. *If $\lim_{n \rightarrow \infty} \sqrt{H_{\epsilon_{\mathcal{X}}, n}} / H_{\tilde{\eta}_q, n} = \infty$, then*

$$\lim_{n \rightarrow \infty} \frac{H_n}{C_{\gamma, \eta} \sqrt{H_{\epsilon_{\mathcal{X}}, n}}} \nu_n^\eta(Af) \stackrel{law}{=} \mathcal{N}(0, \nu(\mathfrak{D}f)).$$

B. *If $\lim_{n \rightarrow \infty} \sqrt{H_{\epsilon_{\mathcal{X}}, n}} / H_{\tilde{\eta}_q, n} = \hat{l} \in \mathbb{R}_+^*$, then*

$$\lim_{n \rightarrow \infty} \frac{H_n}{C_{\gamma, \eta} \sqrt{H_{\epsilon_{\mathcal{X}}, n}}} \nu_n^\eta(Af) \stackrel{law}{=} \mathcal{N}(\hat{l}^{-1} \nu(\mathfrak{M}_q f), \nu(\mathfrak{D}f)).$$

C. *If $\lim_{n \rightarrow \infty} \sqrt{H_{\epsilon_{\mathcal{X}}, n}} / H_{\tilde{\eta}_q, n} = 0$, then*

$$\lim_{n \rightarrow \infty} \frac{H_n}{C_{\gamma, \eta} H_{\tilde{\eta}_q, n}} \nu_n^\eta(Af) \stackrel{\mathbb{P}}{=} \nu(\mathfrak{M}_q f).$$

Proof. Let $n \in \mathbb{N}$. By definition of η (see (69)), we have the following decomposition

$$\begin{aligned} (q = 1) \quad & \nu_n^\eta(Af) = \frac{1}{H_n} \sum_{k=1}^n \eta_k Af(\bar{X}_{\Gamma_{k-1}}) = \frac{C_{\gamma, \eta}}{H_n} \sum_{k=1}^n \gamma_k Af(\bar{X}_{\Gamma_{k-1}}), \\ (q = 2) \quad & \nu_n^\eta(Af) = \frac{C_{\gamma, \eta}}{H_n} \sum_{k=1}^n \frac{\gamma_k}{2} (Af(\bar{X}_{\Gamma_k}) + Af(\bar{X}_{\Gamma_{k-1}})) + \frac{C_{\gamma, \eta} \gamma_0}{2H_n} Af(\bar{X}_0) - \frac{C_{\gamma, \eta} \gamma_n}{2H_n} Af(\bar{X}_{\Gamma_n}). \end{aligned}$$

Since Af is a bounded function, the last two terms of the *r.h.s.* of the equation for $q = 2$ converge to zero when multiplying the corresponding coefficient. We study the first term of the *r.h.s.* of the above equation.

Step 1. Using Proposition 5, we have for $q \in \{1, 2\}$,

$$(q = 1) \quad \sum_{k=1}^n \gamma_k Af(\bar{X}_{\Gamma_{k-1}}) = \sum_{k=0}^{n-1} f(\bar{X}_{\Gamma_{k+1}}) - f(\bar{X}_{\Gamma_k}) + B_1 f(\bar{X}_{\Gamma_k}, \bar{X}_{\Gamma_{k+1}}, \gamma_{k+1}) + \mathcal{R}_1(\bar{X}_{\Gamma_k}, \gamma_{k+1}),$$

$$(q = 2) \quad \sum_{k=1}^n \frac{\gamma_k}{2} (Af(\bar{X}_{\Gamma_k}) + Af(\bar{X}_{\Gamma_{k-1}})) = \sum_{k=0}^{n-1} f(\bar{X}_{\Gamma_{k+1}}) - f(\bar{X}_{\Gamma_k}) + B_2 f(\bar{X}_{\Gamma_k}, \bar{X}_{\Gamma_{k+1}}, \gamma_{k+1}) + \mathcal{R}_2(\bar{X}_{\Gamma_k}, \gamma_{k+1}).$$

Step 2. For $q = 1, 2$, we will prove

$$\lim_{n \rightarrow \infty} \frac{1}{\sqrt{H_{\epsilon_{\mathcal{X}}, n}}} \sum_{k=0}^{n-1} B_q f(\bar{X}_{\Gamma_k}, \bar{X}_{\Gamma_{k+1}}, \gamma_{k+1}) \stackrel{law}{=} \mathcal{N}(0, \nu(\mathfrak{V}f)).$$

By applying Proposition 4, it suffices to show that $\mathbb{P} - a.s.$, for every $f \in F$,

$$\lim_{n \rightarrow +\infty} \frac{1}{\sqrt{H_{\epsilon_{\mathcal{X}}, n}}} \sum_{k=0}^{n-1} B_q f(\bar{X}_{\Gamma_k}, \bar{X}_{\Gamma_{k+1}}, \gamma_{k+1}) - \mathcal{X}_{f, k+1} = 0.$$

It then follows Kronecker's Lemma as soon as we prove the *a.s.* convergence of the martingale $(M_n)_{n \in \mathbb{N}^*}$ defined by $M_0 := 0$ and

$$M_n := \sum_{k=0}^{n-1} \frac{B_q f(\bar{X}_{\Gamma_k}, \bar{X}_{\Gamma_{k+1}}, \gamma_{k+1}) - \mathcal{X}_{f, k+1}}{\sqrt{H_{\epsilon_{\mathcal{X}}, k}}}.$$

From $\mathcal{GC}_{Q, q}(F, g, \rho, \epsilon_{\mathcal{X}}, \epsilon_{\mathcal{GC}}, \mathfrak{V})$ (see (66)) and $\mathcal{SW}_{\mathcal{GC}, \gamma}(g, \rho, \epsilon_{\mathcal{X}}, \epsilon_{\mathcal{GC}})$ (see (68)), we obtain the *a.s.* finiteness of the series

$$\sum_{k=1}^n \mathbb{E}[|M_k - M_{k-1}|^\rho | \bar{X}_{\Gamma_{k-1}}],$$

then the *a.s.* convergence of the martingale $(M_n)_{n \in \mathbb{N}^*}$ follows the Chow's theorem (see Theorem 4).

Step 3. To complete the proof, we will show that

$$\mathbb{P} - a.s. \quad \forall f \in F, \quad \lim_{n \rightarrow \infty} \frac{1}{H_{\tilde{\epsilon}_q, n}} \sum_{k=0}^{n-1} \mathcal{R}_q f(\bar{X}_{\Gamma_k}, \gamma_{k+1}) = \nu(\mathfrak{M}_q f).$$

As a direct consequence of $\mathcal{E}_q(F, \tilde{A}, A, \mathfrak{M}_q, \tilde{\eta}_q)$ (see (65)), since $\mathbb{P} - a.s.$, $\lim_{n \rightarrow \infty} \nu_n^{\tilde{\eta}_q}(\mathfrak{M}_q f) = \nu(\mathfrak{M}_q f)$ for every $f \in F$, we only have to prove that

$$\mathbb{P} - a.s. \quad \forall f \in F, \quad \lim_{n \rightarrow \infty} \frac{1}{H_{\tilde{\epsilon}_q, n}} \sum_{k=0}^{n-1} \mathcal{R}_q f(\bar{X}_{\Gamma_k}, \gamma_{k+1}) - \tilde{\eta}_{q, k+1} \mathfrak{M}_q f(\bar{X}_{\Gamma_e}) = 0.$$

which holds as soon as

$$\mathbb{P} - a.s. \quad \forall f \in F, \quad \lim_{n \rightarrow \infty} \frac{1}{H_{\tilde{\eta}_q, n}} \sum_{k=1}^n \tilde{\eta}_{q, k} \Lambda_{f, q}(\bar{X}_{\Gamma_{k-1}}, \gamma_k) = 0.$$

This follows the same arguments as in the proof of Theorem 2 (B). For details, we refer to the proof of Theorem 2.4 in [1]. $\square$

5

RATE OF CONVERGENCE: A GENERAL APPROACH

In the context of the q -order weak approximation $(\bar{X}_{\Gamma_n})_{n \in \mathbb{N}}$ of a Feller process $(X_t)_{t \geq 0}$, the initial approach introduced in Section 4 illustrates that the convergence rate improves as q increases. In this section, we aim to extend this result to cases where $q \geq 3$. As highlighted in the proof of Theorem 5, a key aspect of the analysis involves decomposing the empirical measure into a martingale component and a residual term, as established in Proposition 5.

We begin by presenting a generalization of Proposition 5 introduced in [2], which applies to the case $q = 3$. While more intricate techniques can extend this approach to higher values of q , the complexity of the computations increases as q grows (see [8], the revised version of [2]). To address this challenge, we propose an alternative framework for computing the weight sequence, which works for all $q \geq 3$. This new approach offers a more straightforward and computationally efficient implementation. However, as a trade-off, it results in a loss of one order in the convergence rate. In the following sections, we provide a detailed exposition of the novel assumptions underlying this framework, along with the corresponding convergence results.

5.1 A GENERALIZATION FOR $q = 3$

A generalization of Proposition 5 has been proposed in [2], aiming to incrementally add one point to the left side of the equation at each step, resulting in a total of q points for a q -order weak approximation. Moreover, for each point, a different linear combination of basic terms can be used. Formally, an extended version of $\mathcal{E}_q(F, \tilde{A}, A, \mathfrak{M}_q, \tilde{\eta}_q)$ (see (65)) is proposed for higher order q by introducing some additional degrees of freedom $e \in \{0, \dots, (q-2)_+\}$:

$$\begin{aligned} \mathcal{E}_q(F, \tilde{A}, A, \mathfrak{M}_q, \tilde{\eta}_q) &\equiv \quad \forall f \in F, \forall x \in E, \forall \gamma \in (0, \bar{\gamma}], \forall e \in \{0, \dots, (q-2)_+\}, \\ &\quad \left| \mathcal{R}_q f(x, \gamma, e) - \tilde{\eta}_q(\gamma) \mathfrak{M}_q f(x) \right| \leq \tilde{\eta}_q(\gamma) \Lambda_{f,q}(x, \gamma), \end{aligned}$$

with

$$\begin{aligned} \mathcal{R}_1 f(x, \gamma, e) &:= -\tilde{\mathcal{R}}_1 f(x, \gamma), \quad \text{and for } q \in \mathbb{N}^*, \\ \mathcal{R}_q f(x, \gamma, e) &:= -\tilde{\mathcal{R}}_q f(x, \gamma) - \sum_{l=1}^{q-2} \gamma^{q-l} \sum_{u=(e-(l-2)_+)_+}^{e \wedge (q-l)} \zeta_u^{q, q-l+1} \mathcal{R}_l A^{q-l} f(x, \gamma, e-u) \\ &\quad - \gamma \sum_{u=0}^1 \zeta_u^{q,2} \mathcal{R}_{q-1} A f(x, \gamma, e-u) \mathbb{1}_{u \leq e \leq (q-3)_+ + u}. \end{aligned} \tag{70}$$

where, for every $m \in \{1, \dots, q\}$, the measurable functions defined by

$$\begin{aligned} \tilde{\mathcal{R}}_m f &: E \times \mathbb{R}_+ \rightarrow \mathbb{R} \\ (x, \gamma) &\mapsto \gamma \tilde{A}_\gamma f(x) - \sum_{i=1}^m \frac{\gamma^i}{i!} A^i f(x), \end{aligned}$$

with $p \in \mathbb{N}, 2 \leq p \leq q$ and the quantity $\zeta^{p,q} = (\zeta_0^{p,q}, \dots, \zeta_{p-1}^{p,q}) \in \mathbb{R}^p$ to be defined. Similarly, the assumption $\mathcal{GC}_{Q,q}(F, g, \rho, \epsilon_{\mathcal{X}}, \epsilon_{\mathcal{GC}}, \mathfrak{D})$ (see (66)) should be rewritten as

$$\begin{aligned} \mathcal{GC}_{Q,q}(F, g, \rho, \epsilon_{\mathcal{X}}, \epsilon_{\mathcal{GC}}, \mathfrak{D}) &\equiv \quad \mathbb{P} - a.s. \quad \forall f \in F, \exists \mathcal{X}_f \in \mathcal{P}_{\bar{X},2} \\ \mathbb{E} \left[\left| B_q f(\bar{X}_{\Gamma_n}, \bar{X}_{\Gamma_{n+1}}, \gamma_{n+1}, n \bmod (q-1)) - \mathcal{X}_{f,n+1} \right|^\rho \middle| \bar{X}_{\Gamma_n} \right] &\leq C_f \epsilon_{\mathcal{GC}}(\gamma_{n+1}) g(\bar{X}_{\Gamma_n}). \end{aligned}$$

with, for every $x, y \in \mathbb{E}$, $\gamma \geq 0$, $e \in \{0, (q-2)_+\}$,

$$\begin{aligned} B_1 f(x, y, \gamma, e) &:= B_1 f(x, y, \gamma) := \mathcal{Q}_\gamma f(x) - f(y), \quad \text{and for } q \in \mathbb{N}^*, \\ B_q f(x, y, \gamma, e) &:= B_1 f(x, y, \gamma) - \sum_{l=1}^{q-2} \gamma^{q-l} \sum_{u=(e-(l-2)_+)_+}^{e \wedge (q-l)} \zeta_u^{q, q-l+1} B_l A^{q-l} f(x, y, \gamma, e-u) \\ &\quad - \gamma \sum_{u=0}^1 \zeta_u^{q, 2} B_{q-1} A f(x, y, \gamma, e-u) \mathbb{1}_{u \leq e \leq (q-3)_+ + u}. \end{aligned} \quad (71)$$

Notice that when $q = 1, 2$ there is only one possible choice of e and the terms $\mathcal{R}_q f(x, \gamma, e)$ and $B_q f(x, y, \gamma, e)$ coincide with the definitions given in Section 4. We can now state a generalized version of Proposition 5:

Proposition 6. *Suppose that $q \geq 2$, $n_q \in \mathbb{N}$ s.t. $\gamma_{n_q+1+e} = \gamma_{n_q+1}$ for $e \in \{0, \dots, (q-2)_+\}$. Let $2 \leq p \leq q$, $\zeta^{p, q} = (\zeta_0^{q, p}, \dots, \zeta_{p-1}^{q, p}) \in \mathbb{R}^p$ with $\zeta_u^{q, q} = (q!)^{-1}$ for every $u \in \{0, \dots, q-2\}$ and $\zeta_{q-1}^{q, q} = 0$. If for $p \in \{3, \dots, q\}$, the equation*

$$M^{p, q} \zeta^{q, p-1} = N^{q, p}(\zeta^{q, p}) \quad (72)$$

is satisfied where $N^{q, p} : \mathbb{R}^p \rightarrow \mathbb{R}^q$, $\zeta \mapsto (\frac{1}{(p-1)!} \mathbb{1}_{j < q-1} + \zeta_{j-(q-p-1)_+ - 1} \mathbb{1}_{p \geq j-(q-p-1)_+ \geq 1} - \zeta_j \mathbb{1}_{j < p})_{j \in \{0, \dots, q-1\}}$ and $M^{q, p} \in \mathbb{R}^{q \times (p-1)}$ defined by $M_{j, u}^{q, p} = c_{j-u}^{q, p+2} \mathbb{1}_{j-u \in \{0, \dots, q-p+1\}}$ with, for every $j \in \{0, \dots, q-1\}$

$$c_j^q = \mathbb{1}_{j \leq q-2} + \sum_{u=0}^1 \zeta_u^{q, 2} (\mathbb{1}_{j=(q-3)_+ + u+1} - \mathbb{1}_{j=u}), \quad (73)$$

with convention $c_0^1 = 1$. Then we have

$$\begin{aligned} \gamma_{n_q+1} \sum_{e=0}^{q-1} c_e^q A f(\bar{X}_{\Gamma_{n_q+e}}) &= \sum_{e=0}^{(q-2)_+} f(\bar{X}_{\Gamma_{n_q+e+1}}) - f(\bar{X}_{\Gamma_{n_q+e}}) \\ &\quad + B_q f(\bar{X}_{\Gamma_{n_q+e}}, \bar{X}_{\Gamma_{n_q+e+1}}, \gamma_{n_q+1}, e) \\ &\quad + \mathcal{R}_q f(\bar{X}_{\Gamma_{n_q+e}}, \gamma_{n_q+1}, e). \end{aligned}$$

Proof. See proof of Proposition 3.2 in [2]. □

Remark 4. *We will show in Section 6.3 that the linear system of equations (72) has solution for $q = 2, 3$, but does not admit a solution for $q = 4$. To address this issue, one solution is to consider $\mathcal{C}(q) \geq q$ points instead of just q points for q -order approximation in Proposition 5. Additionally, an extra variable $\lambda = (\lambda^1, \dots, \lambda^{\mathcal{C}(q)-2}) \in \mathbb{R}^{\mathcal{C}(q)-2} \times \mathbb{R}^{\mathcal{C}(q)-3} \times \dots \times \mathbb{R}$ is included, such that the quantity $\zeta^{q, p}$ now depends on λ : $\zeta^{q, p}(\lambda) = (\zeta_0^{q, p}(\lambda), \dots, \zeta_{p-1}^{q, p}(\lambda)) \in \mathbb{R}^{\mathcal{C}(q)-\mathcal{C}(q-p+1)+1}$. For further details, we refer to [8].*

5.2 FRAMEWORK FOR THE COMPUTATION

In this part, we propose an alternative framework for computing the weight sequence $(c_e^q)_{e \in \{0, \dots, q-1\}}$, $q \geq 3$, which offers a more straightforward and computationally efficient implementation. However, it results in a loss of one order in the convergence rate. We will provide the detailed convergence results in the next parts. We first denote

$$\begin{aligned} \mathcal{B} f(x, y, \gamma) &= \mathcal{Q}_\gamma f(x) - f(y), \\ \tilde{\mathcal{R}}_m f(x, \gamma) &= - \sum_{j=1}^m \frac{\gamma^j}{j!} A^j f(x) + \gamma \tilde{A}_\gamma f(x). \end{aligned}$$

Suppose that $q \geq 3$, $n_q \in \mathbb{N}$ s.t. $\gamma_{n_q+1+e} = \gamma_{n_q+1}$ for $e \in \{0, \dots, q-2\}$. We consider the expression

$$\begin{aligned} \mathcal{MR}(\zeta, \eta, \gamma, A, \tilde{A}, n_q) &= f(\bar{X}_{\Gamma_{n_q+q-1}}) - f(\bar{X}_{\Gamma_{n_q}}) + \sum_{l=0}^{q-1} \sum_{i=1}^{q-1} \zeta_i^l \gamma_{n_q+1}^l \mathcal{B}A^l f(\bar{X}_{\Gamma_{n_q+i-1}}, \bar{X}_{\Gamma_{n_q+i}}, \gamma_{n_q+1}) \\ &\quad - \sum_{l=0}^{q-1} \sum_{i=1}^{q-1} \eta_i^l \gamma_{n_q+1}^l \tilde{\mathcal{R}}_{q-l} A^l f(\bar{X}_{\Gamma_{n_q+i-1}}, \gamma_{n_q+1}), \end{aligned} \quad (74)$$

with $(\zeta_i^l, \eta_i^l)_{i=1, \dots, q-1}^{l=0, \dots, q-1}$ to be determined.

Remark 5. We observe that $\left(\gamma_{n_q+1}^l \mathcal{B}A^l f(\bar{X}_{\Gamma_{n_q+i-1}}, \bar{X}_{\Gamma_{n_q+i}}, \gamma_{n_q+1})\right)_{l=0, \dots, q-1, i=1, \dots, q-1}$ represent all terms of type martingale of order less than q , and $\left(\gamma_{n_q+1}^l \tilde{\mathcal{R}}_{q-l} A^l f(\bar{X}_{\Gamma_{n_q+i-1}}, \gamma_{n_q+1})\right)_{l=0, \dots, q-1, i=1, \dots, q-1}$ represents all terms of type residual of order $q+1$. For $e \in \{0, \dots, q-2\}$, the term $B_q f(\bar{X}_{\Gamma_{n_q+e}}, \bar{X}_{\Gamma_{n_q+e+1}}, \gamma_{n_q+1}, e)$ (see (71)) is indeed the linear combination of $\left(\gamma_{n_q+1}^l \mathcal{B}A^l f(\bar{X}_{\Gamma_{n_q+e}}, \bar{X}_{\Gamma_{n_q+e+1}}, \gamma_{n_q+1})\right)_{l=0, \dots, q-1}$ (resp. $\mathcal{R}_q f(\bar{X}_{\Gamma_{n_q+e}}, \gamma_{n_q+1}, e)$ (see (70)) is the linear combination of $\left(\gamma_{n_q+1}^l \tilde{\mathcal{R}}_{q-l} A^l f(\bar{X}_{\Gamma_{n_q+e}}, \gamma_{n_q+1})\right)_{l=0, \dots, q-1}$). Thus, the conditions in Proposition 6 (see (72)) are equivalent to the existence of some $(\zeta_i^l, \eta_i^l)_{i=1, \dots, q-1}^{l=0, \dots, q-1}$ such that $\mathcal{MR}(\zeta, \eta, \gamma, A, \tilde{A}, n_q)$ can be expressed as a linear combination of $(Af(\bar{X}_{\Gamma_{n_q+e}}))_{e=0, \dots, q-1}$. Precisely, we can write the coefficient of each term in $\mathcal{MR}(\zeta, \eta, \gamma, A, \tilde{A}, n_q)$:

$$\text{Coefficient of } \gamma_{n_q+1}^k \mathcal{Q}_\gamma A^k f(\bar{X}_{\Gamma_{n_q+i-1}}) : \zeta_i^k - \eta_i^k, \quad \forall k = 0, \dots, q-1, \forall i = 1, \dots, q-1.$$

$$\text{Coefficient of } \gamma_{n_q+1}^k A^k f(\bar{X}_{\Gamma_{n_q+i-1}}) : (-\zeta_{i-1}^k \mathbb{1}_{i \geq 2} + \eta_i^k \mathbb{1}_{i \leq q-1}) \mathbb{1}_{k \leq q-1} + \sum_{j=1}^k \frac{\eta_i^{k-j}}{j!} \mathbb{1}_{i \leq q-1},$$

$$\forall k = 0, \dots, q, \forall i = 1, \dots, q.$$

If set $\zeta_i^k = \eta_i^k$ for all k, i , the conditions (72) are equivalent to the existence of solution of

$$(-\eta_{i-1}^k \mathbb{1}_{i \geq 2} + \eta_i^k \mathbb{1}_{i \leq q-1}) \mathbb{1}_{k \leq q-1} + \sum_{j=1}^k \frac{\eta_i^{k-j}}{j!} \mathbb{1}_{i \leq q-1} = \mathbb{1}_{k=0, i=1} - \mathbb{1}_{k=0, i=q} \text{ for } k = 0, 2, \dots, q, i = 1, \dots, q.$$

We can then derive the non-existence of a solution to (72) for $q \geq 4$ through an alternative approach.

To address this issue, we allow the term of the form $\gamma_{n_q+1}^q A^q f(\bar{X}_{\Gamma_{n_q+i-1}})$ to appear in $\mathcal{MR}(\zeta, \eta, \gamma, A, \tilde{A}, n_q)$. In this way, if we denote $(\eta_i^l)_{i=1, \dots, q-1}^{l=0, \dots, q-1}$ solution of

$$-\eta_{i-1}^k \mathbb{1}_{i \geq 2} + \sum_{j=0}^k \frac{\eta_i^{k-j}}{j!} \mathbb{1}_{i \leq q-1} = \mathbb{1}_{k=0, i=1} - \mathbb{1}_{k=0, i=q} \text{ for } k = 0, 2, \dots, q-1, i = 1, \dots, q, \quad (75)$$

and let $(c_i^q)_{i=1, \dots, q}$ be defined as

$$c_1^q = \eta_1^1, \quad c_i^q = -\eta_{i-1}^1 + \eta_i^1 + \eta_i^0 \text{ for } 2 \leq i \leq q-1, \quad c_q^q = -\eta_{q-1}^1. \quad (76)$$

Then

$$\begin{aligned} \mathcal{MR}(\zeta, \eta, \gamma, A, \tilde{A}, n_q) &= \sum_{i=1}^q \gamma_{n_q+1} c_i^q A f(\bar{X}_{\Gamma_{n_q+i-1}}) + \gamma_{n_q+1}^q \sum_{i=1}^q \left(\sum_{j=1}^q \frac{\eta_i^{q-j}}{j!} \right) A^q f(\bar{X}_{\Gamma_{n_q+i-1}}) \\ &= f(\bar{X}_{\Gamma_{n_q+q-1}}) - f(\bar{X}_{\Gamma_{n_q}}) + \sum_{l=0}^{q-1} \sum_{i=1}^{q-1} \zeta_i^l \gamma_{n_q+1}^l \mathcal{B}A^l f(\bar{X}_{\Gamma_{n_q+i-1}}, \bar{X}_{\Gamma_{n_q+i}}, \gamma_{n_q+1}) \\ &\quad - \sum_{l=0}^{q-1} \sum_{i=1}^{q-1} \eta_i^l \gamma_{n_q+1}^l \tilde{\mathcal{R}}_{q-l} A^l f(\bar{X}_{\Gamma_{n_q+i-1}}, \gamma_{n_q+1}). \end{aligned} \quad (77)$$

The next proposition ensures the existence of $(\eta_i^l)_{i=1,\dots,q-1}^{l=0,\dots,q-1}$ defined by (75).

Proposition 7. *For $q \geq 3$, The linear system of equations (75) has a solution. Moreover, the rank of (75) is $q(q-1) - 1$.*

Proof. By taking $i = q$ in (75), we set $\eta_{q-1}^k = 0, \forall k = 2, \dots, q-1$; by taking $k = 0$ in (75), we set $\eta_i^0 = 1, \forall i = 1, \dots, q-1$.

For all $\eta_{q-1}^1 \in \mathbb{R}$, by Lemma 6, let $(\eta_i^k)_{i=1,\dots,q-2}^{k=1,\dots,q-1}$ be the unique solution of

$$-\eta_{i-1}^k \mathbb{1}_{i \geq 2} + \sum_{j=0}^{k-1} \frac{\eta_i^{k-j}}{j!} \mathbb{1}_{i \leq q-2} = -\frac{1}{k!} - \frac{\eta_{q-1}^1}{(k-1)!} \mathbb{1}_{i=q-1} \text{ for } k = 2, \dots, q-1, i = 1, \dots, q-1.$$

We can verify easily that $(\eta_i^k)_{i=1,\dots,q-1}^{k=0,\dots,q-1}$ is indeed solution of (75). Moreover, since we have introduced exactly one free variable η_{q-1}^1 , the rank of the system is $q(q-1) - 1$. $\square$

Lemma 6. *Let $n \geq 1$, the linear system of equations about $(\eta_i^k)_{i=1,\dots,n}^{k=1,\dots,n+1}$*

$$-\eta_{i-1}^k \mathbb{1}_{i \geq 2} + \mathbb{1}_{i \leq n} \sum_{j=0}^{k-1} \frac{\eta_i^{k-j}}{j!} = 0, \text{ for } k = 2, \dots, n+1, i = 1, \dots, n+1 \quad (78)$$

has only zero solution.

Proof. First, we prove by induction that if (η_i^k) is a solution of (78), then

$$\eta_{n-m}^k = \sum_{j=1}^m \frac{j^{k-1} - (j-1)^{k-1}}{(k-1)!} \eta_{n-m+j}^1, \forall k \geq 2 \quad (79)$$

for $m = 1, \dots, n-1$.

- Case $m = 1$: we begin by taking $i = n+1$ in (78), we obtain that $\eta_n^k = 0, \forall k = 2, \dots, n+1$; by taking $i = n$, we have $\eta_{n-1}^k = \frac{1}{(k-1)!} \eta_n^1$.
- Suppose that (79) is true for $m \leq n-2$, then by taking $i = n-m$,

$$\begin{aligned} \eta_{n-m-1}^k &= \sum_{j=0}^{k-1} \frac{\eta_{n-m}^{k-j}}{j!} = \sum_{j=0}^{k-2} \frac{1}{j!} \sum_{l=1}^m \frac{l^{k-j-1} - (l-1)^{k-j-1}}{(k-j-1)!} \eta_{n-m+l}^1 + \frac{\eta_{n-m}^1}{(k-1)!} \\ &= \sum_{l=1}^m \frac{1}{(k-1)!} \left(\sum_{j=0}^{k-2} \binom{k-1}{k-j-1} (l^{k-j-1} - (l-1)^{k-j-1}) \right) \eta_{n-m+l}^1 + \frac{\eta_{n-m}^1}{(k-1)!} \\ &= \sum_{l=1}^m \frac{1}{(k-1)!} ((l+1)^{k-1} - l^{k-1}) \eta_{n-m+l}^1 + \frac{\eta_{n-m}^1}{(k-1)!} \\ &= \sum_{l=1}^{m+1} \frac{l^{k-1} - (l-1)^{k-1}}{(k-1)!} \eta_{n-m-1+l}^1 \end{aligned}$$

i.e. (79) is true for $m+1$.

Let $m = n-1$ in (79), we have $\eta_1^k = \sum_{j=1}^{n-1} \frac{j^{k-1} - (j-1)^{k-1}}{(k-1)!} \eta_{n-m+j}^1, \forall k \geq 2$. Take $i = 1$ in (78), we obtain

$$0 = \sum_{j=0}^{k-1} \frac{\eta_1^{k-j}}{j!} = \sum_{j=0}^{k-2} \frac{1}{j!} \sum_{l=1}^{n-1} \frac{l^{k-j-1} - (l-1)^{k-j-1}}{(k-j-1)!} \eta_{1+l}^1 + \frac{\eta_1^1}{(k-1)!} = \sum_{l=1}^n \frac{l^{k-1} - (l-1)^{k-1}}{(k-1)!} \eta_l^1,$$

$\forall k = 2, \dots, n+1$. But the matrix $\left(\frac{l^{k-1} - (l-1)^{k-1}}{(k-1)!}\right)_{l,k}$ for $l = 1, \dots, n$ and $k = 2, \dots, n+1$ satisfies

$$\left(\frac{l^{k-1} - (l-1)^{k-1}}{(k-1)!}\right)_{l,k} = \text{diag}\left(1, \frac{1}{2!}, \dots, \frac{1}{n!}\right) (l^{k-1} - (l-1)^{k-1})_{l,k}$$

and $\det[(l^{k-1} - (l-1)^{k-1})_{l,k}] = \det[(l^{k-1})_{l,k}] = \text{Vandermonde}(1, \dots, n) \neq 0$. It is invertible so that $\eta_l^1 = 0$, $\forall l = 1, \dots, n$. And we conclude by (79) that $(\eta_i^k) = 0$. $\square$

Example 1 (For $q = 4$). We have

$$\begin{aligned} f(\bar{X}_{\Gamma_{n_3+2}}) - f(\bar{X}_{\Gamma_{n_3}}) &+ \sum_{l=0}^3 \sum_{i=1}^3 \eta_i^l \gamma_{n_3+1}^l \mathcal{B} A^l f(\bar{X}_{\Gamma_{n_3+i-1}}, \bar{X}_{\Gamma_{n_3+i}}, \gamma_{n_3+1}) + \sum_{l=0}^3 \sum_{i=1}^3 \eta_i^l \gamma_{n_3+1}^l \tilde{\mathcal{R}}_{3-l} A^l f(\bar{X}_{\Gamma_{n_3+i-1}}, \gamma_{n_3+1}) \\ &= \gamma_{n_3+1} \left((2\lambda + \frac{3}{8}) A f(\bar{X}_{\Gamma_{n_3}}) + \frac{3}{4} A f(\bar{X}_{\Gamma_{n_3+1}}) + (-6\lambda + \frac{9}{8}) A f(\bar{X}_{\Gamma_{n_3+2}}) + (-2\lambda + \frac{3}{8}) A f(\bar{X}_{\Gamma_{n_3+4}}) \right) \\ &+ \gamma_{n_3+1}^4 \left((\frac{\lambda}{3} + \frac{1}{48}) A^4 f(\bar{X}_{\Gamma_{n_3}}) + \frac{4\lambda}{3} A^4 f(\bar{X}_{\Gamma_{n_3+1}}) + (\frac{\lambda}{3} - \frac{1}{48}) A^4 f(\bar{X}_{\Gamma_{n_3+2}}) \right), \end{aligned}$$

where

$$\begin{pmatrix} \eta_1^0 & \eta_1^1 & \eta_1^2 & \eta_1^3 \\ \eta_2^0 & \eta_2^1 & \eta_2^2 & \eta_2^3 \\ \eta_3^0 & \eta_3^1 & \eta_3^2 & \eta_3^3 \end{pmatrix} = \begin{pmatrix} 1 & -\frac{5}{8} & \frac{5}{4} & \frac{1}{48} \\ 1 & -\frac{1}{2} & \frac{5}{4} & -\frac{1}{48} \\ 1 & -\frac{3}{8} & 0 & 0 \end{pmatrix} + \lambda \begin{pmatrix} 0 & 2 & -2 & 1 \\ 0 & -4 & 2 & 1 \\ 0 & 2 & 0 & 0 \end{pmatrix}, \quad \lambda \in \mathbb{R}.$$

We are now ready to state the hypotheses we need to impose and prove the convergence rate results for all $q \geq 3$.

5.3 ASSUMPTIONS ON THE RANDOM MEASURE

Weak Approximation Assumption Let $q \geq 3$ be the weak order of the approximation, $F \subset \{f, f : (E, \mathcal{B}(E)) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))\}$ and the linear operator $\mathfrak{M}_q$ defined on F . Let $\tilde{\eta}_q : \mathbb{R}_+ \times \{0, \dots, (q-2)_+\} \rightarrow \mathbb{R}_+$ be an increasing function such that the weight sequence $(\tilde{\eta}_{q,n})_{n \in \mathbb{N}^*} = (\tilde{\eta}_q(\gamma_n, n \bmod (q-1)))_{n \in \mathbb{N}^*}$ is decreasing, satisfies (2) and such that $\mathbb{P}$ -a.s., $\lim_{n \rightarrow \infty} \nu_n^{\tilde{\eta}_q}(\mathfrak{M}_q f) = \nu(\mathfrak{M}_q f)$ for every $f \in F$. We suppose that

$$\begin{aligned} \mathcal{E}_q(F, \tilde{A}, A, \mathfrak{M}_q, \tilde{\eta}_q) &\equiv \quad \forall f \in F, \forall x \in E, \forall \gamma \in (0, \tilde{\gamma}], \forall e \in \{0, \dots, q-2\}, \\ &\left| \mathcal{R}_q f(x, \gamma, e) - \tilde{\eta}_q(\gamma, e) \mathfrak{M}_q f(x) \right| \leq \tilde{\eta}_q(\gamma, e) \Lambda_{f,q}(x, \gamma), \end{aligned} \quad (80)$$

where $\mathcal{R}_q f(x, \gamma, e) = -\sum_{l=0}^{q-1} \eta_{e+1}^l \gamma^l \tilde{\mathcal{R}}_{q-l} A^l f(x, \gamma)$, with $(\eta_i^l)_{i=1, \dots, q-1}^{l=0, \dots, q-1}$ defined by (75). In addition, we also assume that $\Lambda_{f,q} : E \times \mathbb{R}_+ \rightarrow \mathbb{R}_+$ can be represented in the same way as in Section 4.1.

Growth Assumption We denote by $\mathcal{P}_{\bar{X},2}$ the set of $\mathcal{F}_n^{\bar{X}} := \sigma(\bar{X}_{\Gamma_k}, k \leq n)$ -progressively measurable processes $(\mathcal{X}_n)_{n \in \mathbb{N}^*}$ with $\mathcal{X}_{n+1} \in L^2(\mathcal{F}_n^{\bar{X}})$ and $\mathbb{E}[\mathcal{X}_{n+1} | \bar{X}_{\Gamma_n}] = 0$ for every $n \in \mathbb{N}$. Let $\rho \in [1, 2]$ and let $\epsilon_{\mathcal{X}}, \epsilon_{\mathcal{GC}} : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ be two increasing functions such that the weight sequence $(\epsilon_{\mathcal{X},n})_{n \in \mathbb{N}^*} = (\epsilon_{\mathcal{X}}(\gamma_n))_{n \in \mathbb{N}^*}$ satisfies (2). Let $F \subset \{f, f : (E, \mathcal{B}(E)) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))\}$, $g : E \rightarrow \mathbb{R}_+$ be a Borel measurable function and $\mathfrak{D}$ be a linear operator defined on F . We assume that

$$\begin{aligned} \mathcal{GC}_{Q,q}(F, g, \rho, \epsilon_{\mathcal{X}}, \epsilon_{\mathcal{GC}}, \mathfrak{D}) &\equiv \quad \mathbb{P} - a.s. \quad \forall f \in F, \exists \mathcal{X}_f \in \mathcal{P}_{\bar{X},2} \\ &\mathbb{E} \left[\left| B_q f(\bar{X}_{\Gamma_n}, \bar{X}_{\Gamma_{n+1}}, \gamma_{n+1}, n \bmod (q-1)) - \mathcal{X}_{f,n+1} \right|^\rho \middle| \bar{X}_{\Gamma_n} \right] \leq C_f \epsilon_{\mathcal{GC}}(\gamma_{n+1}) g(\bar{X}_{\Gamma_n}). \end{aligned} \quad (81)$$

with $C_f > 0$ a finite constant that may depend on f and for every $x, y \in E$, $\gamma \geq 0$, $e \in \{0, \dots, q-2\}$

$$B_q f(x, y, \gamma, e) = \sum_{l=0}^{q-1} \eta_{e+1}^l \gamma^l \mathcal{B} A^l f(x, y, \gamma),$$

and also $\mathbb{E}[|\mathcal{X}_{f,n+1}|^2 \mid \bar{X}_{\Gamma_n}] = \epsilon_{\mathcal{X}}(\gamma_{n+1}) \mathfrak{D} f(\bar{X}_{\Gamma_n})$ with for every $f \in F$, $\lim_{n \in \mathbb{N}^*} \nu_n^{\epsilon_{\mathcal{X}}}(\mathfrak{D} f, w) = \nu(\mathfrak{D} f)$, $\mathbb{P}$ -a.s., and

$$\forall \mathfrak{C} > 0, \lim_{n \rightarrow \infty} \frac{1}{H_{\epsilon_{\mathcal{X}},n}} \sum_{k=0}^{n-1} \mathbb{E}[|\mathcal{X}_{f,k+1}|^2 \mathbb{1}_{|\mathcal{X}_{f,k+1}| > \sqrt{H_{\epsilon_{\mathcal{X}},n}} \mathfrak{C}} \mid \bar{X}_{\Gamma_k}] \stackrel{\mathbb{P}}{=} 0. \quad (82)$$

Finally, we combine this assumption with assumption (68).

5.4 CONVERGENCE RATE RESULTS

We first introduce a proposition concerning the choice of weight sequence and the decomposition of type martingale - residual like Proposition 5.

Proposition 8. *Suppose that $q \geq 2$, $n_q \in \mathbb{N}$ s.t. $\gamma_{n_q+1+e} = \gamma_{n_q+1}$ for $e \in \{0, \dots, (q-2)_+\}$. Let $(\eta_i^l)_{i=1, \dots, q-1}^{l=0, \dots, q-1}$ be defined in (75) and $(c_i^q)_{i=1, \dots, q}$ be defined in (76), then for every $f \in \mathcal{D}(A)$,*

$$\begin{aligned} \gamma_{n_q+1} \sum_{e=0}^{q-1} c_e^q A f(\bar{X}_{\Gamma_{n_q+e}}) &= \sum_{e=0}^{q-2} f(\bar{X}_{\Gamma_{n_q+e+1}}) - f(\bar{X}_{\Gamma_{n_q+e}}) \\ &\quad + B_q f(\bar{X}_{\Gamma_{n_q+e}}, \bar{X}_{\Gamma_{n_q+e+1}}, \gamma_{n_q+1}, e) \\ &\quad + \mathcal{R}_q f(\bar{X}_{\Gamma_{n_q+e}}, \gamma_{n_q+1}, e) - \gamma_{n_q+1}^q \left(\sum_{j=1}^q \frac{\eta_{e+1}^{k-j}}{j!} \right) A^q f(\bar{X}_{\Gamma_{n_q+e}}). \end{aligned} \quad (83)$$

Proof. This is a direct consequence of (77). $\square$

We can now state the q -order ergodic CLT. We introduce the step size and the weight sequences:

$$\begin{aligned} \forall n \in \mathbb{N}, e \in \{0, \dots, q-2\}, \gamma_{(q-1)n+1+e} &= \gamma_{(q-1)n+1}, \\ \eta_{(q-1)n+1+e} &= C_{\gamma, \eta} (c_{e+1}^q \gamma_{(q-1)n+1} + \mathbb{1}_{e=0} c_q^q \gamma_{(q-1)(n-1)+1}), \end{aligned} \quad (84)$$

with $C_{\gamma, \eta} \in \mathbb{R}^*$ and the convention $\gamma_{-l} = 0$ for $l \in \mathbb{N}^*$.

Theorem 6. *Let $q \geq 3$, $F \subset \{f, f : (E, \mathcal{B}(E)) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R})), Af \in \mathcal{C}_b(E)\}$, $g : E \rightarrow \mathbb{R}_+$ a Borel function, $\epsilon_{\mathcal{X}}, \epsilon_{\mathcal{GC}} : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ be two increasing functions, let $\tilde{\eta}_q : \mathbb{R}_+ \times \{0, \dots, q-2\} \rightarrow \mathbb{R}_+$ and let $\mathfrak{M}_q$ and $\mathfrak{D}$ be two linear operators defined on F . Finally let γ_n and η_n , $n \in \mathbb{N}$, be the time step and the weight sequences satisfying (84). Assume that $\mathcal{E}_q(F, \tilde{A}, A, \mathfrak{M}_q, \tilde{\eta}_q)$ (see (80)), $\mathcal{GC}_{Q,q}(F, g, \rho, \epsilon_{\mathcal{X}}, \epsilon_{\mathcal{GC}}, \mathfrak{D})$ (see (81)) and $\mathcal{SW}_{\mathcal{GC}, \gamma}(g, \rho, \epsilon_{\mathcal{X}}, \epsilon_{\mathcal{GC}})$ (see (68)) hold.*

Assume also $\lim_{n \rightarrow \infty} H_{\tilde{\eta}_q, n} / \tilde{H}_{q, n} = 0$ where $\tilde{H}_{q, n} = \sum_{k=1}^n \gamma_k^q$. Then, for every $f \in F$, we have the following properties:

A. *If $\lim_{n \rightarrow \infty} \sqrt{H_{\epsilon_{\mathcal{X}}, n}} / \tilde{H}_{q, n} > 0$ ($+\infty$ included), then*

$$\lim_{n \rightarrow \infty} \frac{H_n}{C_{\gamma, \eta} \sqrt{H_{\epsilon_{\mathcal{X}}, n}}} \nu_n^{\eta}(Af) \stackrel{law}{=} \mathcal{N}(0, \nu(\mathfrak{D} f)).$$

B. *If $\lim_{n \rightarrow \infty} \sqrt{H_{\epsilon_{\mathcal{X}}, n}} / \tilde{H}_{q, n} = 0$, then*

$$\lim_{n \rightarrow \infty} \frac{H_n}{C_{\gamma, \eta} \tilde{H}_{q, n}} \nu_n^{\eta}(Af) \stackrel{\mathbb{P}}{=} 0.$$

Remark 6. The assumption $\lim_{n \rightarrow \infty} H_{\tilde{\eta}_q, n} / \tilde{H}_{q, n} = 0$ is generally true when considering a q -order approximation, since in the most common case, $H_{\tilde{\eta}_q, n} \sim \sum_{k=1}^n \gamma_k^{q+1}$. Thus, we have lost one order of convergence.

Proof. Step 1. Let $n \in \mathbb{N}$. We begin by noticing that

$$\begin{aligned} \nu_n^\eta(Af) &= \frac{1}{H_n} \sum_{k=1}^n \eta_k Af(\bar{X}_{\Gamma_{k-1}}) = \frac{C_{\gamma, \eta}}{H_n} \sum_{k=0}^{N_{q, n}} \gamma_{(q-1)k+1} \sum_{e=0}^{q-1} c_{e+1}^q Af(\bar{X}_{\Gamma_{(q-1)k+e}}) \\ &\quad + \frac{1}{H_n} \sum_{k=(q-1)(N_{q, n}+1)+1}^n \eta_k Af(\bar{X}_{\Gamma_{k-1}}) - \frac{C_{\gamma, \eta} \gamma_{(q-1)N_{q, n}+1} c_q^q}{H_n} Af(\bar{X}_{\Gamma_{(q-1)(N_{q, n}+1)}}) \end{aligned}$$

with the notation $N_{q, n} = \lfloor (n-1)/(q-1) \rfloor - 1$. Since Af is a bounded function, we only need to study the first term of the *r.h.s.* of the above equation. By Proposition 8,

$$\begin{aligned} \gamma_{(q-1)k+1} \sum_{e=0}^{q-1} c_e^q Af(\bar{X}_{\Gamma_{(q-1)k+e}}) &= \sum_{e=0}^{q-2} f(\bar{X}_{\Gamma_{(q-1)k+e+1}}) - f(\bar{X}_{\Gamma_{(q-1)k+e}}) \\ &\quad + B_q f(\bar{X}_{\Gamma_{(q-1)k+e}}, \bar{X}_{\Gamma_{(q-1)k+e+1}}, \gamma_{(q-1)k+1}, e) \\ &\quad + \mathcal{R}_q f(\bar{X}_{\Gamma_{(q-1)k+e}}, \gamma_{(q-1)k+1}, e) - \gamma_{(q-1)k+1}^q \left(\sum_{j=1}^q \frac{\eta_{e+1}^{k-j}}{j!} \right) A^q f(\bar{X}_{\Gamma_{(q-1)k+e}}). \end{aligned}$$

Step 2. We have (see Step 2 in the proof of Theorem 3.2 in [2])

$$\lim_{n \rightarrow \infty} \frac{1}{\sqrt{H_{\epsilon_X, n}}} \sum_{k=0}^{N_{q, n}} \sum_{e=0}^{q-2} B_q f(\bar{X}_{\Gamma_{(q-1)k+e}}, \bar{X}_{\Gamma_{(q-1)k+e+1}}, \gamma_{(q-1)k+1}, e) \stackrel{law}{=} \mathcal{N}(0, \nu(\mathfrak{B}f)).$$

Step 3. It can be shown that (see Step 3 in the proof of Theorem 3.2 in [2])

$$\mathbb{P} - a.s., \forall f \in F, \lim_{n \rightarrow \infty} \frac{1}{H_{\tilde{\eta}_q, n}} \sum_{k=0}^{N_{q, n}} \sum_{e=0}^{q-2} \mathcal{R}_q f(\bar{X}_{\Gamma_{(q-1)k+e}}, \gamma_{(q-1)k+1}, e) - \tilde{\eta}_{q, (q-1)k+1+e} \mathfrak{M}_q f(\bar{X}_{\Gamma_{(q-1)k+e}}) = 0.$$

Moreover, if we denote $\xi_e^q = -\sum_{j=1}^q \frac{\eta_{e+1}^{k-j}}{j!}$, we have

$$\mathbb{P} - a.s., \lim_{n \rightarrow \infty} \frac{1}{\sum_{k=0}^{N_{q, n}} \gamma_{(q-1)k+1}^q} \sum_{k=0}^{N_{q, n}} \sum_{e=0}^{q-2} \xi_e^q \gamma_{(q-1)k+1}^q A^q f(\bar{X}_{\Gamma_{(q-1)k+e}}) = 0,$$

since for all $e \in \{0, \dots, q-2\}$,

$$\lim_{n \rightarrow \infty} \frac{1}{\sum_{k=0}^{N_{q, n}} (q \gamma_{(q-1)k+1})^q} \sum_{k=0}^{N_{q, n}} \xi_e^q (q \gamma_{(q-1)k+1})^q A^q f(\bar{X}_{\Gamma_{(q-1)k+e}}) = \lim_{n \rightarrow \infty} \nu_{N_{q, n}}^{\eta'} (\xi_e^q A^q f) = 0,$$

where $\eta'_m = (q \gamma_{(q-1)m+1})^q$. For conclusion, if $\lim_{n \rightarrow \infty} H_{\tilde{\eta}_q, n} / \tilde{H}_{q, n} = 0$, we have

$$\mathbb{P} - a.s., \lim_{n \rightarrow \infty} \frac{1}{\tilde{H}_{q, n}} \sum_{k=0}^{N_{q, n}} \sum_{e=0}^{q-2} \mathcal{R}_q f(\bar{X}_{\Gamma_{(q-1)k+e}}, \gamma_{(q-1)k+1}, e) - \gamma_{(q-1)k+1}^q \left(\sum_{j=1}^q \frac{\eta_{e+1}^{k-j}}{j!} \right) A^q f(\bar{X}_{\Gamma_{(q-1)k+e}}) = 0.$$

We gather all the terms together and the proof is completed. $\square$

6 NUMERICAL EXPERIMENTS

In the previous section, we have examined the theoretical perspective. In the following, we will utilize `Python` to conduct numerical experimental simulations to verify the theoretical results.

We first need to choose a suitable process which should have an invariant distribution as time t tends to infinity. By writing our process in the following form

$$dX_t = b(t, X_t)dt + \sigma(t, X_t)dW_t,$$

we have to pick appropriate functions σ and b . One option is the Ornstein–Uhlenbeck process. Let σ be a positive constant and $b(x, y) = \theta(\mu - y)$ with $\theta > 0$. Then, we may have

$$X_t \sim \mathcal{N}(\mu + (x_0 - \mu)e^{-\theta t}, \frac{\sigma^2}{2\theta}(1 - e^{-2\theta t})),$$

where we mention that its invariant distribution is $\mathcal{N}(\mu, \sigma^2/(2\theta))$. Thereafter, for the numerical simulation, we take $\theta = 2$, $\sigma = 2$ and $\mu = 1$. Thus, the invariant distribution becomes $\mathcal{N}(1, 1)$.

To get a basic idea, we visualize the general properties of the Ornstein–Uhlenbeck process by showing the following two figures.

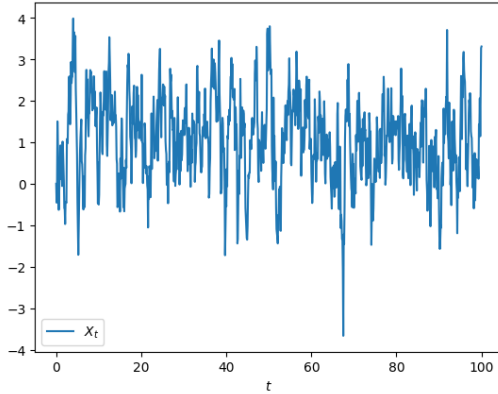


Figure 1: Trajectory of O-U process

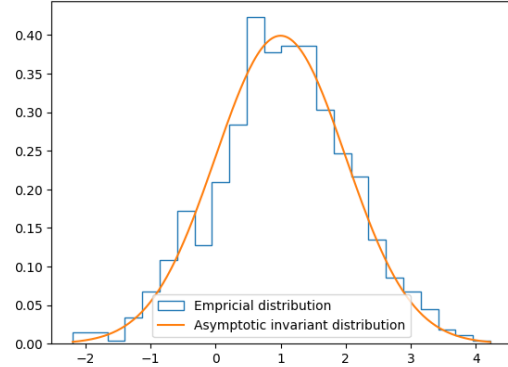


Figure 2: Distribution of O-U process

6.1 CONVERGENCE OF THE EULER SCHEME

We may now verify that, given f enough regular, by noting ν the invariant distribution, one can have $\nu_n^\gamma(f) \rightarrow \nu(f)$ almost surely, where

$$\nu_n^\gamma(f) := \frac{1}{\Gamma_n} \sum_{k=1}^n \gamma_k f(\overline{X_{\Gamma_{k-1}}})$$

with $\Gamma_n = \sum_{k=1}^n \gamma_k$ and $\overline{X_t}$ the Euler Scheme, which is the theorem that we have shown.

For the numerical computation of X_t , for general functions b and σ , we use the so-called Euler Scheme. Additionally, in order to consider γ , we define here a stochastic difference equation:

$$\overline{X_{\Gamma_{i+1}}} - \overline{X_{\Gamma_i}} = b(\Gamma_i, \overline{X_{\Gamma_i}})\gamma_{i+1} + \sigma(\Gamma_i, \overline{X_{\Gamma_i}})\sqrt{\gamma_{i+1}}Z_i,$$

where $(Z_i)_{i \geq 0}$ are standard Gaussian.

Finally, consider γ_k in order to satisfy the conditions of the theorem, we take here $\gamma_k = k^{-\alpha}$ with $\alpha \in (0, 1)$, such that $\gamma_k \rightarrow 0$ and $\Gamma_k \rightarrow +\infty$ when $k \rightarrow +\infty$. Taking $\alpha = 0.3, 0.5$ and 0.7 , we can show the histogram at $n = 10000$, with $f(x) = x$ which gives us the expectation, in the figure 3.

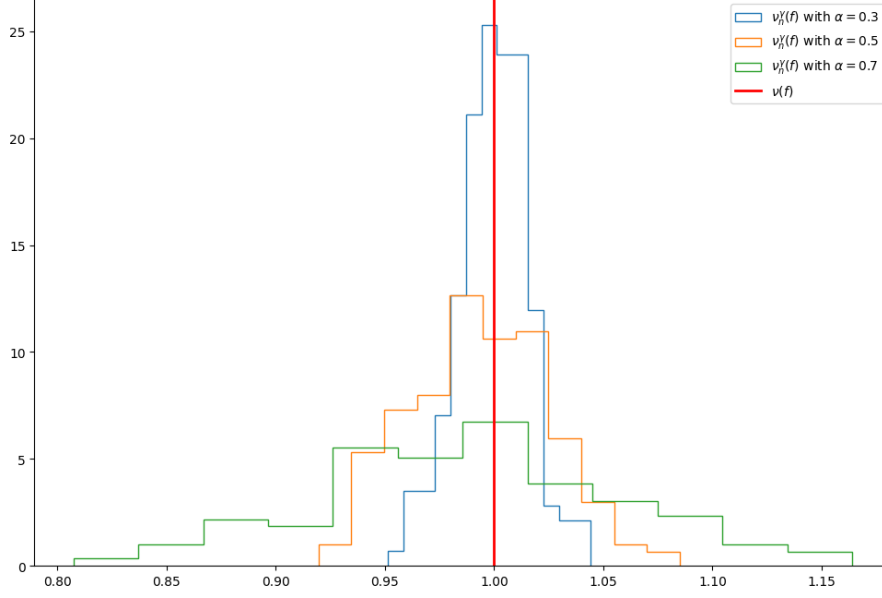


Figure 3: Histogram of $\nu_n^\gamma(f)$ for different α

We may mention that its histogram looks like the normal distribution, because this process, at the time t fix, is a Gaussian. Then, we show its convergence by increasing n in the figure 4.

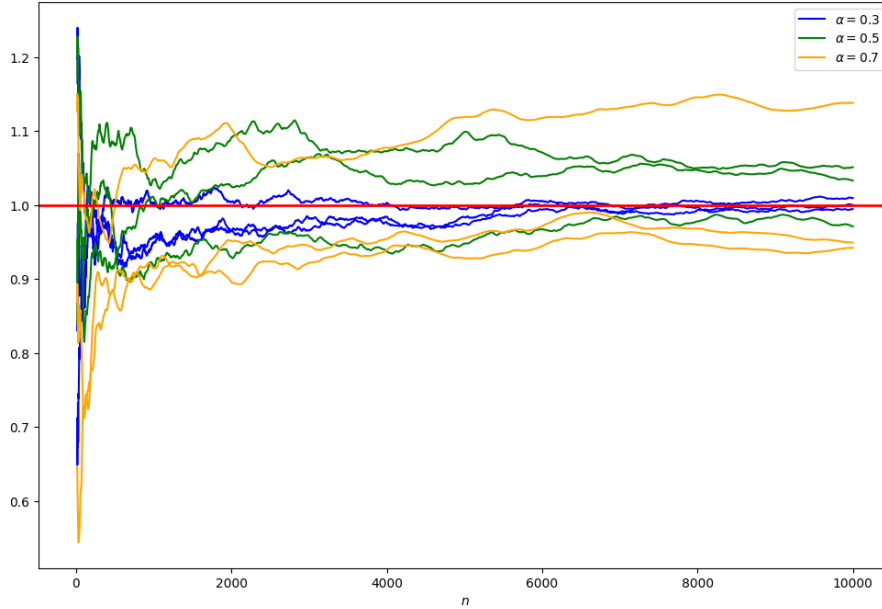


Figure 4: Values of $\nu_n^\gamma(f)$ for different α

With these two figures, we note that $\alpha = 0.3$ seems to be the best case. From the Remark 3.4 of reference

[2], we know that, under certain conditions, the optimum is obtained at $\alpha = 1/3$, which we will discuss in the following subsection.

6.1.1 • RELATED APPLICATION

An application of this framework interesting is to prove the almost sure Central Limit Theorem, which we can find as the Theorem 7 of reference [3]. To simplify the conditions, we may consider the one-dimensional case. Let (U_n) be a sequence of independent and identically distributed square-integrable random variables, satisfying $\mathbb{E}[U_1] = 0$ and $\text{Var}(U_1) = 1$. Then, we have

$$\frac{1}{\log n} \sum_{k=1}^n \frac{1}{k} \delta_{(U_1 + \dots + U_k)/\sqrt{k}} \rightarrow \mathcal{N}(0, 1)$$

in distribution almost surely.

We now want to verify numerically this theorem. By denoting $\nu = \mathcal{N}(0, 1)$ and $\nu_k = \delta_{(U_1 + \dots + U_k)/\sqrt{k}}$, we want to show that

$$\frac{1}{\log n} \sum_{k=1}^n \frac{1}{k} \nu_k(f) \rightarrow \nu(f)$$

almost surely, whose form is very similar to what we have concerned if we take $\gamma_k = 1/k$.

We now want to verify numerically this theorem. In the numerical simulation, we may use $U_1 \sim \mathcal{U}(-\sqrt{3}, \sqrt{3})$ the uniform distribution, and take $f(x) = x$ for the expectation and $f(x) = x^2$ for the variance.

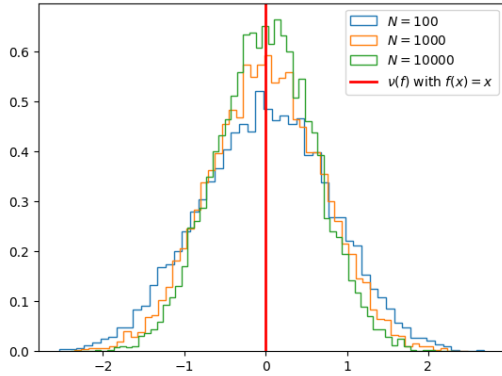


Figure 5: Convergence of expectation

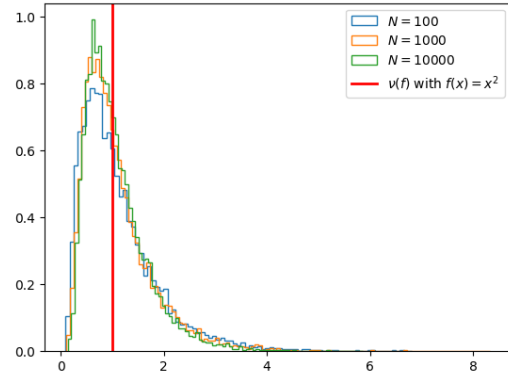


Figure 6: Convergence of variance

We can see that the distribution becomes concentrated as N becomes larger. Since logarithmic functions do not grow as fast as power functions, the rate of its convergence is not very fast even for large N .

6.2 RATE OF CONVERGENCE: A DIRECT ROUGH STUDY

When we wish to accelerate the rate of convergence, for specific problems, there are two possible cases to analyse: adjusting γ_k , in particular the exponent α ; and for the specific Ornstein–Uhlenbeck process, a better approach than the Euler Scheme can be used.

6.2.1 • CHOICE OF γ_k

Under the assumption that $\gamma_k = k^{-\alpha}$, we may note numerically that although α can take on all values between $(0, 1)$, once α takes on a value close to 0, it will lead to an explosion of constants.

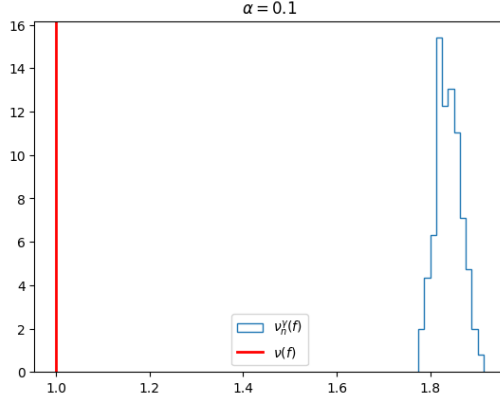


Figure 7: Histogram of $\nu_n^\gamma(f)$ for $\alpha = 0.1$

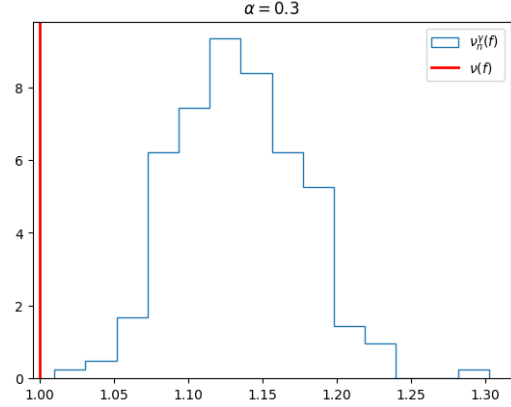


Figure 8: Histogram of $\nu_n^\gamma(f)$ for $\alpha = 0.3$

We used here $f(x) = (x - \mu)^2$ which gives us the variance. There may be two factors taken into account, one is that the value of Γ_n will become smaller when α becomes larger, which will contribute less to X_T which is close to the invariant distribution, for T given and large; on the other hand, γ_k will become larger when α becomes smaller, which will reduce the fineness and cause the Euler Scheme to be different from the real one.

The solution to this problem can actually be very simple, and in order to ensure that γ_k is not too large when α is small, it is sufficient to perform a translation. Specifically, we can define

$$\gamma_k = (k + n_0)^{-\alpha},$$

where n_0 is a fixed constant. Our next step will be about the choice of α .

According to the Theorem 3.2 of reference [2], in case of diffusion process, for f of the form Ag , where A is the infinitesimal generator (see (17)) and g is a function, the optimal α will be $1/3$. More precisely, a process is a diffusion process when both b and σ are functions of X_t itself. Taking $g(x) = (x - 1)/(1 + (x - 1)^2)$, we calculate then

$$\begin{aligned} f(x) &= Ag(x) = b(x)g'(x) + \frac{1}{2}\sigma^2(x)g''(x) \\ &= 2(1-x) \cdot \frac{1-(x-1)^2}{(1+(x-1)^2)^2} + 2 \cdot \frac{-4(x-1)}{(1+(x-1)^2)^3} = \frac{2(1-x)(5-(x-1)^4)}{(1+(x-1)^2)^3}. \end{aligned}$$

We mention that $x \mapsto f(x+1)$ is an odd function, hence $\nu(f)$ is exactly 0. We may take here $n = 10000$ and show the normalized absolute errors in heat-map form.

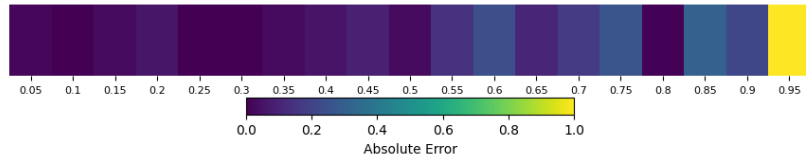


Figure 9: Normalized absolute errors of $\nu_n^\gamma(f)$ for different α

This heat-map figure 9 shows that the error is significantly lower around the theoretical optimal choice of α .

6.2.2 • RECURSIVE FORMULA OF THE ORNSTEIN–UHLENBECK PROCESS

The Ornstein–Uhlenbeck process can actually be simulated directly because we have a concrete solution:

$$X_t = \mu + (x_0 - \mu) \exp(-\theta t) + \sigma \int_0^t \exp(-\theta(t-s)) dW_s.$$

While, it may be emphasised that since the γ_k is not homogeneous (it gets smaller as k gets larger), we cannot use the conventional simulation method of dividing the time equally. Obviously, the detail we want to deal with here is the treatment of the integral, so we want to compute the integral numerically in steps of γ_k .

By denoting

$$I_i = \exp(-\Gamma_i) \int_0^{\Gamma_i} \exp(\theta s) dW_s,$$

we will have

$$\exp(\theta \Gamma_i) I_i - \exp(\theta \Gamma_{i-1}) I_{i-1} = \int_{\Gamma_{i-1}}^{\Gamma_i} \exp(\theta s) dW_s.$$

So, if the integral is replaced by $\exp(\theta \Gamma_{i-1}) \sqrt{\gamma_i} Z_i$, we may have the discretised recursive formula for I_i :

$$\bar{I}_i = \exp(-\theta \gamma_i) \bar{I}_{i-1} + \exp(-\theta \gamma_i) \sqrt{\gamma_i} Z_i.$$

We can notice here that Z_i is preceded by an exponent in addition to $\sqrt{\gamma_i}$, which increases a little the speed of convergence somewhat compared to the Euler Scheme, as we will show in the following numerical simulations. We fix here the $f : x \mapsto (x - \mu)^2$ giving the variance.

We also want to verify that $\nu_n^\gamma(f)$ will converge to $\nu(f)$ almost surely, and compare to the convergence of the Euler Scheme, where $\bar{X}_t$ is replaced by X_t in the expression of $\nu_n^\gamma(f)$. This is proved in the previous section: for diffusion process, there can be convergence for the direct use of the exact X_t when right conditions are satisfied.

As our functions b and σ are extremely simple affine or constant functions, the conditions for regularity are satisfied. It is sufficient to take $V(x) = x^2$ as the Lyapunov function to satisfy the rest of the conditions. Thus, the theoretical proof tells us that there is indeed an almost sure convergence, which we are going to verify by numerical simulation.

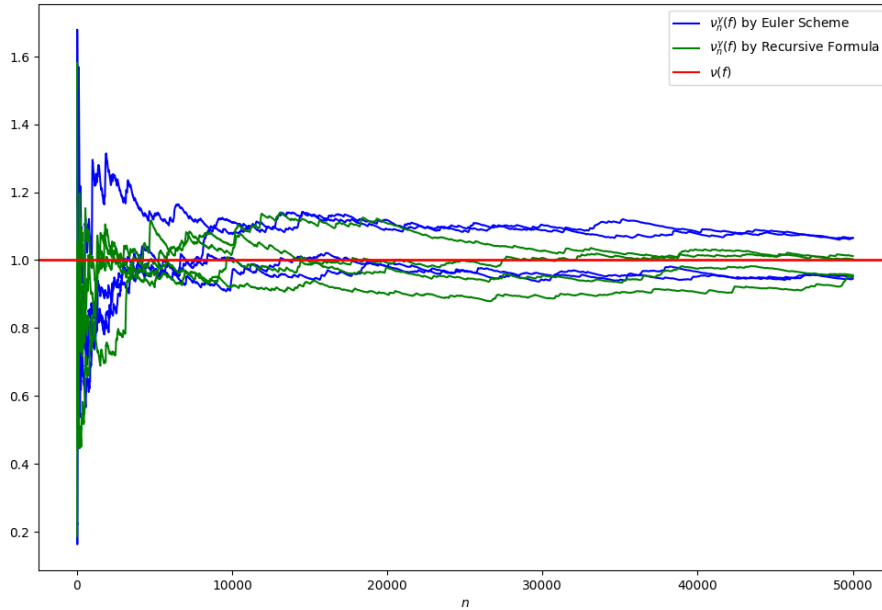


Figure 10: Values of $\nu_n^\gamma(f)$ for different approach

We can notice a slight advantage of using recursive formula over Euler Scheme, most likely due to the factor $\exp(-\theta \gamma_i)$. It is clear that this advantage is not obvious. This is probably because both the Euler Scheme and the recursive formula have the same order, first-order error.

6.3 RATE OF CONVERGENCE: AN IN-DEPTH STUDY USING CLT

In this section, we introduce numerical methods to verify our results about functional central limit theorem and explore their connections with the Newton-Cotes formula. As a starting point, we provide a detailed computation of the coefficients c_q as presented in [2], complementing our discussion in Proposition 6 in Section 5.1. Furthermore, we explain the necessity of proposing a generalized approach for $q \geq 4$.

To simplify the calculations, we set $\mu = 0$ from now on, which ensures that the invariant distribution is $\mathcal{N}(0, 1)$. The function under consideration is

$$f(x) = Ag(x) = A\left(\frac{x}{1+x^2}\right) = -\frac{2x(5-x^4)}{(1+x^2)^3},$$

which is required to take the form $f = Ag$ with $g(x) = x/(1+x^2)$. According to [2], the quantity $\nu(\mathfrak{D}f)$ in Theorem 5, as being examined in our numerical experiments, is given by

$$\nu(\mathfrak{D}f) = \nu(|\sigma f'|^2) = \int |2\frac{1-x^2}{(1+x^2)^2}|^2 \nu(dx) = \frac{4}{\sqrt{2\pi}} \int \frac{(1-x^2)^2}{(1+x^2)^4} \exp(-\frac{x^2}{2}) dx.$$

Since obtaining a closed-form solution for this expression is challenging, we instead rely on numerical approximations in the subsequent analysis, about 1.1258.

6.3.1 • COMPUTATION OF COEFFICIENTS

We begin by addressing the tasks outlined in Remark 4. It will be evident that for $q = 1, 2$, the computations remain straightforward. However, for $q = 3$, a complication already arises, and for $q = 4$, a significant difficulty emerges. This observation motivates us to seek a generalized solution. We have firstly defined that $c_0^1 = 1$.

For $q = 2$, we have

$$\xi^{2,2} = \left(\frac{1}{2}, 0\right)^\top.$$

Hence

$$c_0^2 = 1 + \xi_0^{2,2} \cdot (-1) + \xi_1^{2,2} \cdot 1 = \frac{1}{2},$$

and

$$c_1^2 = \xi_0^{2,2} \cdot 1 + \xi_1^{2,2} \cdot (-1) = \frac{1}{2}.$$

For $q = 3$, we have firstly

$$\xi^{3,3} = \left(\frac{1}{6}, \frac{1}{6}, 0\right)^\top.$$

We obtain then

$$N^{3,3}(\xi^{3,3}) = \left(\frac{1}{2} - \xi_0^{3,3}, \frac{1}{2} + \xi_0^{3,3} - \xi_1^{3,3}, \xi_1^{3,3} - \xi_2^{3,3}\right)^\top = \left(\frac{1}{3}, \frac{1}{2}, \frac{1}{6}\right)^\top,$$

and thanks to c^2 , one may deduce

$$M^{3,3} = \begin{pmatrix} 1/2 & 0 \\ 1/2 & 1/2 \\ 0 & 1/2 \end{pmatrix}.$$

We observe that it is hard to directly obtain $\xi^{3,3}$ through the pseudo-inverse of the matrix $M^{3,3}$. Therefore, we introduce the coefficients of the Newton-Cotes formula as a reference. One can obtain that

$$c^3 = (1 - \xi_0^{3,2}, 1 + \xi_0^{3,2} - \xi_1^{3,2}, \xi_1^{3,2})^\top.$$

Since the Newton-Cotes formula gives

$$c^3 \propto \left(\frac{1}{6}, \frac{2}{3}, \frac{1}{6}\right)^\top,$$

we deduce from mentioning the sum of all its components is 2 that

$$\xi^{3,2} = \left(\frac{2}{3}, \frac{1}{3}\right)^\top \text{ and } c^3 = \left(\frac{1}{3}, \frac{4}{3}, \frac{1}{3}\right)^\top.$$

Finally, one may conclude that

$$M^{3,3}\xi^{3,2} = \begin{pmatrix} 1/2 & 0 \\ 1/2 & 1/2 \\ 0 & 1/2 \end{pmatrix} \begin{pmatrix} 2/3 \\ 1/3 \end{pmatrix} = \begin{pmatrix} 1/3 \\ 1/2 \\ 1/6 \end{pmatrix} = N^{3,3}(\xi^{3,3}).$$

For $q = 4$, we consider directly c^4 . One has

$$c^4 = (1 - \xi_0^{4,2}, 1 - \xi_1^{4,2}, 1 + \xi_0^{4,2}, \xi_1^{4,2})^\top.$$

We notice the problem that $\xi^{4,2}$ is only of dimension 2, while c^4 is of dimension 4 or affinely of dimension 3, which means that normally this system of equations has no solution. In particular, there is no solution when applying the coefficients of Newton-Cotes. We return to dealing with the general equation

$$M^{q,p}\xi^{q,p-1} = N^{q,p}(\xi^{q,p}).$$

Since $q \geq p$, one may have the same problem that it may have no solution. We show then that this problem will soon appear. We have firstly

$$\xi^{4,4} = \left(\frac{1}{24}, \frac{1}{24}, \frac{1}{24}, 0\right)^\top,$$

hence

$$N^{4,4}(\xi^{4,4}) = \begin{pmatrix} 1/6 - \xi_0^{4,4} \\ 1/6 + \xi_0^{4,4} - \xi_1^{4,4} \\ 1/6 + \xi_1^{4,4} - \xi_2^{4,4} \\ \xi_2^{4,4} - \xi_3^{4,4} \end{pmatrix} = \begin{pmatrix} 1/8 \\ 1/6 \\ 1/6 \\ 1/24 \end{pmatrix}.$$

On the other hand, we have

$$M^{4,4} = \begin{pmatrix} 1/2 & 0 & 0 \\ 1/2 & 1/2 & 0 \\ 0 & 1/2 & 1/2 \\ 0 & 0 & 1/2 \end{pmatrix}.$$

We mention that $N^{4,4}(\xi^{4,4}) \notin \text{Im}(M^{4,4})$, i.e. $\xi^{4,3}$ is not defined.

6.3.2 • NUMERICAL CLT STUDY

In the previous subsection, we have identified the difficulties in computing c_q for $q \geq 4$ as presented in [2], while also establishing a connection with the Newton-Cotes formula. Therefore we first examine and analyze the three cases for $q \leq 3$, verifying the optimal convergence behavior and assessing whether the convergence results are reasonable. Furthermore, we may conjecture whether the coefficients of the Newton-Cotes formula can be directly applied to the case of $q = 4$.

We first recall the standard results. Previously, we have proven the convergence of $\nu_n(Af)$. To characterize its convergence rate, we introduce a scaling factor $H_n/\sqrt{H_{\epsilon_x,n}}$ such that it converges in distribution to a normal distribution. As before, we consider $\gamma_n = O(1/n^\alpha)$, and from the proof of the backward induction theorem, we observe that this scaling factor is determined by the dominant term between $\sqrt{n^{1-\alpha}}$ and $n^{q\alpha}$. Consequently, a natural choice is $\alpha = 1/(2q+1)$, leading to a convergence rate of $n^{q/(2q+1)}$. In what follows, we shall refer to this rate as the q -th order convergence rate.

We first present the case of $q = 1$. In fact, this is nearly identical to the previously studied case, since here $\eta_n \propto \gamma_n$. Using the previously established recursive formula, we can implement numerical experiments for convergence at any order. Based on theoretical proofs and computations, we can also construct the target distribution. Finally, in addition to verifying the standard theoretical results, we illustrate that an incorrect

convergence rate can lead to a histogram that significantly deviates from the standard normal shape. To recall, the convergence rate here is $O(n^{1/3})$.

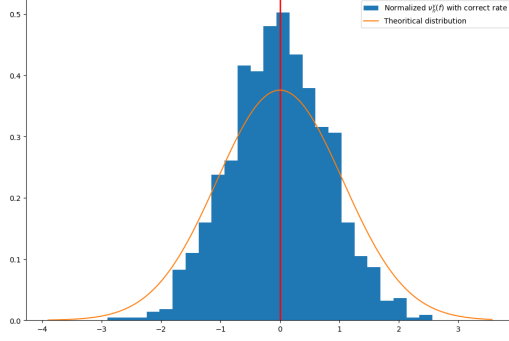


Figure 11: Simulation of case $q = 1$

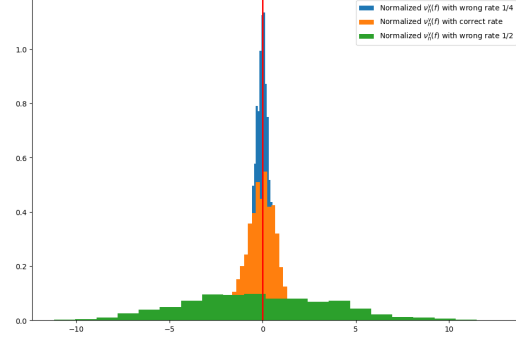


Figure 12: Different convergence rates

To employ a more statistical approach in testing for normality, we use the Shapiro-Wilk normality test. For the case of $q = 1$, we obtain a p -value of 0.9829.

For the case of $q = 2$, the results are highly similar. In particular, the article [3] explicitly mentions that the target distribution of this theory, especially its variance, is independent of the convergence order. Therefore, we present again two figures: one demonstrating the strong agreement with the theoretical convergence, and the other illustrating the accuracy of the convergence order.

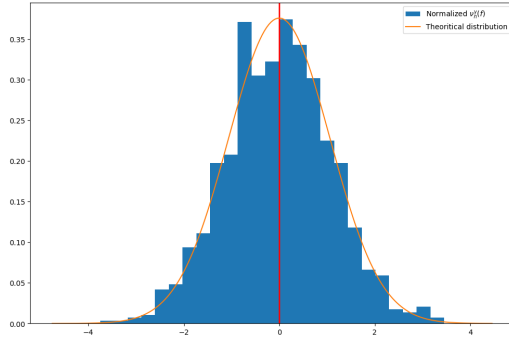


Figure 13: Simulation of case $q = 2$

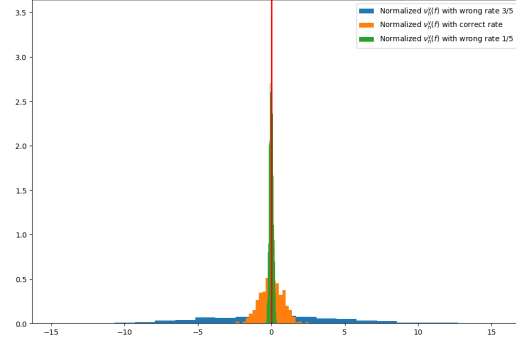


Figure 14: Different convergence rates

In the numerical experiments, these two cases were conducted with similar choices of N as the length of the sequence (γ_n) , yet the convergence effects exhibit noticeable differences, which is a consequence of varying q . On the other hand, applying the Shapiro-Wilk normality test yields a p -value of 0.8777, indicating a strong agreement with the Gaussian distribution. To recall, the convergence rate here is $O(n^{2/5})$.

For $q = 3$, an essential difference arises. In this case, γ_n has now become a sequence with two repeating components, and the computation of η_n becomes more intricate. However, the convergence rate is $O(n^{3/7})$, which represents a small improvement compared to the previous cases.

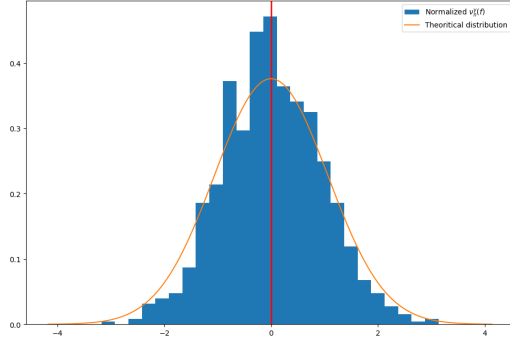


Figure 15: Simulation of case $q = 3$

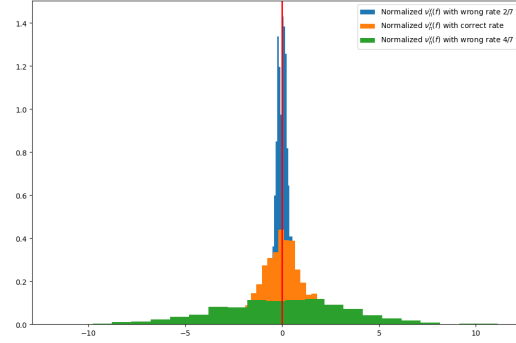


Figure 16: Different convergence rates

This time, the Shapiro-Wilk normality test yields a p -value of 0.9205. On the other hand, an issue that requires attention gradually emerges: as q increases, the change in the convergence rate $q/(2q+1)$ becomes noticeably smaller. For example, for $q = 3$, in the right-hand figure, a difference of $O(n^{1/7})$ in the convergence rate is evident, meaning that for $N = 3^7 = 2187$, a discernible difference in the histogram appears. However, when using the convergence rate corresponding to $q = 2$, the difference is about $O(n^{1/35})$, which is extremely subtle. In our simulation, with $N \sim 6000$, the corresponding histogram undergoes at most a horizontal stretching by a factor of 1.2822, which remains relatively minor.

Although there is no solution for $q = 4$, we can still forcibly apply the Newton-Cotes formula and conduct simulations, which will significantly amplify the aforementioned issue. In this case, one has $c'_4 = [3/8, 9/8, 9/8, 3/8]$.

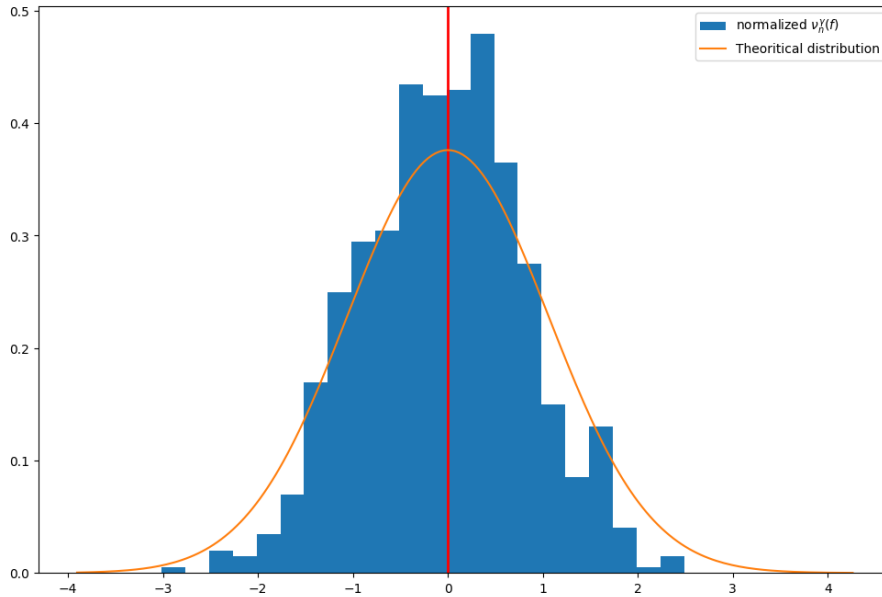


Figure 17: Simulation of case $q = 4$

This time, the p -value of the normality test remains at 0.7116. However, we do not have a theoretical justification to assert that its convergence rate is exactly $O(n^{4/9})$. In fact, considering the difference in convergence rates between $q = 3$ and $q = 4$, it is only $O(n^{1/63})$. If we take $N = 6000$, the horizontal scaling factor is only 1.1481, which is almost imperceptible.

In reality, this highlights the limitations or practical constraints of numerical methods: in actual applications, using $q = 2$ or $q = 3$ already provides sufficiently good results, especially when the exact solution is not known,

unlike in the case of the Ornstein-Uhlenbeck process. Instead, one must rely on numerical schemes such as the Euler Scheme or the Talay Second Weak Order Scheme.

Nonetheless, if we still wish to roughly observe the convergence rate, we propose a log-log method and employ linear fitting to estimate it.

6.3.3 • LOG-LOG METHOD

We here introduce the basic idea of the log-log method. The main consideration is the behavior of $\nu_n(f) - \nu(f)$ as a random variable. If we denote the convergence rate by r , then by definition, we can express it as $n^r(\nu_n(f) - \nu(f)) \rightarrow \mathcal{N}(0, D_f)$, where D_f is a fixed constant depending on f and the original process. When n is sufficiently large, we may assume that both sides are approximately equal. Taking the variance on both sides yields

$$n^{2r} \text{Var}(\nu_n(f) - \nu(f)) \approx D_f.$$

Applying the logarithm to both sides and rearranging, we obtain

$$\log \text{Var}(\nu_n(f) - \nu(f)) \approx -2r \cdot \log(n) + \log(D_f).$$

Thus, by plotting $\log(n)$ against $\log \text{Var}(\nu_n(f) - \nu(f))$ and performing a linear fit at sufficiently large n , we can obtain an approximate estimate of the convergence rate.

In practical implementation, ensuring accuracy requires balancing two aspects: on one hand, selecting an excessively long segment at the tail may lead to the failure of the approximation, while on the other hand, choosing too few data points may result in excessive fluctuations, degrading the quality of the linear fit. As a compromise, we roughly use the last one-third of the data for fitting. It is important to note that, due to the logarithmic transformation, the displayed segment will actually be quite short. Next, we present examples for $q = 2$ and $q = 3$.

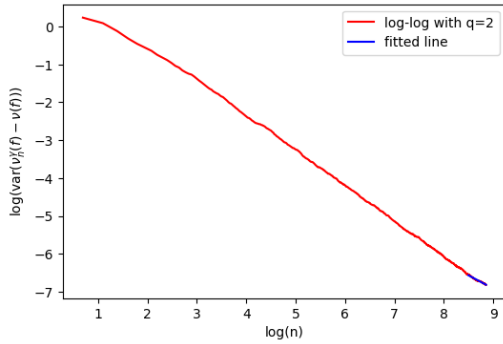


Figure 18: log-log Method of case $q = 2$

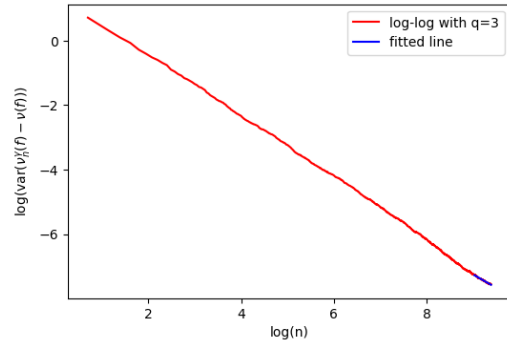


Figure 19: log-log Method of case $q = 3$

It can be observed that, overall, the plot is approximately linear, demonstrating the effectiveness of this approach. Here, the fitted convergence rate for $q = 2$ is 0.3494, while for $q = 3$, it is 0.4146, which is close to the theoretical values.

For the case of $q = 4$, where Newton-Cotes is applied and the theoretical rate is unknown, we use the same method to generate a plot for rough estimation.

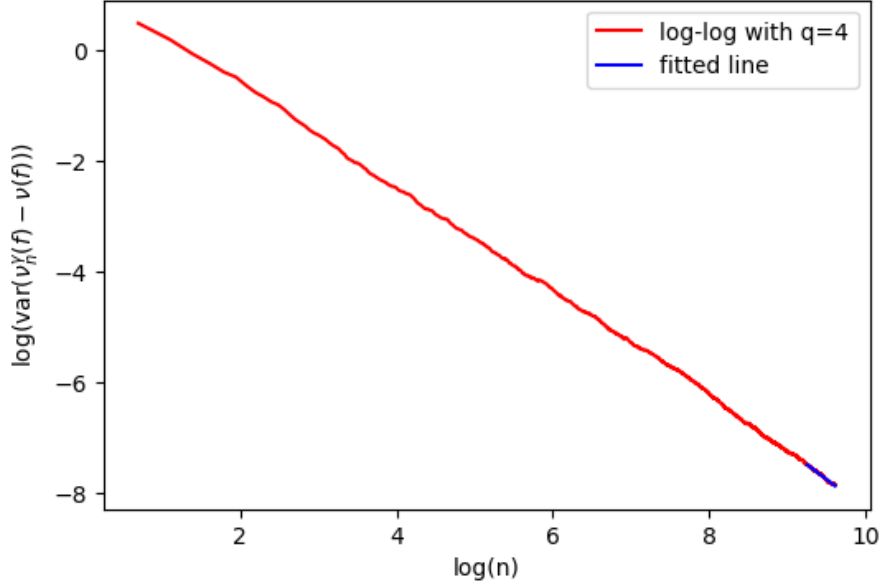


Figure 20: Simulation of case $q = 4$

It can be seen that the result is also quite satisfactory, with the fitted estimate indicating a convergence rate of 0.4846, suggesting a certain level of effectiveness. However, since a complete theoretical framework, particularly an accompanying error analysis, has not been established for this method, it should only be regarded as a rough reference.

6.4 RATE OF CONVERGENCE: A GENERALIZED STUDY

Due to the theoretical difficulty encountered for $q \geq 4$ in [2], we have developed a new and more general approach in the previous sections to address similar problems. However, since the theoretical framework is evidently less complete than that in [2], our investigation here is limited to numerical exploration in this section.

6.4.1 • NUMERICAL APPROACH OF CONVERGENCE RATE

In this section, we first demonstrate the applicability of the new method to the case of $q = 3$, providing partial validation of its effectiveness. We then extend our analysis to the cases of $q = 4$ and $q = 5$. Subsequently, we employ the previously introduced log-log method to estimate the convergence rate. Here, the exponent α in γ_n is still of the form $1/(2q - 1)$.

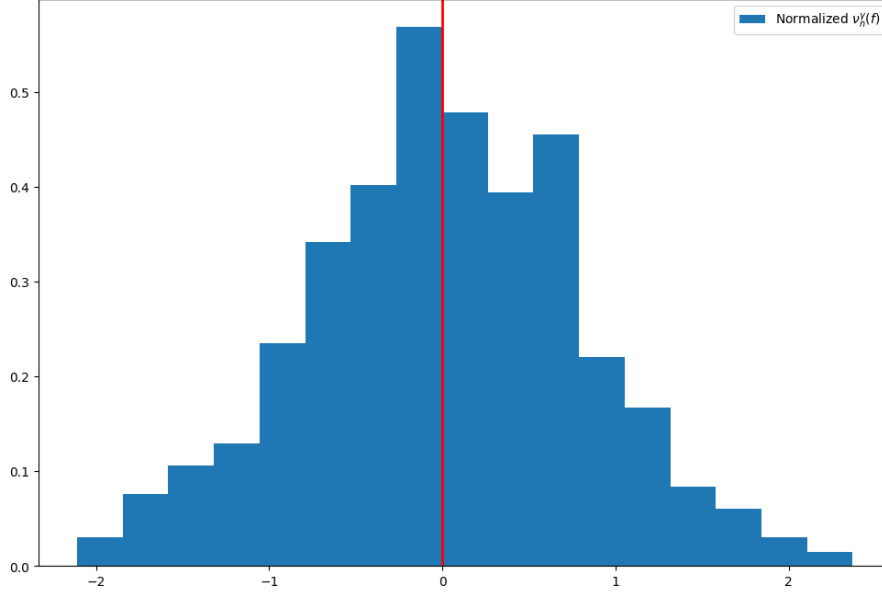


Figure 21: Simulation of case $q = 3$

The corresponding normality test yields a p -value of 0.6840, which remains highly reliable. However, it is important to note that although $q = 3$, the convergence rate is only $O(n^{2/5})$. Below, we present two improved convergence results, particularly for $q = 5$, where the theoretical convergence rate can reach $O(n^{4/9})$.

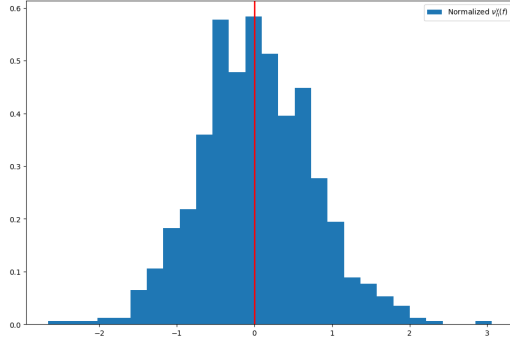


Figure 22: Simulation of case $q = 4$

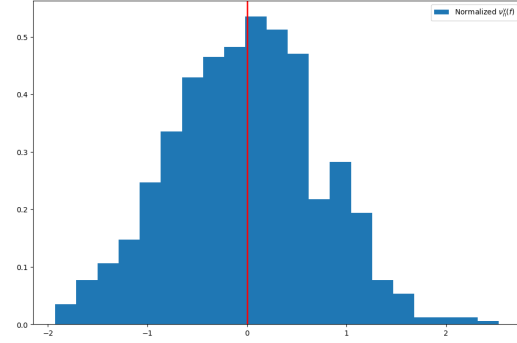


Figure 23: Simulation of case $q = 5$

On one hand, both cases pass the normality test, confirming the reliability of the results. On the other hand, it can be observed that the general distribution range remains fixed, indicating that the convergence rate is reasonably valid. To further examine the convergence rate, we introduce the previously proposed log-log method. For simplicity, we directly consider the two key cases of $q = 4$ and $q = 5$.

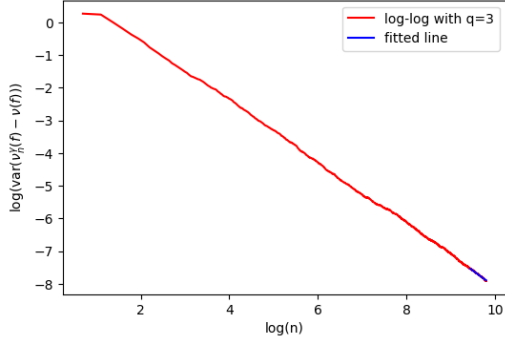


Figure 24: log-log Method of case $q = 4$

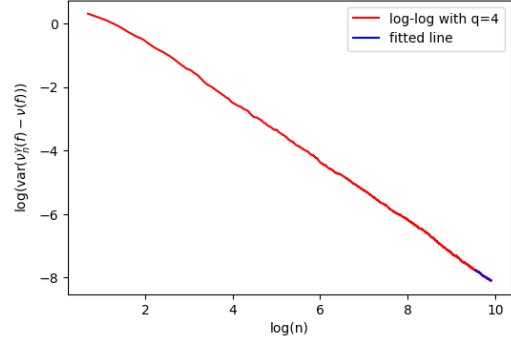


Figure 25: log-log Method of case $q = 5$

It is important to first emphasize that in the code, the q represents the order of the theoretical convergence rate. For convenience in coding, the q shown in the figure should be understood as corresponding to our $(q+1)$ -th order method.

The fitting results indicate here that the estimated convergence rate for $q = 4$ is 0.5006, while for $q = 5$, it is 0.4671, both of which are close to $1/2$. From one perspective, this demonstrates the numerical effectiveness of the method in theoretical analysis. However, from another perspective, the proximity of these values suggests that the numerical estimation error might even exceed the theoretical error under consideration, such as the discrepancy of $1/63 \approx 0.0158$ in this case. This discrepancy will become more pronounced in the analysis of the following section.

6.4.2 • RELATED COMPARISON

In this section, we focus on supplementary comparative analyses. Firstly, we extend the previous comparisons between different values of q by considering cases with larger differences. Additionally, to explore the possibility of finding a better exponent α in γ_n , we utilize the results from the log-log method for a simple comparative analysis. Finally, through these numerical simulations, we illustrate both the partial effectiveness and the evident limitations of numerical methods in this context.

As previously mentioned multiple times, numerical methods have inherent limitations. In this case, the finiteness of N results in an indistinct separation of convergence rates, and numerical errors can significantly exceed the theoretical differences between rates. Next, we compare the cases of $q = 4$ and $q = 10$ to illustrate this phenomenon.

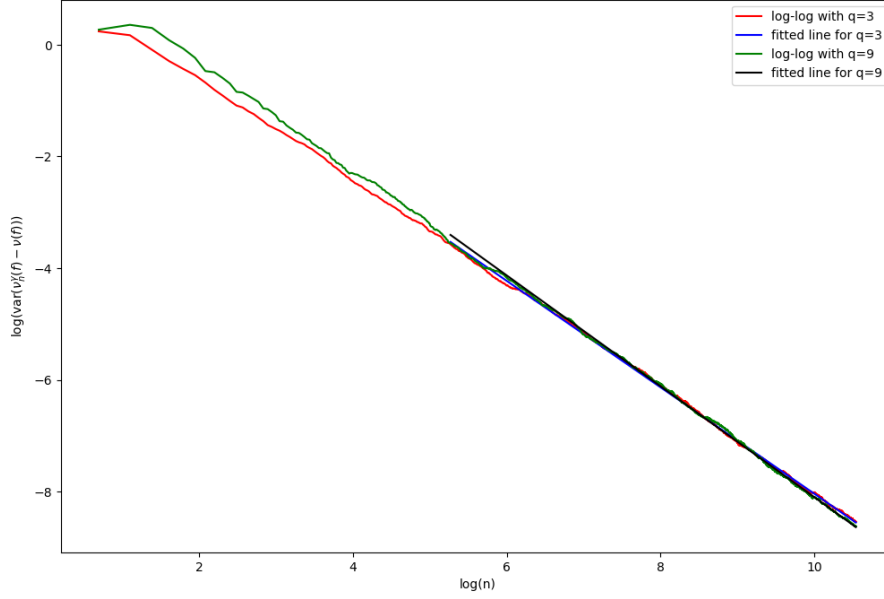


Figure 26: Comparison of $q = 4$ and $q = 10$

This figure illustrates why, in practice, smaller values of q are often sufficient. On one hand, it is evident that $q = 10$ results in a significantly larger constant, meaning that despite its theoretically faster rate, it remains inferior to $q = 4$ for smaller values of n . On the other hand, the black fitted line and the blue fitted line in the figure are very close to each other. This is because the theoretical difference in convergence rates is quite small, approximately 0.0451, implying that $q = 10$ only surpasses $q = 4$ when n is sufficiently large. Considering these two aspects, it follows that using $q = 10$ is not practically advantageous.

Next, we discuss a problem that was partially explored in the previous section—verifying the optimal choice for the convergence rate. Previously, we employed the normalized L^∞ -error for this purpose. Now, one revisits the question using the log-log method. Specifically, we vary the exponent α in γ_n and plot the corresponding convergence rates.

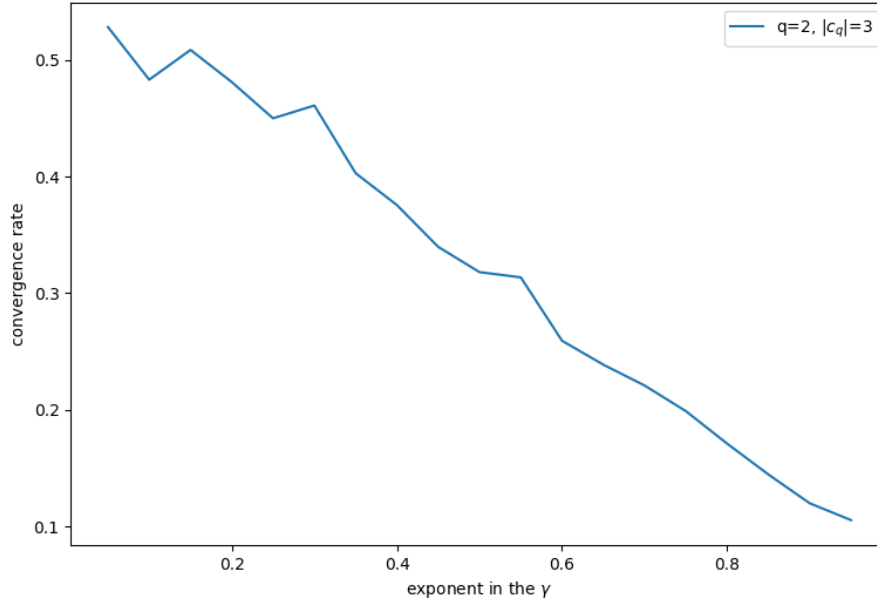


Figure 27: Different convergence of case $q = 3$

It can be observed that the convergence rate decreases significantly as the exponent α increases, indicating the rationality of choosing $\alpha = 1/(2q - 1)$. However, when we focus on a smaller interval such as $(0, 0.3)$ or even narrower ranges, the limitations of numerical methods become evident, preventing us from determining the optimal exponent precisely.

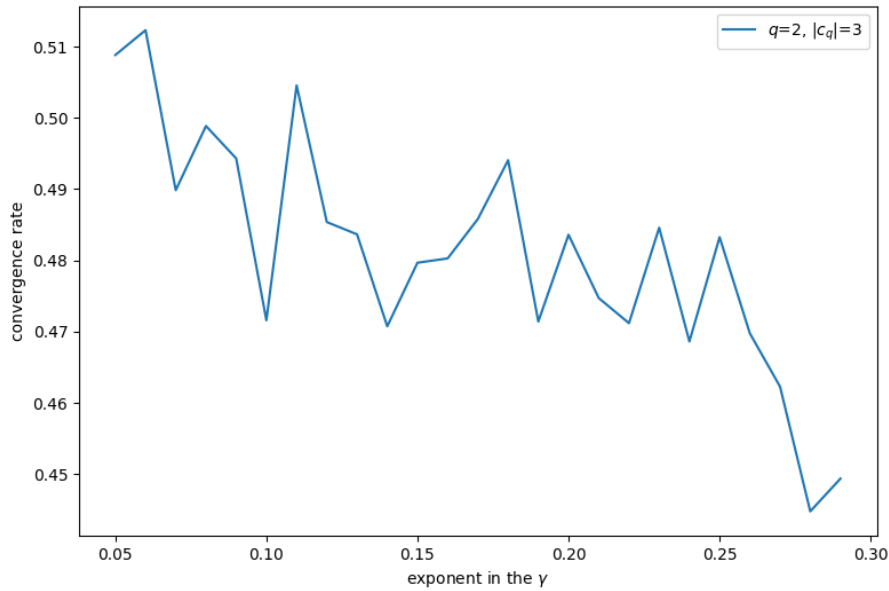


Figure 28: Different convergence of case $q = 3$ on $(0, 0.3)$

In this figure, we can barely observe the overall trend; however, it is evidently insufficient to determine the optimal exponent. Next, we present the cases for $q = 4$ and $q = 5$, where greater fluctuations will be observed.

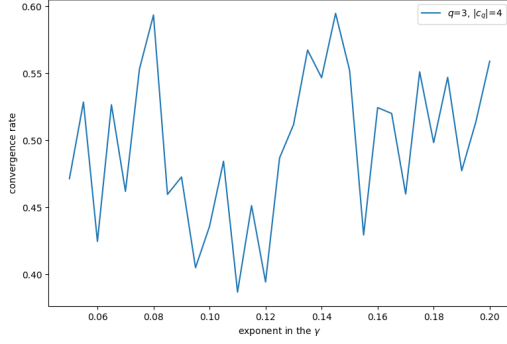


Figure 29: Differences of case $q = 4$

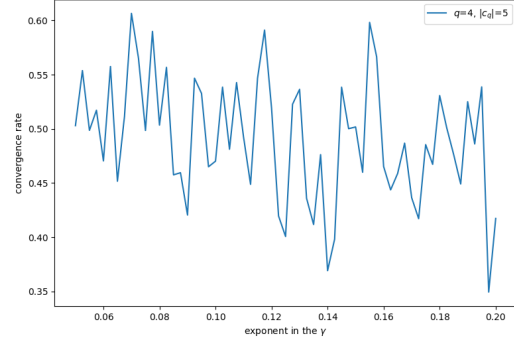


Figure 30: Differences of case $q = 5$

In these two cases, the numerical errors themselves are significantly exceed the theoretical difference between the two cases. This is also one of the core ideas in our theoretical analysis in second part of numerical experiments. On the other hand, it highlights the robustness of this methodological approach: regardless of variations in q or γ_n , slight deviations do not lead to fundamental changes, and the resulting convergence rate remains close to $1/2$. This is already an improvement over the trivial $1/3$ -convergence rate observed when $q = 1$.

7 CONCLUSION

In this article, our primary focus is to comprehend a series of foundational works, apply the key theorems presented to derive new results, and subsequently verify these results through numerical simulations. The central objective of this study is to develop a general methodology for determining the invariant measure of a stochastic differential equation, which enables us to achieve highly accurate approximations with the simulation of a single trajectory, significantly reducing the space-time complexity of the algorithm. A major component of our work also involves applying the proposed method to an alternative scheme, rigorously validating all the assumptions to deepen our understanding of the approach. Furthermore, we have numerically investigated optimization possibilities, such as identifying the optimal exponent to enhance convergence. Finally, within the framework of the central limit theorem, we prove and analyze the main results, propose a new general method to handle cases of arbitrary order, and validate and compare it through numerical simulations. We conclude by summarizing its advantages and limitations in practical applications.

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